

Comparison of CNN and Transformers using Brain Tumor Detection using MRI Images

Import All Required Libraries

```
# Basic utilities
import os
import random
import numpy as np
import pandas as pd
from tqdm import tqdm

# Visualization
import matplotlib.pyplot as plt
import seaborn as sns

# Image processing
from PIL import Image
import cv2

# TensorFlow & Keras
import tensorflow as tf
from tensorflow.keras.preprocessing.image import ImageDataGenerator
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import (Conv2D, MaxPooling2D, Flatten,
                                     Dense, Dropout,
                                     BatchNormalization,
                                     GlobalAveragePooling2D)
from tensorflow.keras.applications import VGG16, ResNet50,
EfficientNetB0
from tensorflow.keras.applications.vgg16 import preprocess_input as
vgg_preprocess
from tensorflow.keras.callbacks import EarlyStopping, ModelCheckpoint
from tensorflow.keras.utils import to_categorical

# Transformers
from tensorflow.keras import layers
from tensorflow.keras.models import Model

# Evaluation
from sklearn.metrics import classification_report, confusion_matrix,
roc_auc_score

# Ignore warnings
import warnings
warnings.filterwarnings("ignore")
```

```

# Set random seeds
SEED = 42
np.random.seed(SEED)
tf.random.set_seed(SEED)
random.seed(SEED)

2025-05-01 15:12:42.696575: E
external/local_xla/xla/stream_executor/cuda/cuda_fft.cc:477] Unable to
register cuFFT factory: Attempting to register factory for plugin
cuFFT when one has already been registered
WARNING: All log messages before absl::InitializeLog() is called are
written to STDERR
E0000 00:00:1746112362.865672      31 cuda_dnn.cc:8310] Unable to
register cuDNN factory: Attempting to register factory for plugin
cuDNN when one has already been registered
E0000 00:00:1746112362.912023      31 cuda_blas.cc:1418] Unable to
register cuBLAS factory: Attempting to register factory for plugin
cuBLAS when one has already been registered

```

Prepare Data Loaders for Training and Validation

```

# Define paths
train_dir = "/kaggle/input/brain-tumor-mri-dataset/Training"

# Constants
IMG_SIZE = (224, 224)
BATCH_SIZE = 32

# ImageDataGenerator with validation split
train_datagen = ImageDataGenerator(
    rescale=1./255,
    rotation_range=15,
    zoom_range=0.1,
    horizontal_flip=True,
    validation_split=0.2 # 20% for validation
)

# Training generator (80%)
train_generator = train_datagen.flow_from_directory(
    train_dir,
    target_size=IMG_SIZE,
    batch_size=BATCH_SIZE,
    class_mode='categorical',
    subset='training',
    shuffle=True,
    seed=42
)

```

```

# Validation generator (20%)
val_generator = train_datagen.flow_from_directory(
    train_dir,
    target_size=IMG_SIZE,
    batch_size=BATCH_SIZE,
    class_mode='categorical',
    subset='validation',
    shuffle=False,
    seed=42
)

# View class indices
print("Class mapping:", train_generator.class_indices)

Found 4571 images belonging to 4 classes.
Found 1141 images belonging to 4 classes.
Class mapping: {'glioma': 0, 'meningioma': 1, 'notumor': 2,
'pituitary': 3}

```

Dataset Exploration & Statistical Analysis Plan

Count Number of Images per Class

```

from collections import defaultdict

def count_images_per_class(folder_path):
    class_counts = defaultdict(int)
    for class_folder in os.listdir(folder_path):
        class_path = os.path.join(folder_path, class_folder)
        if os.path.isdir(class_path):
            class_counts[class_folder] = len(os.listdir(class_path))
    return class_counts

# Count images
train_counts = count_images_per_class(train_dir)

# Convert to DataFrame for plotting
train_df = pd.DataFrame(list(train_counts.items()), columns=['Class',
'Image_Count'])
train_df = train_df.sort_values(by='Image_Count', ascending=False)

# Plot
plt.figure(figsize=(8,5))
sns.barplot(data=train_df, x='Class', y='Image_Count')
plt.title("Image Count per Class in Training Set")
plt.xticks(rotation=45)

```

```
plt.tight_layout()
plt.show()
```

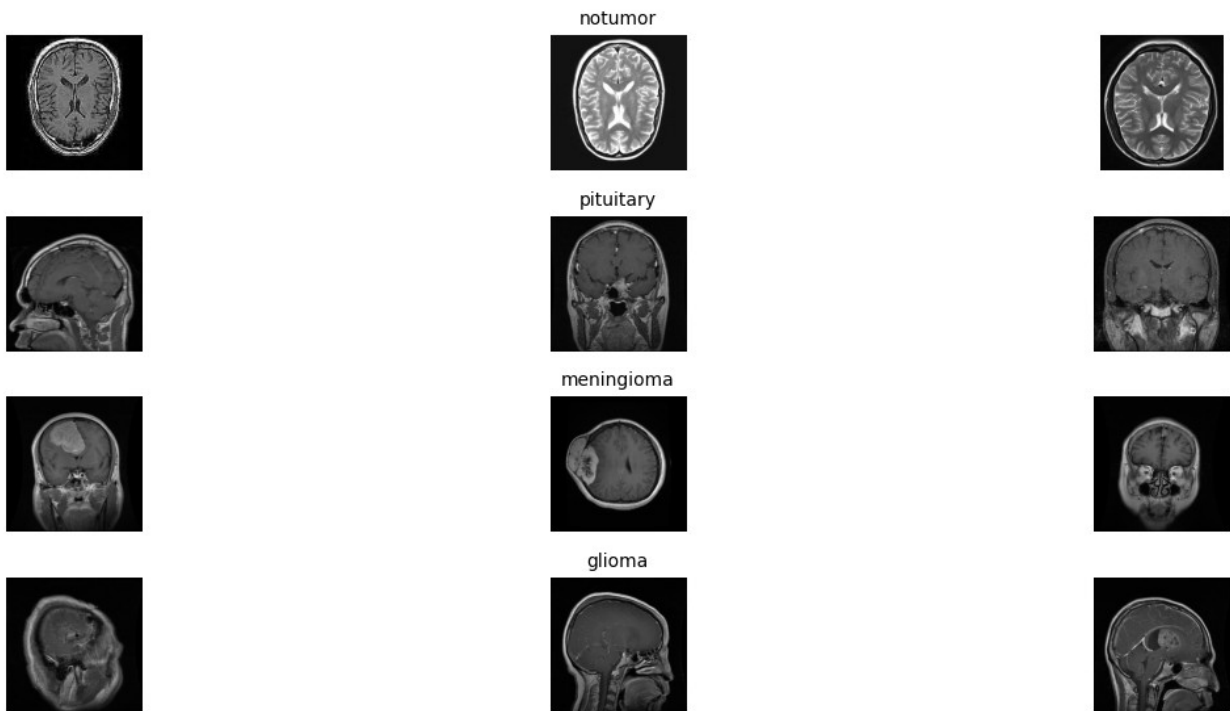


Show Sample Images from Each Class

```
def show_sample_images(folder_path, classes, n_images=3):
    plt.figure(figsize=(15, 6))
    for i, cls in enumerate(classes):
        class_folder = os.path.join(folder_path, cls)
        image_files = os.listdir(class_folder)[:n_images]
        for j, img_file in enumerate(image_files):
            img_path = os.path.join(class_folder, img_file)
            img = Image.open(img_path)
            plt.subplot(len(classes), n_images, i*n_images + j + 1)
            plt.imshow(img, cmap='gray')
            plt.axis('off')
            if j == 1:
                plt.title(cls, fontsize=10)
    plt.suptitle("Sample Images from Each Class", fontsize=16)
    plt.tight_layout()
    plt.show()

# Show 3 images from each class
show_sample_images(train_dir, train_df['Class'].tolist(), n_images=3)
```

Sample Images from Each Class



Analyze Image Dimensions

```
def get_image_shapes(folder_path):
    widths, heights = [], []
    for class_folder in os.listdir(folder_path):
        class_path = os.path.join(folder_path, class_folder)
        for img_file in os.listdir(class_path):
            img_path = os.path.join(class_path, img_file)
            try:
                with Image.open(img_path) as img:
                    w, h = img.size
                    widths.append(w)
                    heights.append(h)
            except:
                continue
    return widths, heights

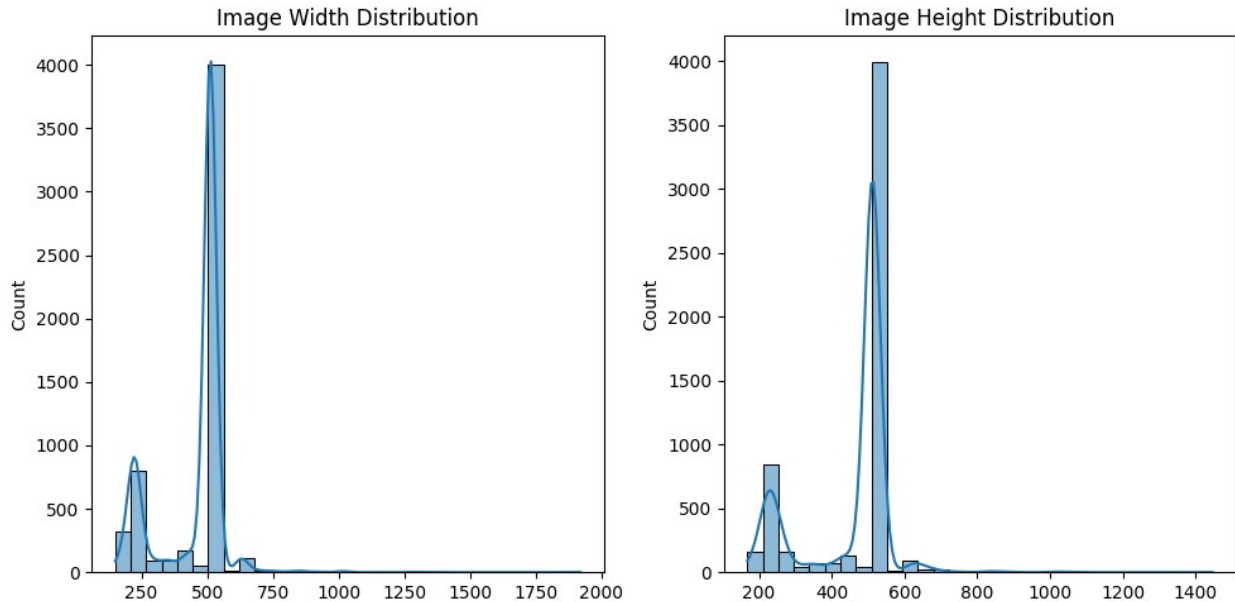
widths, heights = get_image_shapes(train_dir)

# Plot dimensions
plt.figure(figsize=(10, 5))
plt.subplot(1,2,1)
sns.histplot(widths, bins=30, kde=True)
plt.title("Image Width Distribution")

plt.subplot(1,2,2)
sns.histplot(heights, bins=30, kde=True)
```

```
plt.title("Image Height Distribution")

plt.tight_layout()
plt.show()
```



Pixel Intensity Stats

```
def compute_pixel_stats(folder_path, n_images=500):
    means = []
    stds = []
    count = 0
    for class_folder in os.listdir(folder_path):
        class_path = os.path.join(folder_path, class_folder)
        for img_file in os.listdir(class_path):
            img_path = os.path.join(class_path, img_file)
            img = cv2.imread(img_path, cv2.IMREAD_GRAYSCALE)
            if img is not None:
                means.append(np.mean(img))
                stds.append(np.std(img))
                count += 1
            if count >= n_images:
                break
    return np.mean(means), np.mean(stds)

mean_pixel, std_pixel = compute_pixel_stats(train_dir)
print(f"Average pixel intensity mean: {mean_pixel:.2f}, std: {std_pixel:.2f}")
```

Average pixel intensity mean: 49.46, std: 41.67

Check if the dataset is Balance

```
# Path to full training dataset
train_dir_path = "/kaggle/input/brain-tumor-mri-dataset/Training"

# Count number of images per class
class_counts = {
    class_name: len(os.listdir(os.path.join(train_dir_path,
class_name)))
    for class_name in os.listdir(train_dir_path)
    if os.path.isdir(os.path.join(train_dir_path, class_name))
}

# Convert to DataFrame
class_df = pd.DataFrame(list(class_counts.items()), columns=['Class',
'Image_Count'])

import plotly.express as px

# Create a treemap
fig = px.treemap(
    class_df,
    path=['Class'],
    values='Image_Count',
    color='Image_Count',
    color_continuous_scale='Viridis',
    title='Class Distribution Treemap - Brain Tumor Training Set'
)

fig.update_layout(
    margin=dict(t=50, l=25, r=25, b=25),
    font=dict(size=14)
)

fig.show()

import plotly.express as px

# 1. Count images per class
class_counts = count_images_per_class(train_dir)
class_df = pd.DataFrame(class_counts.items(), columns=["Class",
"Image_Count"])

# 2. Compute mean and standard deviation
mean_count = class_df["Image_Count"].mean()
std_count = class_df["Image_Count"].std()
lower_thresh = mean_count - std_count
```

```

# 3. Determine balance status based only on lower threshold
class_df["Balanced"] = class_df["Image_Count"].apply(
    lambda x: "No" if x < lower_thresh else "Yes"
)

# 4. Print summary
print(f"Mean image count: {mean_count:.2f}")
print(f"Standard deviation: {std_count:.2f}")
print(f"Lower threshold: {lower_thresh:.0f}")
display(class_df)

# 5. Plot treemap
fig = px.treemap(
    class_df,
    path=['Class'],
    values='Image_Count',
    color='Balanced',
    color_discrete_map={'Yes': 'lightgreen', 'No': 'tomato'},
    title='Brain Tumor Dataset Class Distribution (Based on Lower
Threshold)'
)
fig.update_layout(margin=dict(t=50, l=25, r=25, b=25))
fig.show()

```

Mean image count: 1428.00
Standard deviation: 126.62
Lower threshold: 1301

	Class	Image_Count	Balanced
0	pituitary	1457	Yes
1	notumor	1595	Yes
2	meningioma	1339	Yes
3	glioma	1321	Yes

```

import plotly.express as px

# Create a "Balanced" column just for consistency
class_df["Balanced"] = "Yes"

# Plot treemap (all green since it's balanced)
fig = px.treemap(
    class_df,
    path=['Class'],
    values='Image_Count',
    color='Balanced',
    color_discrete_map={'Yes': 'green'},
    title='Class Distribution Treemap - Brain Tumor Training Set
(Balanced)'
)

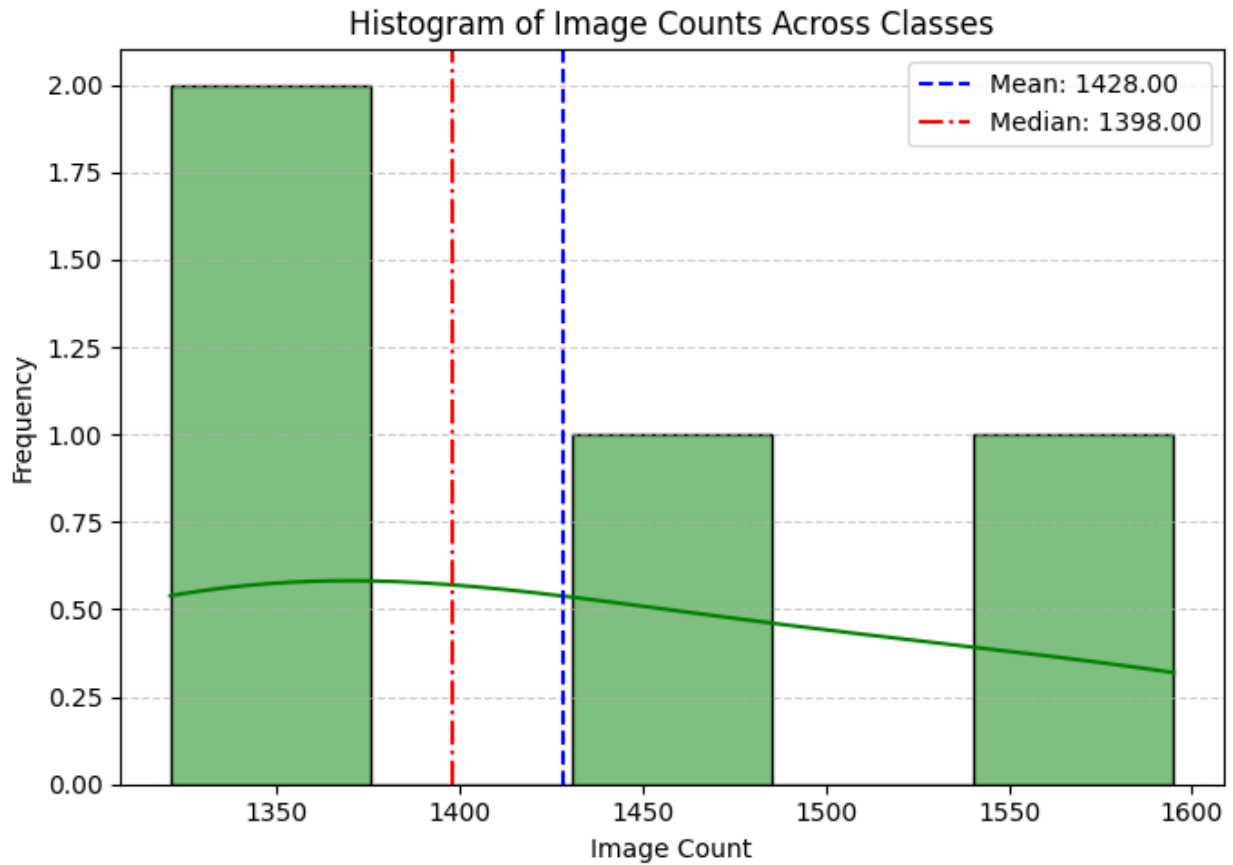
```



```
fig.update_layout(  
    margin=dict(t=50, l=25, r=25, b=25),  
    font=dict(size=14)  
)  
  
fig.show()
```

Histogram of Image Counts per Class (Training Set)

```
# Extract image counts  
counts = class_df["Image_Count"]  
  
# Calculate basic stats  
mean_val = counts.mean()  
median_val = counts.median()  
std_val = counts.std()  
  
# Plot histogram  
plt.figure(figsize=(7, 5))  
sns.histplot(counts, bins=5, kde=True, color='green')  
  
# Add vertical lines for mean and median  
plt.axvline(mean_val, color='blue', linestyle='--', label=f"Mean: {mean_val:.2f}")  
plt.axvline(median_val, color='red', linestyle='-.', label=f"Median: {median_val:.2f}")  
  
# Formatting  
plt.title("Histogram of Image Counts Across Classes")  
plt.xlabel("Image Count")  
plt.ylabel("Frequency")  
plt.legend()  
plt.grid(axis='y', linestyle='--', alpha=0.6)  
plt.tight_layout()  
plt.show()  
  
# Print stats  
print(f"Mean: {mean_val:.2f}")  
print(f"Median: {median_val:.2f}")  
print(f"Standard Deviation: {std_val:.2f}")
```



Mean: 1428.00
Median: 1398.00
Standard Deviation: 126.62

Image Preprocessing

```
from tensorflow.keras.preprocessing.image import ImageDataGenerator

# Constants
IMG_SIZE = (224, 224)
BATCH_SIZE = 32

# Initialize ImageDataGenerator WITHOUT 'color_mode'
train_datagen = ImageDataGenerator(
    rescale=1./255,
    horizontal_flip=True,
    vertical_flip=True,
```

```

        rotation_range=10,
        validation_split=0.2
    )

    # Training generator (80%)
    train_generator = train_datagen.flow_from_directory(
        train_dir,
        target_size=IMG_SIZE,
        batch_size=BATCH_SIZE,
        class_mode='categorical',
        color_mode='grayscale', # Passed here, not in the generator
        subset='training',
        shuffle=True,
        seed=42
    )

    # Validation generator (20%)
    val_generator = train_datagen.flow_from_directory(
        train_dir,
        target_size=IMG_SIZE,
        batch_size=BATCH_SIZE,
        class_mode='categorical',
        color_mode='grayscale',
        subset='validation',
        shuffle=False,
        seed=42
    )

    # Sample batch shapes
    x_batch, y_batch = next(train_generator)
    print("X shape:", x_batch.shape) # Expected: (32, 224, 224, 1)
    print("Y shape:", y_batch.shape)

    Found 4571 images belonging to 4 classes.
    Found 1141 images belonging to 4 classes.
    X shape: (32, 224, 224, 1)
    Y shape: (32, 4)

    # Get one batch of images and labels
    x_batch, y_batch = next(train_generator)

    # Get class labels
    class_labels = list(train_generator.class_indices.keys())
    inv_map = {v: k for k, v in train_generator.class_indices.items()}

    # Show 6 images with their one-hot label + class name
    import matplotlib.pyplot as plt

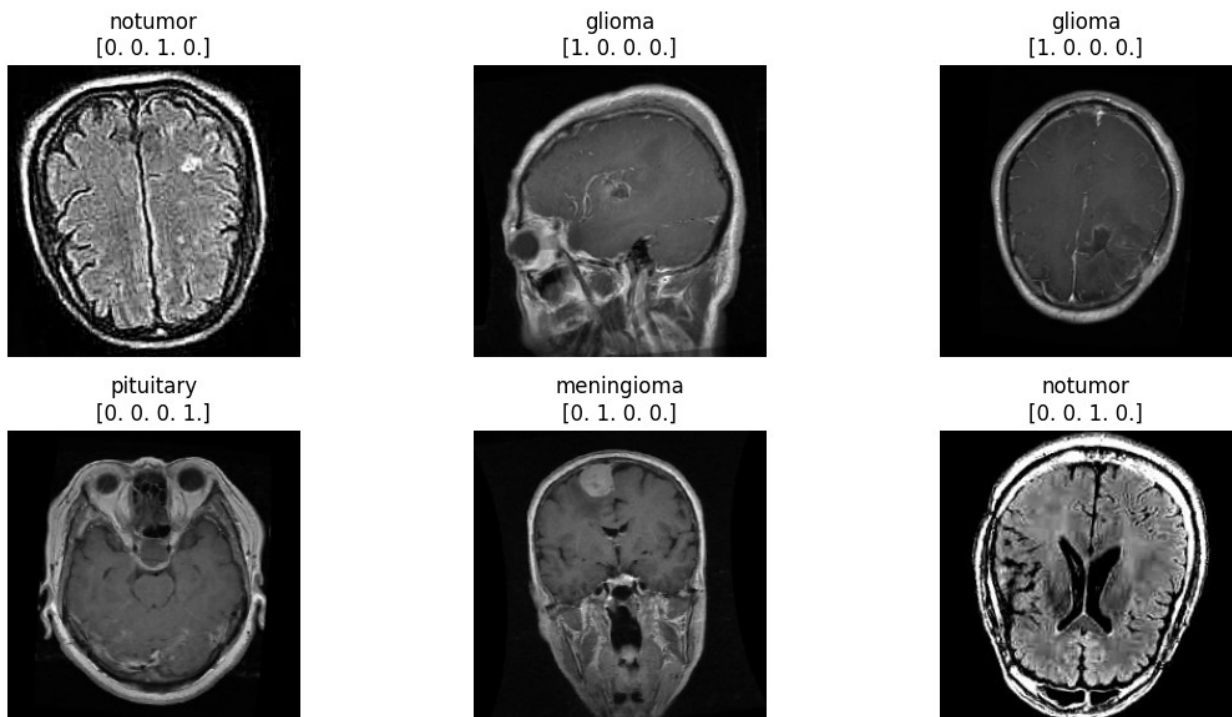
    plt.figure(figsize=(12, 6))
    for i in range(6):

```

```

plt.subplot(2, 3, i+1)
plt.imshow(x_batch[i].squeeze(), cmap='gray') # squeeze removes
extra dim
class_idx = y_batch[i].argmax() # Convert one-hot to index
class_name = inv_map[class_idx]
one_hot = y_batch[i]
plt.title(f"{class_name}\n{one_hot}")
plt.axis('off')
plt.tight_layout()
plt.show()

```



```

# Get one batch
x_batch, y_batch = next(train_generator)

# Select one image from the batch (first one)
image = x_batch[0].squeeze() # shape: (224, 224)

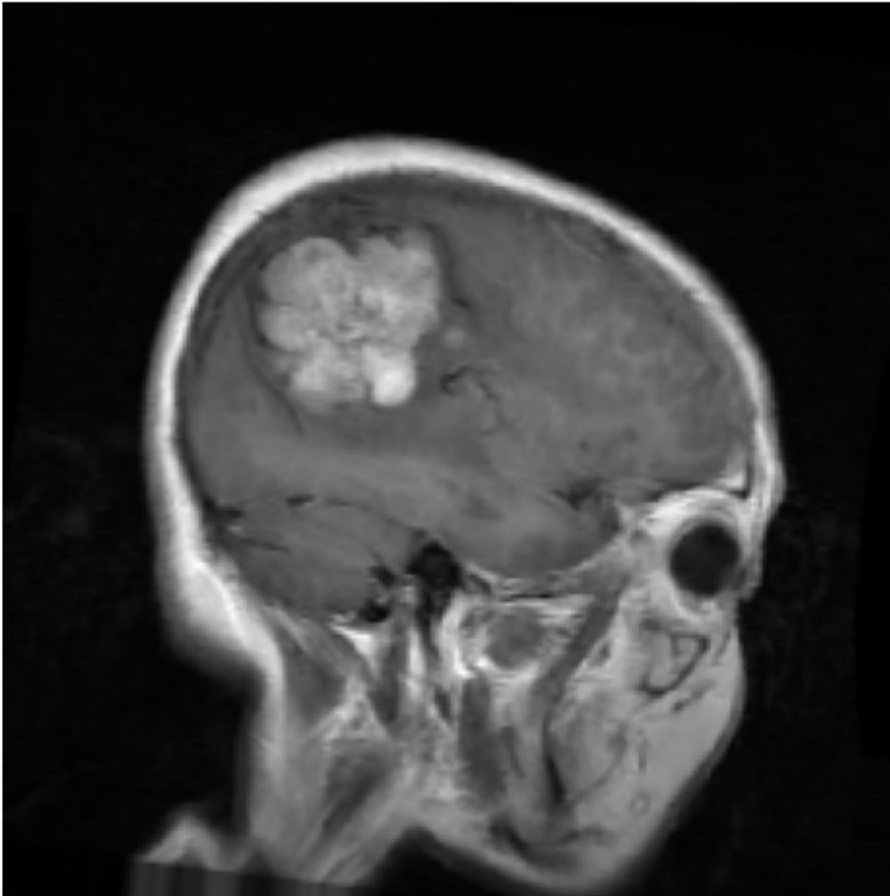
# Display image
plt.figure(figsize=(6, 6))
plt.imshow(image, cmap='gray')
plt.title("Grayscale Image (Normalized)")
plt.axis('off')
plt.show()

# Show array values (trimmed)

```

```
print("Sample of Normalized Pixel Array (shape:  
{}):".format(image.shape))  
print(np.round(image[100:110, 100:110], 3)) # just a 10x10 region for  
clarity
```

Grayscale Image (Normalized)



```
Sample of Normalized Pixel Array (shape: (224, 224)):  
[[0.372 0.333 0.319 0.321 0.315 0.31 0.316 0.324 0.331 0.354]  
 [0.337 0.336 0.331 0.333 0.332 0.332 0.332 0.337 0.345 0.368]  
 [0.326 0.33 0.331 0.336 0.344 0.347 0.346 0.349 0.36 0.369]  
 [0.326 0.321 0.321 0.326 0.335 0.346 0.352 0.353 0.363 0.368]  
 [0.331 0.322 0.319 0.324 0.322 0.339 0.354 0.358 0.363 0.365]  
 [0.332 0.324 0.32 0.323 0.322 0.339 0.354 0.36 0.364 0.368]  
 [0.335 0.324 0.321 0.322 0.324 0.341 0.354 0.358 0.358 0.357]  
 [0.335 0.323 0.322 0.321 0.326 0.344 0.355 0.356 0.353 0.352]  
 [0.332 0.326 0.323 0.329 0.337 0.342 0.343 0.34 0.338 0.338]  
 [0.331 0.34 0.342 0.341 0.338 0.338 0.337 0.331 0.33 0.328]]
```

Heatmap

```
# Get one batch
x_batch, y_batch = next(train_generator)

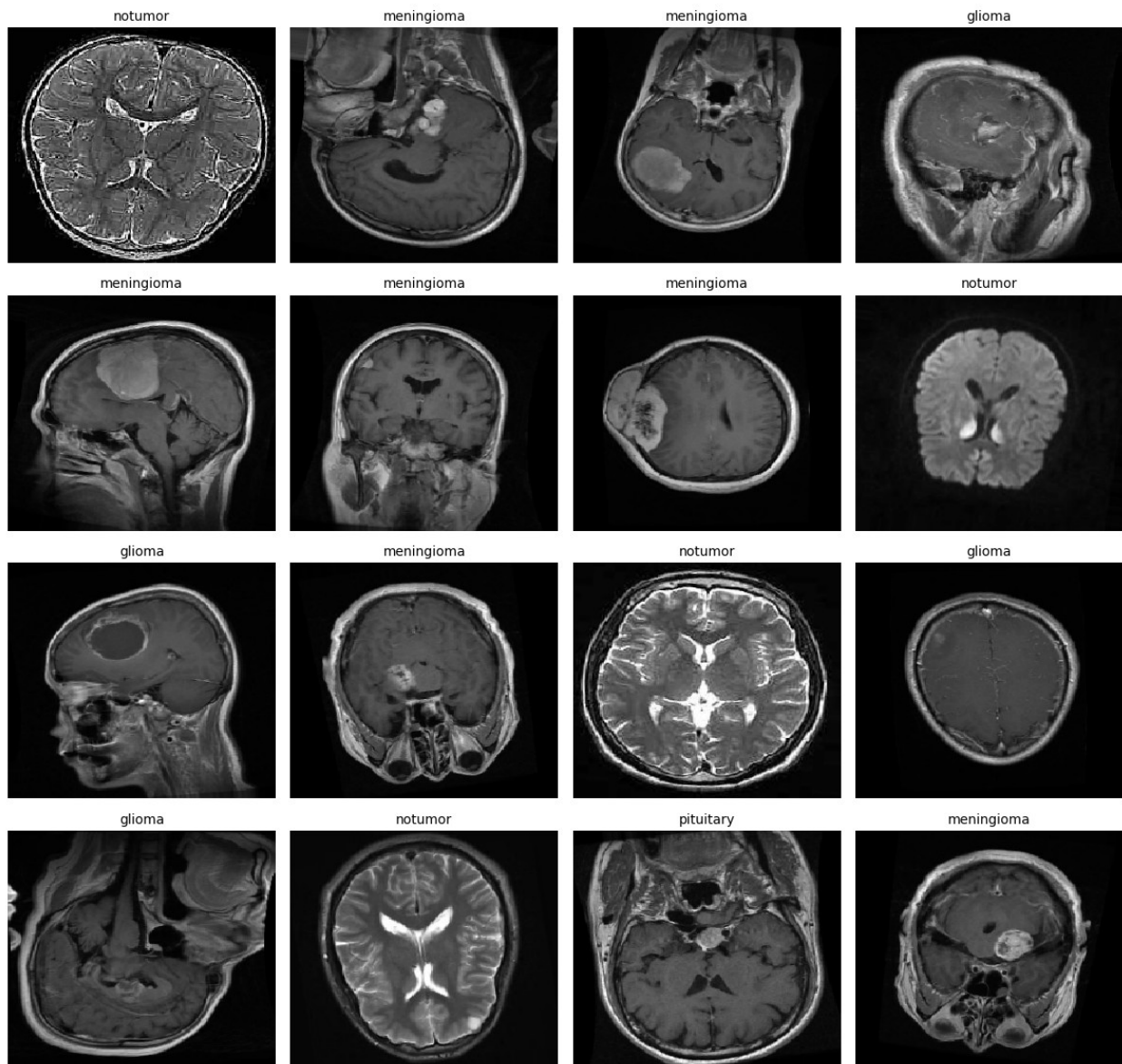
# Reverse class index mapping
class_indices = train_generator.class_indices # e.g., {'glioma':
0, ...}
inv_class_indices = {v: k for k, v in class_indices.items()}

# Plot 4x4 grid
plt.figure(figsize=(12, 12))
for i in range(16):
    image = x_batch[i].squeeze()
    class_idx = y_batch[i].argmax()
    class_name = inv_class_indices[class_idx]

    plt.subplot(4, 4, i + 1)
    sns.heatmap(image, cmap='gray', cbar=False, xticklabels=False,
yticklabels=False)
    plt.title(f"{class_name}", fontsize=10)
    plt.axis('off')

plt.suptitle("4x4 Heatmap Grid of Normalized Grayscale MRI Images with
Class Labels", fontsize=16)
plt.tight_layout()
plt.subplots_adjust(top=0.92)
plt.show()
```

4x4 Heatmap Grid of Normalized Grayscale MRI Images with Class Labels



```
# Get one batch from the generator
x_batch, y_batch = next(train_generator)
image = x_batch[0].squeeze() # shape: (224, 224)
label_idx = y_batch[0].argmax()
label_name = list(train_generator.class_indices.keys())[label_idx]

# Plot side-by-side
plt.figure(figsize=(10, 4))

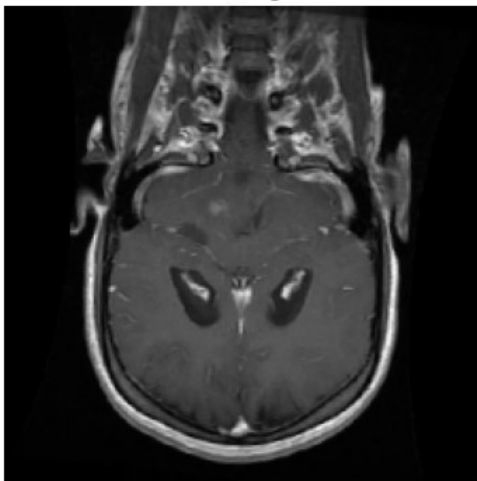
# Using imshow
plt.subplot(1, 2, 1)
plt.imshow(image, cmap='gray')
```

```
plt.title(f"imshow() - {label_name}")
plt.axis('off')

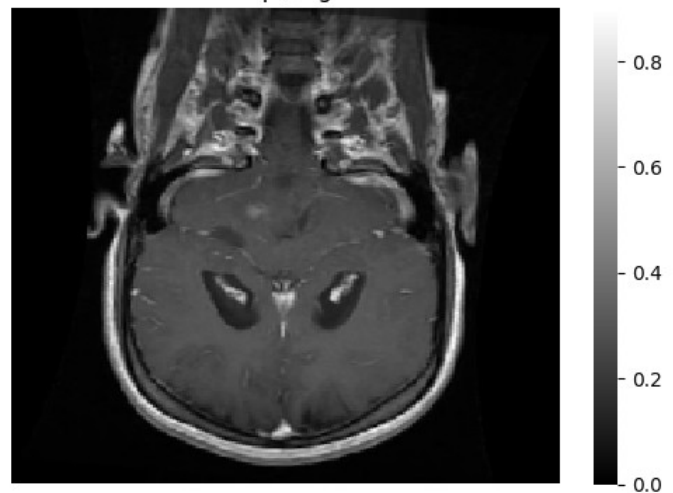
# Using heatmap
plt.subplot(1, 2, 2)
sns.heatmap(image, cmap='gray', cbar=True, xticklabels=False,
yticklabels=False)
plt.title(f"heatmap() - {label_name}")
plt.axis('off')

plt.tight_layout()
plt.show()
```

imshow() - glioma



heatmap() - glioma



Test Generator Data setup

```
# Test directory
test_dir = "/kaggle/input/brain-tumor-mri-dataset/Testing"

# Create a test ImageDataGenerator (no augmentation)
test_datagen = ImageDataGenerator(rescale=1./255)

# Test generator (full batch size to evaluate all at once if needed)
test_generator = test_datagen.flow_from_directory(
    test_dir,
    target_size=IMG_SIZE,           # (224, 224)
    batch_size=BATCH_SIZE,
    class_mode='categorical',        # one-hot labels
    color_mode='grayscale',         # same as training
    shuffle=False                    # important for evaluation and
```



```
prediction alignment
)
```

Found 1311 images belonging to 4 classes.

CNN Models

VGG16

Build VGG16-Based Model

```
from tensorflow.keras.applications import VGG16
from tensorflow.keras.models import Model
from tensorflow.keras.layers import Dense, GlobalAveragePooling2D,
Input, Lambda
from tensorflow.keras.optimizers import Adam
import tensorflow as tf
```

```
def build_vgg16_model():
    input_tensor = Input(shape=(224, 224, 1))
    rgb_tensor = Lambda(lambda x: tf.image.grayscale_to_rgb(x))
    (input_tensor)
```

```
    base_model = VGG16(weights='imagenet', include_top=False,
input_tensor=rgb_tensor)
    for layer in base_model.layers:
        layer.trainable = False
```

```
    x = base_model.output
    x = GlobalAveragePooling2D()(x)
    x = Dense(128, activation='relu')(x)
    output = Dense(4, activation='softmax')(x)
```

```
    model = Model(inputs=input_tensor, outputs=output)
    return model
```

```
model = build_vgg16_model()
model.compile(optimizer=Adam(learning_rate=1e-4),
loss='categorical_crossentropy', metrics=['accuracy'])
```

```
I0000 00:00:1746112406.438569      31 gpu_device.cc:2022] Created
device /job:localhost/replica:0/task:0/device:GPU:0 with 15513 MB
memory: -> device: 0, name: Tesla P100-PCIE-16GB, pci bus id:
0000:00:04.0, compute capability: 6.0
```

```
Downloading data from https://storage.googleapis.com/tensorflow/keras-
applications/vgg16/vgg16_weights_tf_dim_ordering_tf_kernels_notop.h5
58889256/58889256 ————— 3s 0us/step
```

Callbacks and Train the Model

```
from tensorflow.keras.callbacks import EarlyStopping, ModelCheckpoint

early_stop = EarlyStopping(patience=5, restore_best_weights=True)
checkpoint = ModelCheckpoint("vgg16_brain_tumor.keras",
                             save_best_only=True)

history = model.fit(
    train_generator,
    epochs=25,
    validation_data=val_generator,
    callbacks=[early_stop, checkpoint]
)
```

Epoch 1/25

WARNING: All log messages before absl::InitializeLog() is called are written to STDERR

```
I0000 00:00:1746112415.554980      129 service.cc:148] XLA service
0x78b1c800e060 initialized for platform CUDA (this does not guarantee
that XLA will be used). Devices:
I0000 00:00:1746112415.555734      129 service.cc:156]   StreamExecutor
device (0): Tesla P100-PCIE-16GB, Compute Capability 6.0
W0000 00:00:1746112415.670197      129 assert_op.cc:38] Ignoring Assert
operator
functional_1/lambda_1/grayscale_to_rgb/assert_equal_1/Assert/Assert
W0000 00:00:1746112415.671537      129 assert_op.cc:38] Ignoring Assert
operator
functional_1/lambda_1/grayscale_to_rgb/assert_greater_equal/Assert/
Assert
I0000 00:00:1746112415.890217      129 cuda_dnn.cc:529] Loaded cuDNN
version 90300
```

2/143 ————— 10s 74ms/step - accuracy: 0.3125 - loss: 1.5013

```
I0000 00:00:1746112425.116438      129 device_compiler.h:188] Compiled
cluster using XLA! This line is logged at most once for the lifetime
of the process.
```

123/143 ————— 3s 178ms/step - accuracy: 0.3576 - loss: 1.3669

```
W0000 00:00:1746112447.071782      128 assert_op.cc:38] Ignoring Assert
operator
functional_1/lambda_1/grayscale_to_rgb/assert_equal_1/Assert/Assert
W0000 00:00:1746112447.071963      128 assert_op.cc:38] Ignoring Assert
operator
```

functional_1/lambda_1/grayscale_to_rgb/assert_greater_equal/Assert/
Assert

143/143 ————— 0s 217ms/step - accuracy: 0.3700 - loss:
1.3571

W0000 00:00:1746112456.897304 128 assert_op.cc:38] Ignoring Assert
operator

functional_1/lambda_1/grayscale_to_rgb/assert_equal_1/Assert/Assert

W0000 00:00:1746112456.898750 128 assert_op.cc:38] Ignoring Assert
operator

functional_1/lambda_1/grayscale_to_rgb/assert_greater_equal/Assert/
Assert

W0000 00:00:1746112464.092438 129 assert_op.cc:38] Ignoring Assert
operator

functional_1/lambda_1/grayscale_to_rgb/assert_equal_1/Assert/Assert

W0000 00:00:1746112464.092555 129 assert_op.cc:38] Ignoring Assert
operator

functional_1/lambda_1/grayscale_to_rgb/assert_greater_equal/Assert/
Assert

143/143 ————— 57s 312ms/step - accuracy: 0.3706 - loss:
1.3567 - val_accuracy: 0.5741 - val_loss: 1.2317

Epoch 2/25

143/143 ————— 33s 223ms/step - accuracy: 0.6115 - loss:
1.1332 - val_accuracy: 0.6529 - val_loss: 1.1234

Epoch 3/25

143/143 ————— 34s 227ms/step - accuracy: 0.7301 - loss:
0.9831 - val_accuracy: 0.6564 - val_loss: 1.0454

Epoch 4/25

143/143 ————— 33s 226ms/step - accuracy: 0.7577 - loss:
0.8638 - val_accuracy: 0.6880 - val_loss: 0.9876

Epoch 5/25

143/143 ————— 33s 225ms/step - accuracy: 0.7848 - loss:
0.7786 - val_accuracy: 0.6705 - val_loss: 0.9620

Epoch 6/25

143/143 ————— 33s 221ms/step - accuracy: 0.8017 - loss:
0.7149 - val_accuracy: 0.6994 - val_loss: 0.9320

Epoch 7/25

143/143 ————— 33s 224ms/step - accuracy: 0.8327 - loss:
0.6472 - val_accuracy: 0.7029 - val_loss: 0.8927

Epoch 8/25

143/143 ————— 33s 222ms/step - accuracy: 0.8260 - loss:
0.6160 - val_accuracy: 0.7082 - val_loss: 0.8776

Epoch 9/25

143/143 ————— 32s 219ms/step - accuracy: 0.8372 - loss:
0.5786 - val_accuracy: 0.7117 - val_loss: 0.8565

Epoch 10/25

143/143 ————— 33s 221ms/step - accuracy: 0.8452 - loss:
0.5549 - val_accuracy: 0.7213 - val_loss: 0.8403

Epoch 11/25
143/143 _____ 32s 217ms/step - accuracy: 0.8481 - loss: 0.5192 - val_accuracy: 0.7230 - val_loss: 0.8324

Epoch 12/25
143/143 _____ 33s 220ms/step - accuracy: 0.8463 - loss: 0.5073 - val_accuracy: 0.7266 - val_loss: 0.8310

Epoch 13/25
143/143 _____ 33s 220ms/step - accuracy: 0.8560 - loss: 0.4848 - val_accuracy: 0.7318 - val_loss: 0.8012

Epoch 14/25
143/143 _____ 32s 218ms/step - accuracy: 0.8667 - loss: 0.4566 - val_accuracy: 0.7353 - val_loss: 0.8046

Epoch 15/25
143/143 _____ 32s 219ms/step - accuracy: 0.8743 - loss: 0.4460 - val_accuracy: 0.7432 - val_loss: 0.8004

Epoch 16/25
143/143 _____ 32s 218ms/step - accuracy: 0.8724 - loss: 0.4395 - val_accuracy: 0.7248 - val_loss: 0.7895

Epoch 17/25
143/143 _____ 32s 214ms/step - accuracy: 0.8695 - loss: 0.4340 - val_accuracy: 0.7511 - val_loss: 0.7817

Epoch 18/25
143/143 _____ 32s 219ms/step - accuracy: 0.8747 - loss: 0.4195 - val_accuracy: 0.7388 - val_loss: 0.7793

Epoch 19/25
143/143 _____ 32s 218ms/step - accuracy: 0.8808 - loss: 0.3918 - val_accuracy: 0.7415 - val_loss: 0.7702

Epoch 20/25
143/143 _____ 33s 221ms/step - accuracy: 0.8793 - loss: 0.3906 - val_accuracy: 0.7344 - val_loss: 0.7921

Epoch 21/25
143/143 _____ 34s 228ms/step - accuracy: 0.8777 - loss: 0.3930 - val_accuracy: 0.7493 - val_loss: 0.7534

Epoch 22/25
143/143 _____ 34s 227ms/step - accuracy: 0.8779 - loss: 0.3980 - val_accuracy: 0.7441 - val_loss: 0.7487

Epoch 23/25
143/143 _____ 31s 212ms/step - accuracy: 0.8787 - loss: 0.3864 - val_accuracy: 0.7415 - val_loss: 0.7519

Epoch 24/25
143/143 _____ 33s 222ms/step - accuracy: 0.8851 - loss: 0.3848 - val_accuracy: 0.7555 - val_loss: 0.7204

Epoch 25/25
143/143 _____ 32s 216ms/step - accuracy: 0.8875 - loss: 0.3684 - val_accuracy: 0.7520 - val_loss: 0.7360

Test Loss and Accuracy

```
test_loss, test_acc = model.evaluate(test_generator)
print(f"Test Accuracy: {test_acc:.4f}")
print(f"Test Loss: {test_loss:.4f}")

40/41 ————— 0s 165ms/step - accuracy: 0.7026 - loss:
0.8006

W0000 00:00:1746113260.720759      126 assert_op.cc:38] Ignoring Assert
operator
functional_1/lambda_1/grayscale_to_rgb/assert_equal_1/Assert/Assert
W0000 00:00:1746113260.720936      126 assert_op.cc:38] Ignoring Assert
operator
functional_1/lambda_1/grayscale_to_rgb/assert_greater_equal/Assert/
Assert

41/41 ————— 14s 338ms/step - accuracy: 0.7069 - loss:
0.7921
Test Accuracy: 0.7918
Test Loss: 0.6222
```

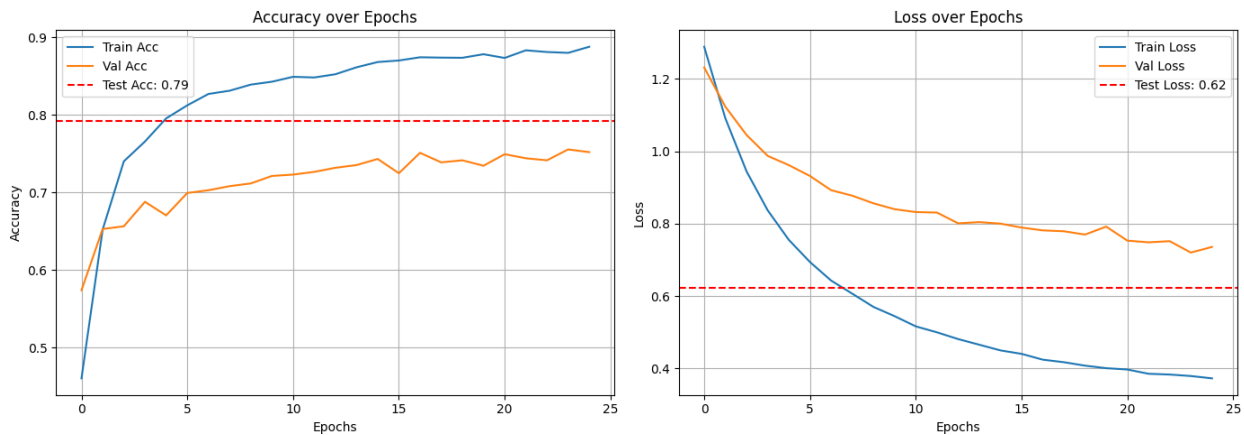
Train Test Val Loss Plot

```
def plot_all_curves(history, test_acc, test_loss):
    plt.figure(figsize=(14, 5))

    # Accuracy plot
    plt.subplot(1, 2, 1)
    plt.plot(history.history['accuracy'], label='Train Acc')
    plt.plot(history.history['val_accuracy'], label='Val Acc')
    plt.axhline(y=test_acc, color='r', linestyle='--', label=f'Test
Acc: {test_acc:.2f}')
    plt.title('Accuracy over Epochs')
    plt.xlabel('Epochs')
    plt.ylabel('Accuracy')
    plt.legend()
    plt.grid(True)

    # Loss plot
    plt.subplot(1, 2, 2)
    plt.plot(history.history['loss'], label='Train Loss')
    plt.plot(history.history['val_loss'], label='Val Loss')
    plt.axhline(y=test_loss, color='r', linestyle='--', label=f'Test
Loss: {test_loss:.2f}')
    plt.title('Loss over Epochs')
    plt.xlabel('Epochs')
    plt.ylabel('Loss')
    plt.legend()
    plt.grid(True)
```

```
plt.tight_layout()
plt.show()
plot_all_curves(history, test_acc, test_loss)
```



ROC - AUC Curve

```
from sklearn.metrics import roc_curve, auc
from sklearn.preprocessing import label_binarize

# Predictions
y_pred = model.predict(test_generator)
y_true = test_generator.classes
n_classes = y_pred.shape[1]

# One-hot encode the true labels
y_true_onehot = label_binarize(y_true, classes=np.arange(n_classes))

# Compute ROC curve and AUC for each class
fpr = {}
tpr = {}
roc_auc = {}

plt.figure(figsize=(10, 8))

for i in range(n_classes):
    fpr[i], tpr[i], _ = roc_curve(y_true_onehot[:, i], y_pred[:, i])
    roc_auc[i] = auc(fpr[i], tpr[i])
    plt.plot(fpr[i], tpr[i], label=f'Class {i} (AUC = {roc_auc[i]:.2f})')

# Plot diagonal reference line
plt.plot([0, 1], [0, 1], 'k--', linewidth=1)

# Labels and title
plt.xlabel("False Positive Rate")
```

```
plt.ylabel("True Positive Rate")
plt.title("ROC Curve per Class")
plt.legend(loc="lower right")
plt.grid(True)
plt.tight_layout()
plt.show()
```

40/41 ————— 0s 67ms/step

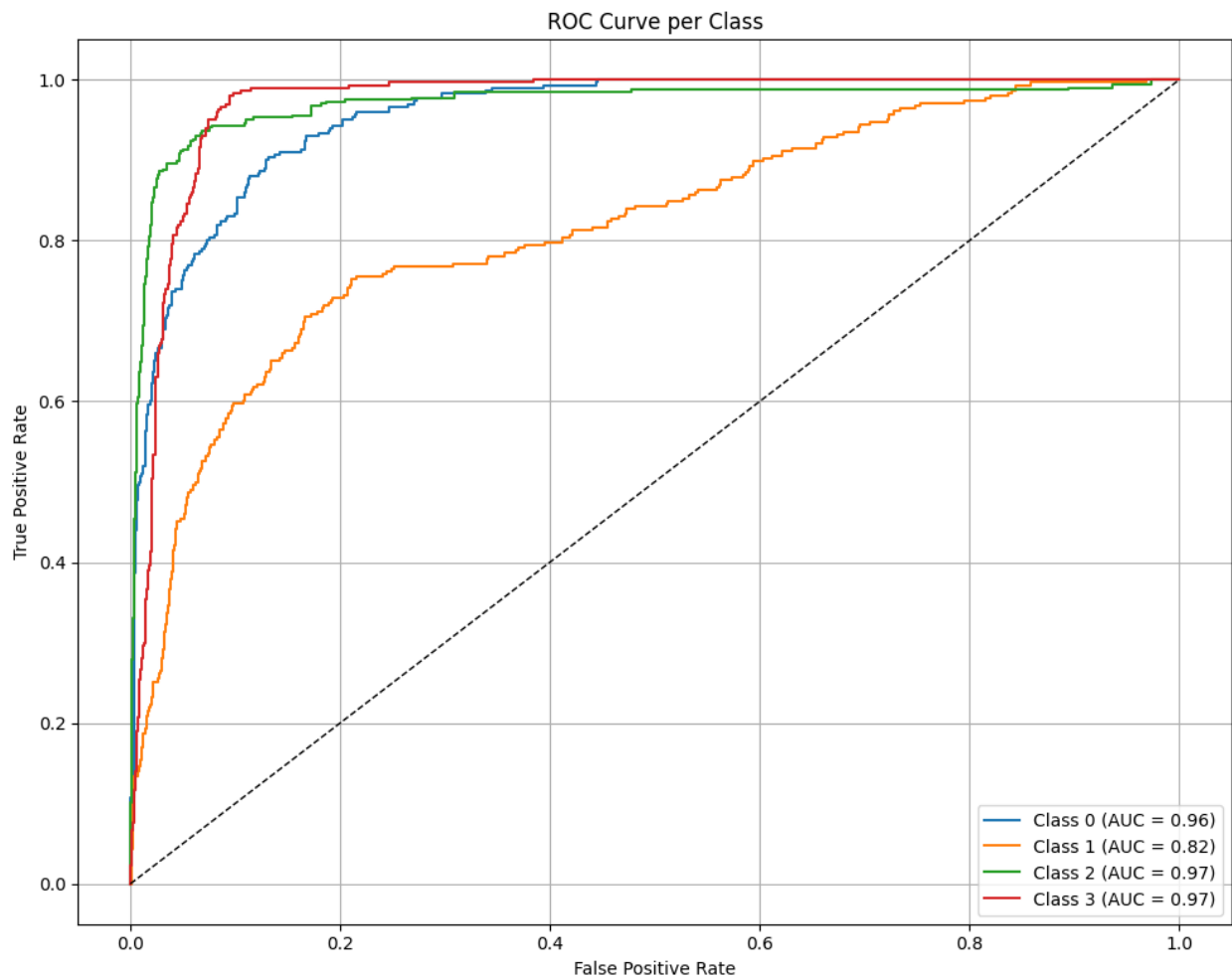
W0000 00:00:1746113271.900545 128 assert_op.cc:38] Ignoring Assert operator

functional_1/lambda_1/grayscale_to_rgb/assert_equal_1/Assert/Assert

W0000 00:00:1746113271.901698 128 assert_op.cc:38] Ignoring Assert operator

functional_1/lambda_1/grayscale_to_rgb/assert_greater_equal/Assert/Assert

41/41 ————— 4s 80ms/step



Confusion Matrix

```
from sklearn.metrics import confusion_matrix, classification_report,
precision_score, recall_score, f1_score
import seaborn as sns
import pandas as pd

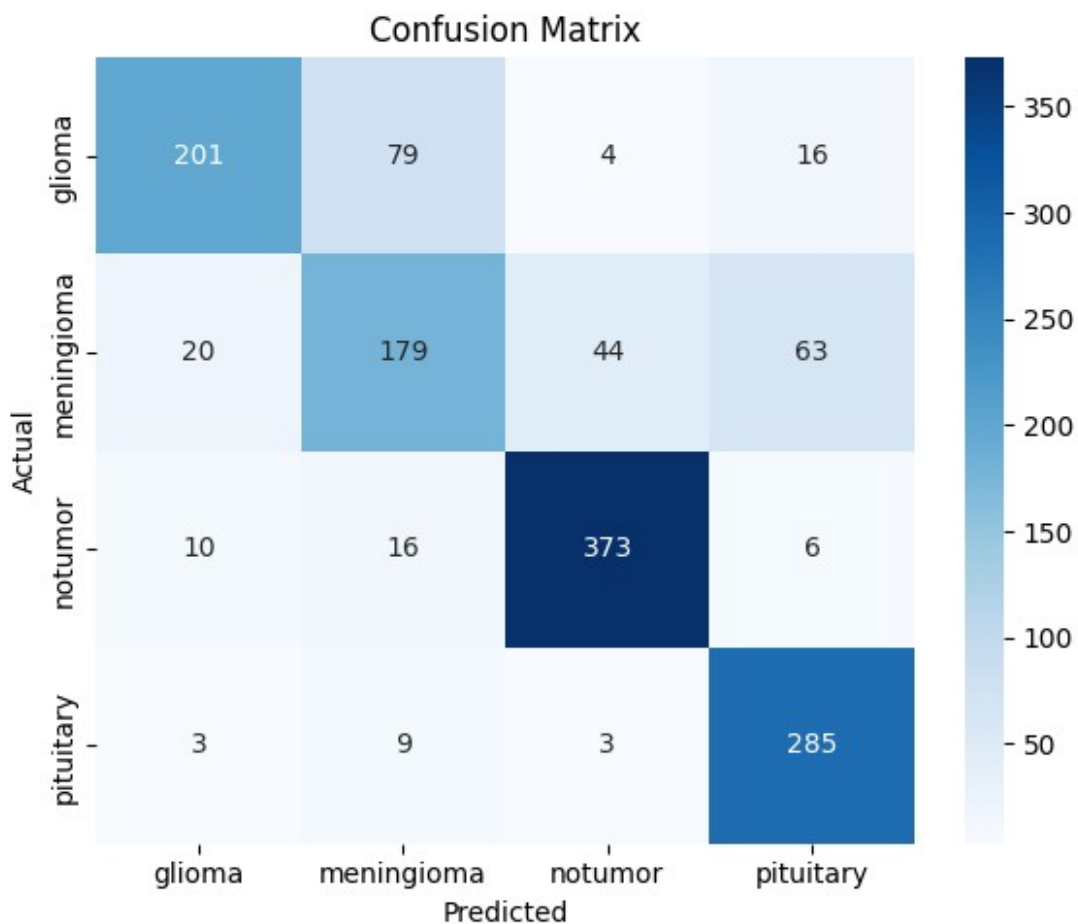
# Get predicted class indices
y_pred_class = np.argmax(y_pred, axis=1)
y_true_class = test_generator.classes
class_labels = list(test_generator.class_indices.keys())

# Confusion matrix
cm = confusion_matrix(y_true_class, y_pred_class)
df_cm = pd.DataFrame(cm, index=class_labels, columns=class_labels)

# Plot the confusion matrix
plt.figure(figsize=(6, 5))
sns.heatmap(df_cm, annot=True, fmt='d', cmap='Blues')
plt.title("Confusion Matrix")
plt.ylabel("Actual")
plt.xlabel("Predicted")
plt.tight_layout()
plt.show()

# Classification report (precision, recall, F1)
report = classification_report(y_true_class, y_pred_class,
target_names=class_labels, output_dict=True)
report_df = pd.DataFrame(report).transpose()

# Round values and print neatly
report_df = report_df.round(3)
display(report_df)
```

	precision	recall	f1-score	support
glioma	0.859	0.670	0.753	300.000
meningioma	0.633	0.585	0.608	306.000
notumor	0.880	0.921	0.900	405.000
pituitary	0.770	0.950	0.851	300.000
accuracy	0.792	0.792	0.792	0.792
macro avg	0.785	0.781	0.778	1311.000
weighted avg	0.792	0.792	0.787	1311.000

```

try:
    results_df
except NameError:
    results_df = pd.DataFrame(columns=["Model", "Test Accuracy", "Test Loss", "Precision", "Recall", "F1-Score", "AUC"])

# Compute macro average metrics
from sklearn.metrics import precision_score, recall_score, f1_score, roc_auc_score

macro_precision = precision_score(y_true_class, y_pred_class,
average='macro')

```

```

macro_recall = recall_score(y_true_class, y_pred_class,
                             average='macro')
macro_f1 = f1_score(y_true_class, y_pred_class, average='macro')
macro_auc = roc_auc_score(label_binarize(y_true_class,
                                           classes=np.arange(n_classes)), y_pred, average='macro',
                             multi_class='ovr')

# Append VGG16 results
results_df.loc[len(results_df)] = [
    "VGG16",
    test_acc,
    test_loss,
    round(macro_precision, 4),
    round(macro_recall, 4),
    round(macro_f1, 4),
    round(macro_auc, 4)
]

# Display results so far
display(results_df)

```

	Model	Test Accuracy	Test Loss	Precision	Recall	F1-Score
AUC						
0	VGG16	0.791762	0.622208	0.7854	0.7815	0.7778
		0.9282				

ResNet50

Build VGG16-Based Model

```

def build_resnet50_model():
    input_tensor = Input(shape=(224, 224, 1))
    rgb_tensor = Lambda(lambda x: tf.image.grayscale_to_rgb(x))
    (input_tensor)

    base_model = ResNet50(weights='imagenet', include_top=False,
                             input_tensor=rgb_tensor)
    for layer in base_model.layers:
        layer.trainable = False

    x = base_model.output
    x = GlobalAveragePooling2D()(x)
    x = Dense(256, activation='relu')(x)
    x = Dropout(0.5)(x)
    output = Dense(4, activation='softmax')(x)

    model = Model(inputs=input_tensor, outputs=output)
    return model

```

```
# Compile the model
```

```
model = build_resnet50_model()  
model.compile(optimizer=Adam(learning_rate=1e-4),  
loss='categorical_crossentropy', metrics=['accuracy'])
```

```
Downloading data from https://storage.googleapis.com/tensorflow/keras-  
applications/resnet/  
resnet50_weights_tf_dim_ordering_tf_kernels_notop.h5  
94765736/94765736 _____ 4s 0us/step
```

Callbacks and Train the Model

```
early_stop = EarlyStopping(patience=5, restore_best_weights=True)  
checkpoint = ModelCheckpoint("resnet50_brain_tumor.keras",  
save_best_only=True)
```

```
history = model.fit(  
    train_generator,  
    epochs=25,  
    validation_data=val_generator,  
    callbacks=[early_stop, checkpoint]  
)
```

Epoch 1/25

```
W0000 00:00:1746113289.583962    127 assert_op.cc:38] Ignoring Assert  
operator
```

```
functional_1_1/lambda_1_1/grayscale_to_rgb/assert_equal_1/Assert/Assert
```

```
W0000 00:00:1746113289.585354    127 assert_op.cc:38] Ignoring Assert  
operator
```

```
functional_1_1/lambda_1_1/grayscale_to_rgb/assert_greater_equal/Assert  
/Assert
```

```
3/143 _____ 14s 107ms/step - accuracy: 0.3299 - loss:  
1.6319
```

```
W0000 00:00:1746113295.065520    128 assert_op.cc:38] Ignoring Assert  
operator
```

```
functional_1_1/lambda_1_1/grayscale_to_rgb/assert_equal_1/Assert/Assert
```

```
W0000 00:00:1746113295.065653    128 assert_op.cc:38] Ignoring Assert  
operator
```

```
functional_1_1/lambda_1_1/grayscale_to_rgb/assert_greater_equal/Assert  
/Assert
```

```
143/143 _____ 0s 189ms/step - accuracy: 0.3030 - loss:  
1.5110
```

```
W0000 00:00:1746113325.060635      126 assert_op.cc:38] Ignoring Assert
operator
functional_1_1/lambda_1_1/grayscale_to_rgb/assert_equal_1/Assert/Assert
W0000 00:00:1746113325.061881      126 assert_op.cc:38] Ignoring Assert
operator
functional_1_1/lambda_1_1/grayscale_to_rgb/assert_greater_equal/Assert
/Assert
W0000 00:00:1746113332.553018      129 assert_op.cc:38] Ignoring Assert
operator
functional_1_1/lambda_1_1/grayscale_to_rgb/assert_equal_1/Assert/Assert
W0000 00:00:1746113332.553223      129 assert_op.cc:38] Ignoring Assert
operator
functional_1_1/lambda_1_1/grayscale_to_rgb/assert_greater_equal/Assert
/Assert

143/143 _____ 58s 297ms/step - accuracy: 0.3034 - loss:
1.5101 - val_accuracy: 0.5162 - val_loss: 1.2538
Epoch 2/25
143/143 _____ 31s 207ms/step - accuracy: 0.4335 - loss:
1.1851 - val_accuracy: 0.5968 - val_loss: 1.1939
Epoch 3/25
143/143 _____ 31s 212ms/step - accuracy: 0.5113 - loss:
1.1083 - val_accuracy: 0.6161 - val_loss: 1.1610
Epoch 4/25
143/143 _____ 31s 206ms/step - accuracy: 0.5646 - loss:
1.0531 - val_accuracy: 0.6231 - val_loss: 1.1361
Epoch 5/25
143/143 _____ 31s 210ms/step - accuracy: 0.5939 - loss:
1.0235 - val_accuracy: 0.6039 - val_loss: 1.1168
Epoch 6/25
143/143 _____ 31s 213ms/step - accuracy: 0.6075 - loss:
1.0056 - val_accuracy: 0.6433 - val_loss: 1.0959
Epoch 7/25
143/143 _____ 31s 210ms/step - accuracy: 0.6204 - loss:
0.9657 - val_accuracy: 0.6564 - val_loss: 1.0637
Epoch 8/25
143/143 _____ 31s 209ms/step - accuracy: 0.6303 - loss:
0.9648 - val_accuracy: 0.6407 - val_loss: 1.0540
Epoch 9/25
143/143 _____ 31s 209ms/step - accuracy: 0.6500 - loss:
0.9546 - val_accuracy: 0.6582 - val_loss: 1.0320
Epoch 10/25
143/143 _____ 31s 210ms/step - accuracy: 0.6602 - loss:
0.9221 - val_accuracy: 0.6792 - val_loss: 1.0098
Epoch 11/25
143/143 _____ 31s 207ms/step - accuracy: 0.6668 - loss:
0.8934 - val_accuracy: 0.6696 - val_loss: 0.9995
Epoch 12/25
```

```
143/143 _____ 31s 212ms/step - accuracy: 0.6626 - loss:
0.8895 - val_accuracy: 0.6757 - val_loss: 0.9830
Epoch 13/25
143/143 _____ 31s 210ms/step - accuracy: 0.6794 - loss:
0.8671 - val_accuracy: 0.6713 - val_loss: 0.9718
Epoch 14/25
143/143 _____ 31s 207ms/step - accuracy: 0.6643 - loss:
0.8712 - val_accuracy: 0.6687 - val_loss: 0.9717
Epoch 15/25
143/143 _____ 31s 207ms/step - accuracy: 0.6829 - loss:
0.8392 - val_accuracy: 0.6784 - val_loss: 0.9506
Epoch 16/25
143/143 _____ 31s 208ms/step - accuracy: 0.6948 - loss:
0.8328 - val_accuracy: 0.6792 - val_loss: 0.9457
Epoch 17/25
143/143 _____ 30s 205ms/step - accuracy: 0.6908 - loss:
0.8365 - val_accuracy: 0.6836 - val_loss: 0.9285
Epoch 18/25
143/143 _____ 31s 211ms/step - accuracy: 0.6747 - loss:
0.8298 - val_accuracy: 0.6810 - val_loss: 0.9202
Epoch 19/25
143/143 _____ 30s 200ms/step - accuracy: 0.6962 - loss:
0.8165 - val_accuracy: 0.6678 - val_loss: 0.9251
Epoch 20/25
143/143 _____ 31s 209ms/step - accuracy: 0.6987 - loss:
0.7877 - val_accuracy: 0.6845 - val_loss: 0.9080
Epoch 21/25
143/143 _____ 30s 200ms/step - accuracy: 0.6955 - loss:
0.7964 - val_accuracy: 0.6801 - val_loss: 0.9189
Epoch 22/25
143/143 _____ 31s 208ms/step - accuracy: 0.6931 - loss:
0.8003 - val_accuracy: 0.6731 - val_loss: 0.9078
Epoch 23/25
143/143 _____ 30s 202ms/step - accuracy: 0.6876 - loss:
0.7886 - val_accuracy: 0.6635 - val_loss: 0.9192
Epoch 24/25
143/143 _____ 30s 203ms/step - accuracy: 0.6914 - loss:
0.7835 - val_accuracy: 0.6617 - val_loss: 0.9113
Epoch 25/25
143/143 _____ 30s 203ms/step - accuracy: 0.6974 - loss:
0.7758 - val_accuracy: 0.6643 - val_loss: 0.9111
```

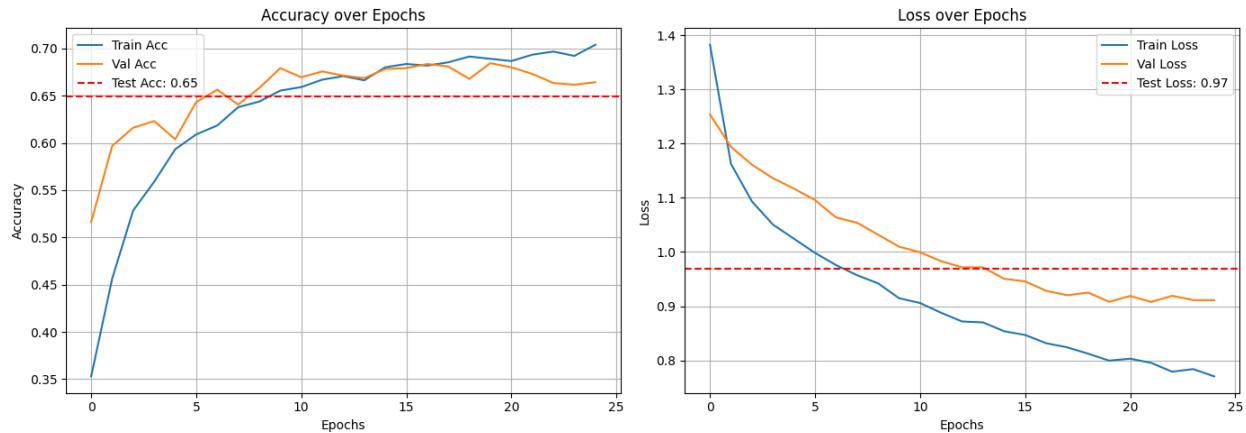
Test Loss and Accuracy

```
test_loss, test_acc = model.evaluate(test_generator)
print(f"Test Accuracy: {test_acc:.4f}")
print(f"Test Loss: {test_loss:.4f}")
```

41/41 ————— 7s 172ms/step - accuracy: 0.5784 - loss: 1.1267
Test Accuracy: 0.6491
Test Loss: 0.9688

Train Test Val Accuracy And Loss Plot

```
def plot_all_curves(history, test_acc, test_loss):  
    plt.figure(figsize=(14, 5))  
  
    # Accuracy plot  
    plt.subplot(1, 2, 1)  
    plt.plot(history.history['accuracy'], label='Train Acc')  
    plt.plot(history.history['val_accuracy'], label='Val Acc')  
    plt.axhline(y=test_acc, color='r', linestyle='--', label=f'Test  
Acc: {test_acc:.2f}')  
    plt.title('Accuracy over Epochs')  
    plt.xlabel('Epochs')  
    plt.ylabel('Accuracy')  
    plt.legend()  
    plt.grid(True)  
  
    # Loss plot  
    plt.subplot(1, 2, 2)  
    plt.plot(history.history['loss'], label='Train Loss')  
    plt.plot(history.history['val_loss'], label='Val Loss')  
    plt.axhline(y=test_loss, color='r', linestyle='--', label=f'Test  
Loss: {test_loss:.2f}')  
    plt.title('Loss over Epochs')  
    plt.xlabel('Epochs')  
    plt.ylabel('Loss')  
    plt.legend()  
    plt.grid(True)  
  
    plt.tight_layout()  
    plt.show()  
plot_all_curves(history, test_acc, test_loss)
```



AUC-ROC Curve

```
from sklearn.metrics import roc_curve, auc
from sklearn.preprocessing import label_binarize

# Predictions
y_pred = model.predict(test_generator)
y_true = test_generator.classes
n_classes = y_pred.shape[1]

# One-hot encode the true labels
y_true_onehot = label_binarize(y_true, classes=np.arange(n_classes))

# Compute ROC curve and AUC for each class
fpr = {}
tpr = {}
roc_auc = {}

plt.figure(figsize=(10, 8))

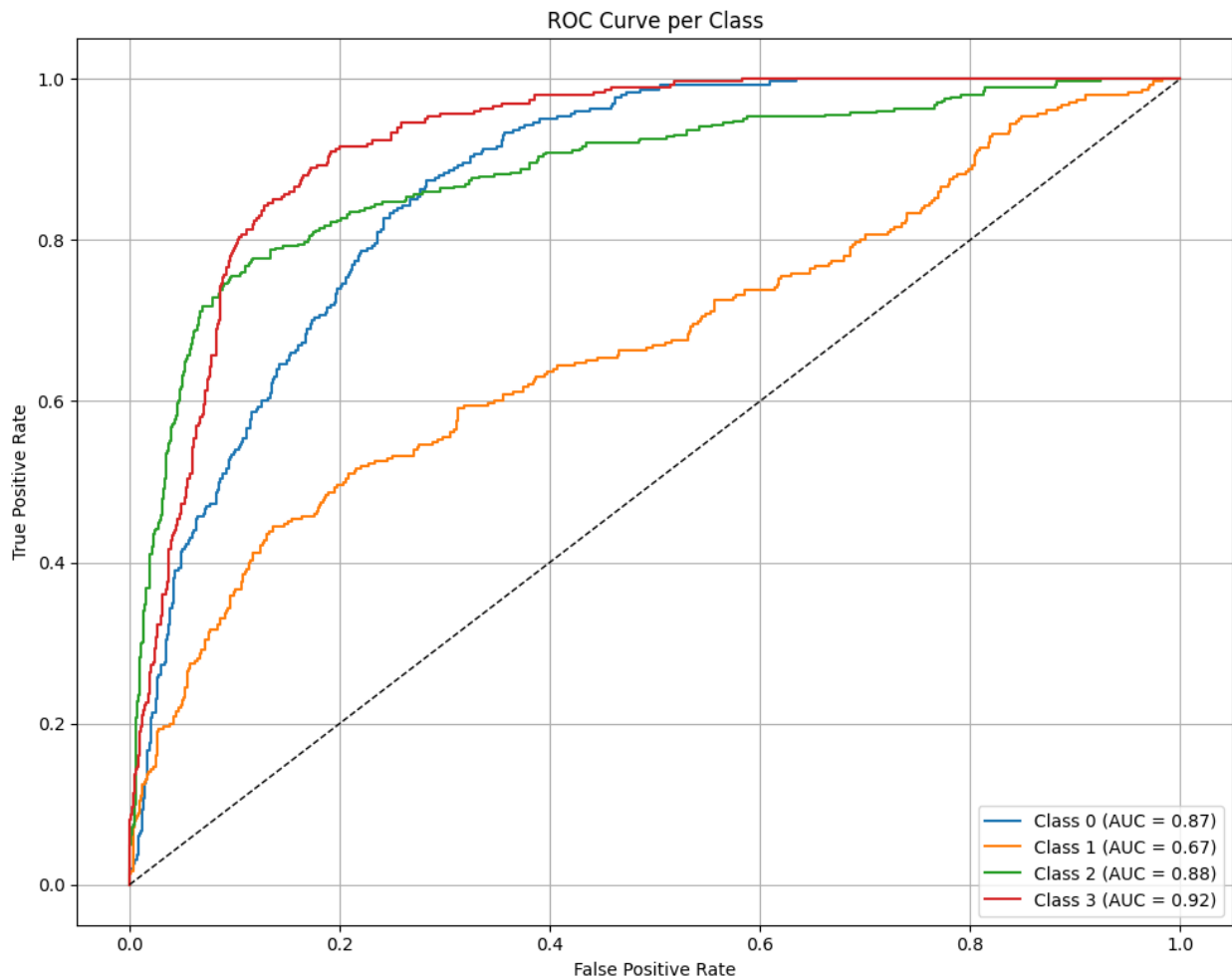
for i in range(n_classes):
    fpr[i], tpr[i], _ = roc_curve(y_true_onehot[:, i], y_pred[:, i])
    roc_auc[i] = auc(fpr[i], tpr[i])
    plt.plot(fpr[i], tpr[i], label=f'Class {i} (AUC = {roc_auc[i]:.2f})')

# Plot diagonal reference line
plt.plot([0, 1], [0, 1], 'k--', linewidth=1)

# Labels and title
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve per Class")
plt.legend(loc="lower right")
```

```
plt.grid(True)
plt.tight_layout()
plt.show()
```

41/41 ————— 11s 168ms/step



Confusiuon Matrix

```
from sklearn.metrics import confusion_matrix, classification_report,
precision_score, recall_score, f1_score
import seaborn as sns
import pandas as pd

# Get predicted class indices
y_pred_class = np.argmax(y_pred, axis=1)
y_true_class = test_generator.classes
class_labels = list(test_generator.class_indices.keys())
```

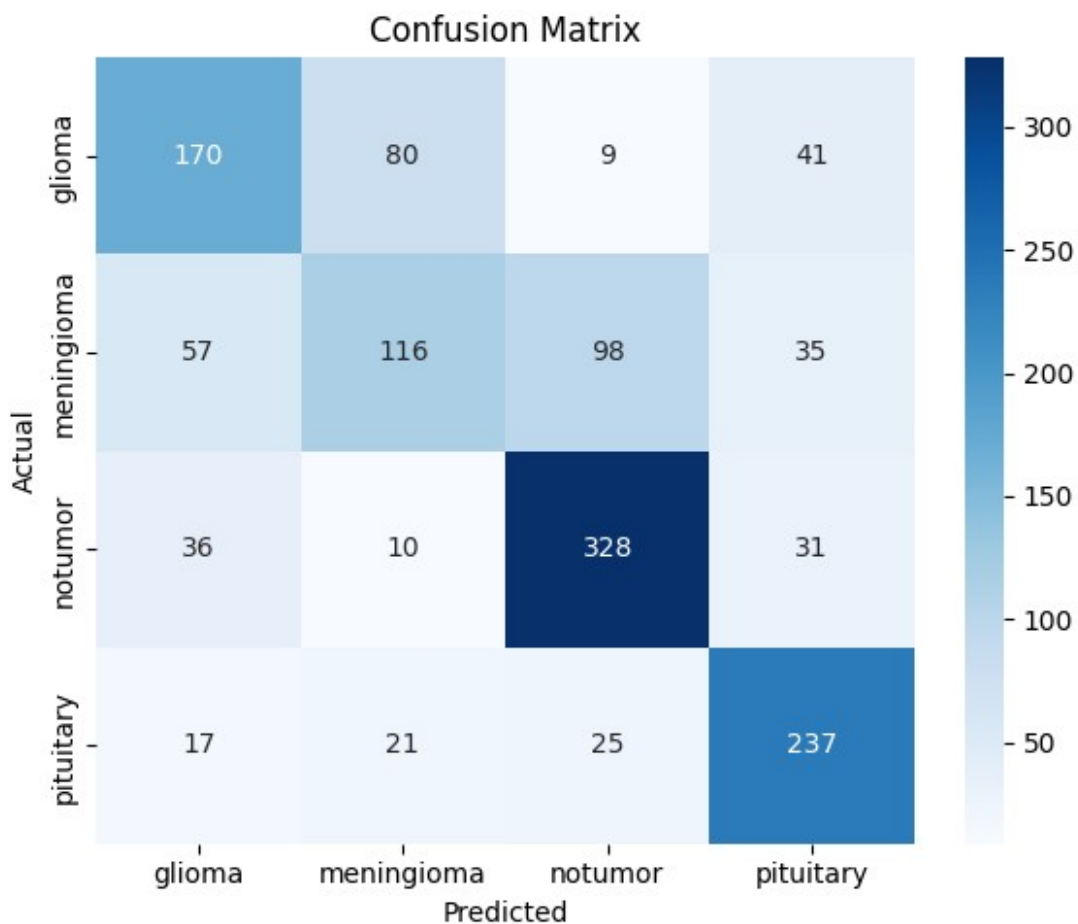


```
# Confusion matrix
cm = confusion_matrix(y_true_class, y_pred_class)
df_cm = pd.DataFrame(cm, index=class_labels, columns=class_labels)

# Plot the confusion matrix
plt.figure(figsize=(6, 5))
sns.heatmap(df_cm, annot=True, fmt='d', cmap='Blues')
plt.title("Confusion Matrix")
plt.ylabel("Actual")
plt.xlabel("Predicted")
plt.tight_layout()
plt.show()

# Classification report (precision, recall, F1)
report = classification_report(y_true_class, y_pred_class,
target_names=class_labels, output_dict=True)
report_df = pd.DataFrame(report).transpose()

# Round values and print neatly
report_df = report_df.round(3)
display(report_df)
```



	precision	recall	f1-score	support
glioma	0.607	0.567	0.586	300.000
meningioma	0.511	0.379	0.435	306.000
notumor	0.713	0.810	0.758	405.000
pituitary	0.689	0.790	0.736	300.000
accuracy	0.649	0.649	0.649	0.649
macro avg	0.630	0.636	0.629	1311.000
weighted avg	0.636	0.649	0.638	1311.000

One-hot encode true labels for AUC

```
y_true_binarized = label_binarize(y_true_class,
classes=np.arange(n_classes))
```

Compute macro average metrics

```
macro_precision = precision_score(y_true_class, y_pred_class,
average='macro')
```

```
macro_recall = recall_score(y_true_class, y_pred_class,
average='macro')
```

```
macro_f1 = f1_score(y_true_class, y_pred_class, average='macro')
```

```
macro_auc = roc_auc_score(y_true_binarized, y_pred, average='macro',
multi_class='ovr')
```

```
# Append ResNet50 results
results_df.loc[len(results_df)] = [
    "ResNet50",
    test_acc,
    test_loss,
    round(macro_precision, 4),
    round(macro_recall, 4),
    round(macro_f1, 4),
    round(macro_auc, 4)
]
```

```
# Display the updated results
display(results_df)
```

	Model	Test Accuracy	Test Loss	Precision	Recall	F1-Score
AUC						
0	VGG16	0.791762	0.622208	0.7854	0.7815	0.7778
0.9282						
1	ResNet50	0.649123	0.968826	0.6300	0.6364	0.6290
0.8358						

DenseNet121

Build DenseNet121 Model

```
from tensorflow.keras.applications import DenseNet121

def build_densenet121_model():
    input_tensor = Input(shape=(224, 224, 1))
    rgb_tensor = Lambda(lambda x: tf.image.grayscale_to_rgb(x))
    (input_tensor)

    base_model = DenseNet121(weights='imagenet', include_top=False,
input_tensor=rgb_tensor)
    for layer in base_model.layers:
        layer.trainable = False

    x = base_model.output
    x = GlobalAveragePooling2D()(x)
    x = Dense(256, activation='relu')(x)
    x = Dropout(0.5)(x)
    output = Dense(4, activation='softmax')(x)

    model = Model(inputs=input_tensor, outputs=output)
    return model
```

```
# Compile the model
```

```
model = build_densenet121_model()  
model.compile(optimizer=Adam(learning_rate=1e-4),  
loss='categorical_crossentropy', metrics=['accuracy'])
```

```
Downloading data from https://storage.googleapis.com/tensorflow/keras-  
applications/densenet/  
densenet121_weights_tf_dim_ordering_tf_kernels_notop.h5  
29084464/29084464 _____ 2s 0us/step
```

Callback and Training

```
early_stop = EarlyStopping(patience=5, restore_best_weights=True)  
checkpoint = ModelCheckpoint("densenet121_brain_tumor.keras",  
save_best_only=True)
```

```
history = model.fit(  
    train_generator,  
    epochs=25,  
    validation_data=val_generator,  
    callbacks=[early_stop, checkpoint]  
)
```

Epoch 1/25

```
143/143 _____ 101s 474ms/step - accuracy: 0.4641 -  
loss: 1.3233 - val_accuracy: 0.6810 - val_loss: 0.9065
```

Epoch 2/25

```
143/143 _____ 31s 212ms/step - accuracy: 0.7571 - loss:  
0.6496 - val_accuracy: 0.7239 - val_loss: 0.8315
```

Epoch 3/25

```
143/143 _____ 31s 209ms/step - accuracy: 0.8047 - loss:  
0.5228 - val_accuracy: 0.7292 - val_loss: 0.7718
```

Epoch 4/25

```
143/143 _____ 32s 215ms/step - accuracy: 0.8488 - loss:  
0.4509 - val_accuracy: 0.7564 - val_loss: 0.7265
```

Epoch 5/25

```
143/143 _____ 31s 210ms/step - accuracy: 0.8458 - loss:  
0.4217 - val_accuracy: 0.7669 - val_loss: 0.6712
```

Epoch 6/25

```
143/143 _____ 32s 214ms/step - accuracy: 0.8534 - loss:  
0.4113 - val_accuracy: 0.7642 - val_loss: 0.6618
```

Epoch 7/25

```
143/143 _____ 31s 208ms/step - accuracy: 0.8800 - loss:  
0.3516 - val_accuracy: 0.7783 - val_loss: 0.6345
```

Epoch 8/25

```
143/143 _____ 32s 214ms/step - accuracy: 0.8718 - loss:
```

0.3467 - val_accuracy: 0.7844 - val_loss: 0.6156
Epoch 9/25
143/143 _____ 30s 202ms/step - accuracy: 0.8876 - loss:
0.3267 - val_accuracy: 0.7739 - val_loss: 0.6315
Epoch 10/25
143/143 _____ 31s 211ms/step - accuracy: 0.8879 - loss:
0.3179 - val_accuracy: 0.7958 - val_loss: 0.5774
Epoch 11/25
143/143 _____ 31s 208ms/step - accuracy: 0.8971 - loss:
0.2910 - val_accuracy: 0.8081 - val_loss: 0.5360
Epoch 12/25
143/143 _____ 30s 201ms/step - accuracy: 0.9061 - loss:
0.2700 - val_accuracy: 0.8046 - val_loss: 0.5560
Epoch 13/25
143/143 _____ 31s 209ms/step - accuracy: 0.8916 - loss:
0.2854 - val_accuracy: 0.8046 - val_loss: 0.5325
Epoch 14/25
143/143 _____ 32s 217ms/step - accuracy: 0.8951 - loss:
0.2817 - val_accuracy: 0.8212 - val_loss: 0.4926
Epoch 15/25
143/143 _____ 31s 209ms/step - accuracy: 0.9106 - loss:
0.2465 - val_accuracy: 0.8221 - val_loss: 0.5022
Epoch 16/25
143/143 _____ 31s 210ms/step - accuracy: 0.9193 - loss:
0.2477 - val_accuracy: 0.8247 - val_loss: 0.4917
Epoch 17/25
143/143 _____ 31s 208ms/step - accuracy: 0.9186 - loss:
0.2347 - val_accuracy: 0.8273 - val_loss: 0.4783
Epoch 18/25
143/143 _____ 30s 204ms/step - accuracy: 0.9066 - loss:
0.2521 - val_accuracy: 0.8195 - val_loss: 0.4894
Epoch 19/25
143/143 _____ 30s 205ms/step - accuracy: 0.9197 - loss:
0.2371 - val_accuracy: 0.8256 - val_loss: 0.4932
Epoch 20/25
143/143 _____ 31s 207ms/step - accuracy: 0.9161 - loss:
0.2424 - val_accuracy: 0.8335 - val_loss: 0.4512
Epoch 21/25
143/143 _____ 32s 216ms/step - accuracy: 0.9134 - loss:
0.2367 - val_accuracy: 0.8449 - val_loss: 0.4494
Epoch 22/25
143/143 _____ 30s 201ms/step - accuracy: 0.9207 - loss:
0.2276 - val_accuracy: 0.8317 - val_loss: 0.4567
Epoch 23/25
143/143 _____ 31s 208ms/step - accuracy: 0.9189 - loss:
0.2137 - val_accuracy: 0.8247 - val_loss: 0.4633
Epoch 24/25
143/143 _____ 32s 217ms/step - accuracy: 0.9227 - loss:
0.2114 - val_accuracy: 0.8309 - val_loss: 0.4439

```
Epoch 25/25
143/143 ————— 32s 216ms/step - accuracy: 0.9241 - loss:
0.2158 - val_accuracy: 0.8519 - val_loss: 0.4091
```

Test Loss And Accuracy

```
test_loss, test_acc = model.evaluate(test_generator)
print(f"Test Accuracy: {test_acc:.4f}")
print(f"Test Loss: {test_loss:.4f}")

41/41 ————— 16s 391ms/step - accuracy: 0.8147 - loss:
0.4412
Test Accuracy: 0.8658
Test Loss: 0.3518
```

Train Test Val Loss PLOT

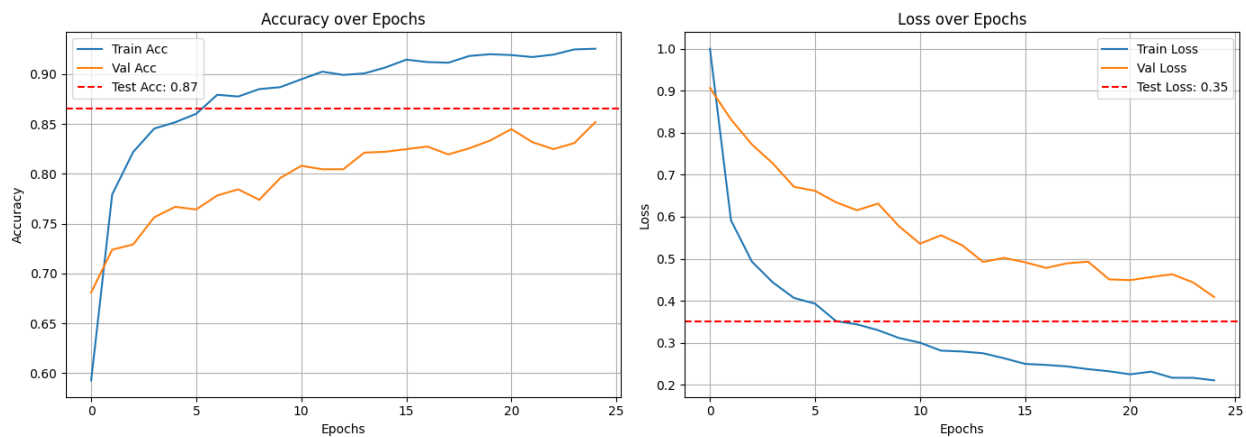
```
def plot_all_curves(history, test_acc, test_loss):
    plt.figure(figsize=(14, 5))

    # Accuracy plot
    plt.subplot(1, 2, 1)
    plt.plot(history.history['accuracy'], label='Train Acc')
    plt.plot(history.history['val_accuracy'], label='Val Acc')
    plt.axhline(y=test_acc, color='r', linestyle='--', label=f'Test
Acc: {test_acc:.2f}')
    plt.title('Accuracy over Epochs')
    plt.xlabel('Epochs')
    plt.ylabel('Accuracy')
    plt.legend()
    plt.grid(True)

    # Loss plot
    plt.subplot(1, 2, 2)
    plt.plot(history.history['loss'], label='Train Loss')
    plt.plot(history.history['val_loss'], label='Val Loss')
    plt.axhline(y=test_loss, color='r', linestyle='--', label=f'Test
Loss: {test_loss:.2f}')
    plt.title('Loss over Epochs')
    plt.xlabel('Epochs')
    plt.ylabel('Loss')
    plt.legend()
    plt.grid(True)

    plt.tight_layout()
```

```
plt.show()
plot_all_curves(history, test_acc, test_loss)
```



AUC - ROC Curve

```
from sklearn.metrics import roc_curve, auc
from sklearn.preprocessing import label_binarize

# Predictions
y_pred = model.predict(test_generator)
y_true = test_generator.classes
n_classes = y_pred.shape[1]

# One-hot encode the true labels
y_true_onehot = label_binarize(y_true, classes=np.arange(n_classes))

# Compute ROC curve and AUC for each class
fpr = {}
tpr = {}
roc_auc = {}

plt.figure(figsize=(10, 8))

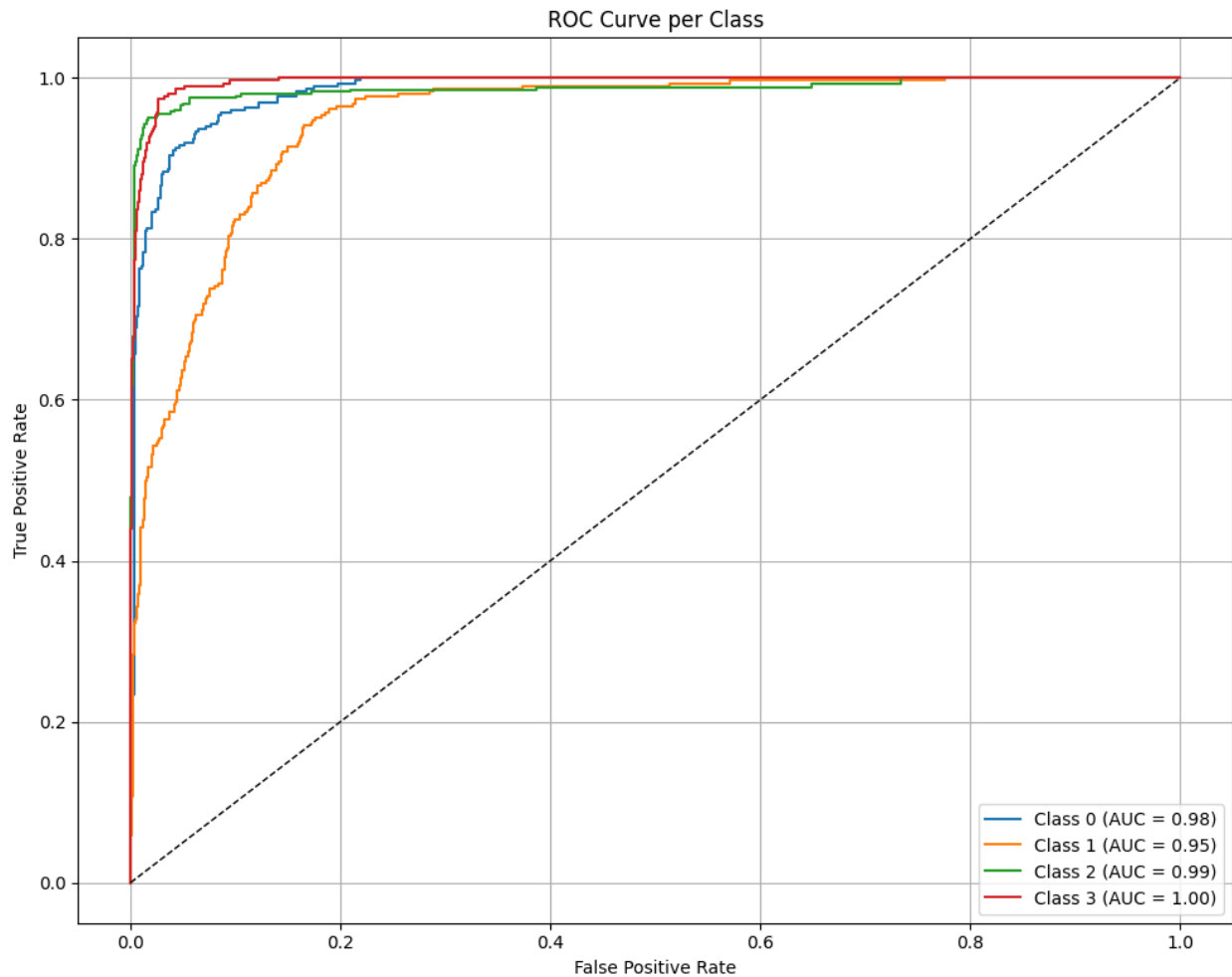
for i in range(n_classes):
    fpr[i], tpr[i], _ = roc_curve(y_true_onehot[:, i], y_pred[:, i])
    roc_auc[i] = auc(fpr[i], tpr[i])
    plt.plot(fpr[i], tpr[i], label=f'Class {i} (AUC = {roc_auc[i]:.2f})')

# Plot diagonal reference line
plt.plot([0, 1], [0, 1], 'k--', linewidth=1)

# Labels and title
```

```
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve per Class")
plt.legend(loc="lower right")
plt.grid(True)
plt.tight_layout()
plt.show()
```

41/41 ————— 26s 372ms/step



Confusion Matrix

```
from sklearn.metrics import confusion_matrix, classification_report,
precision_score, recall_score, f1_score
import seaborn as sns
import pandas as pd
```



```

# Get predicted class indices
y_pred_class = np.argmax(y_pred, axis=1)
y_true_class = test_generator.classes
class_labels = list(test_generator.class_indices.keys())

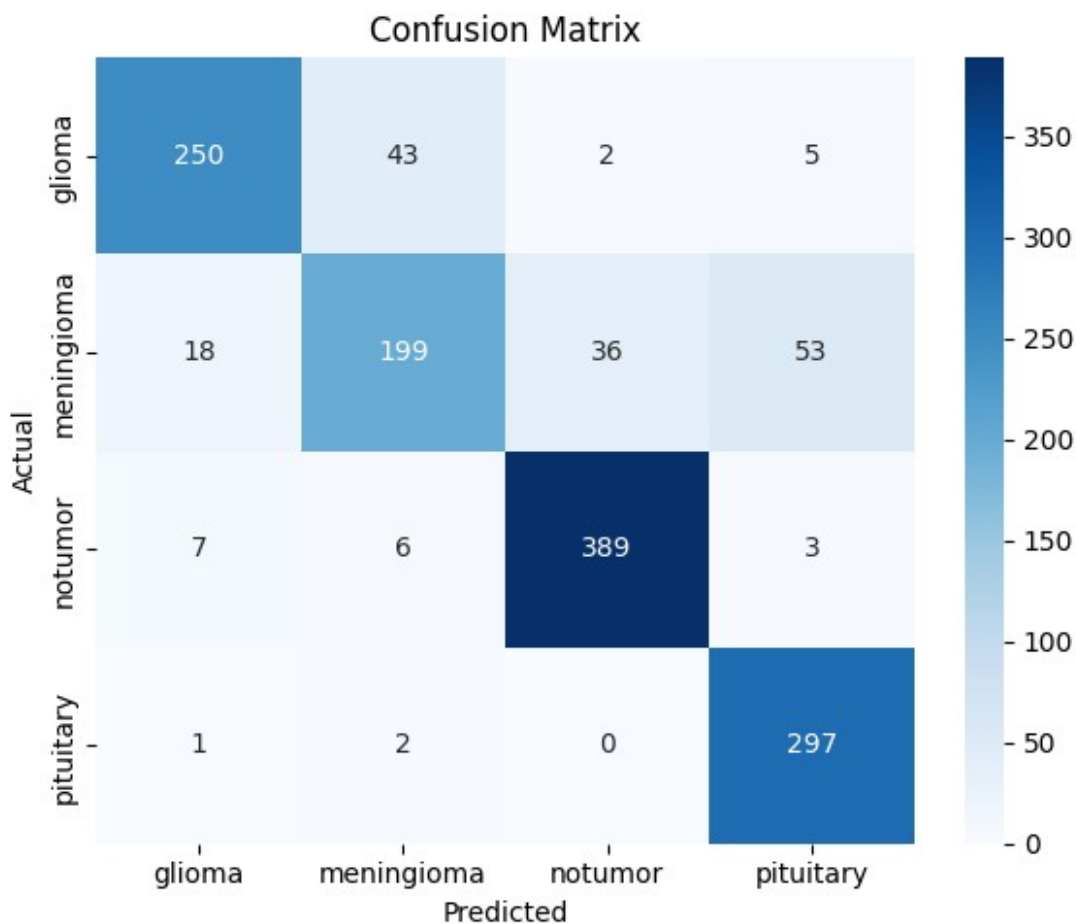
# Confusion matrix
cm = confusion_matrix(y_true_class, y_pred_class)
df_cm = pd.DataFrame(cm, index=class_labels, columns=class_labels)

# Plot the confusion matrix
plt.figure(figsize=(6, 5))
sns.heatmap(df_cm, annot=True, fmt='d', cmap='Blues')
plt.title("Confusion Matrix")
plt.ylabel("Actual")
plt.xlabel("Predicted")
plt.tight_layout()
plt.show()

# Classification report (precision, recall, F1)
report = classification_report(y_true_class, y_pred_class,
target_names=class_labels, output_dict=True)
report_df = pd.DataFrame(report).transpose()

# Round values and print neatly
report_df = report_df.round(3)
display(report_df)

```



	precision	recall	f1-score	support
glioma	0.906	0.833	0.868	300.000
meningioma	0.796	0.650	0.716	306.000
notumor	0.911	0.960	0.935	405.000
pituitary	0.830	0.990	0.903	300.000
accuracy	0.866	0.866	0.866	0.866
macro avg	0.861	0.859	0.855	1311.000
weighted avg	0.864	0.866	0.861	1311.000

One-hot encode true labels for AUC

```
y_true_binarized = label_binarize(y_true_class,
classes=np.arange(n_classes))
```

Compute macro average metrics

```
macro_precision = precision_score(y_true_class, y_pred_class,
average='macro')
```

```
macro_recall = recall_score(y_true_class, y_pred_class,
average='macro')
```

```
macro_f1 = f1_score(y_true_class, y_pred_class, average='macro')
```

```
macro_auc = roc_auc_score(y_true_binarized, y_pred, average='macro',
multi_class='ovr')
```

```
# Append DenseNet121 results
results_df.loc[len(results_df)] = [
    "DenseNet121",
    test_acc,
    test_loss,
    round(macro_precision, 4),
    round(macro_recall, 4),
    round(macro_f1, 4),
    round(macro_auc, 4)
]
```

```
# Display the updated results
display(results_df)
```

	Model	Test Accuracy	Test Loss	Precision	Recall	F1-Score
AUC						
0	VGG16	0.791762	0.622208	0.7854	0.7815	0.7778
0.9282						
1	ResNet50	0.649123	0.968826	0.6300	0.6364	0.6290
0.8358						
2	DenseNet121	0.865751	0.351833	0.8606	0.8585	0.8554
0.9781						

EfficientNetB0

Build Efficientnet Model

```
def build_efficientnetb0_model():
    input_tensor = Input(shape=(224, 224, 1))
    rgb_tensor = Lambda(lambda x: tf.image.grayscale_to_rgb(x))
    (input_tensor)

    base_model = EfficientNetB0(weights='imagenet', include_top=False,
    input_tensor=rgb_tensor)
    for layer in base_model.layers:
        layer.trainable = False

    x = base_model.output
    x = GlobalAveragePooling2D()(x)
    x = Dense(256, activation='relu')(x)
    x = Dropout(0.5)(x)
    output = Dense(4, activation='softmax')(x)

    model = Model(inputs=input_tensor, outputs=output)
    return model
```

```
# Compile the model
```

```
model = build_efficientnetb0_model()  
model.compile(optimizer=Adam(learning_rate=1e-4),  
loss='categorical_crossentropy', metrics=['accuracy'])
```

```
Downloading data from https://storage.googleapis.com/keras-  
applications/efficientnetb0_notop.h5  
16705208/16705208 _____ 1s 0us/step
```

CallBack and Train the model

```
early_stop = EarlyStopping(patience=5, restore_best_weights=True)  
checkpoint = ModelCheckpoint("efficientnetb0_brain_tumor.keras",  
save_best_only=True)
```

```
history = model.fit(  
    train_generator,  
    epochs=25,  
    validation_data=val_generator,  
    callbacks=[early_stop, checkpoint]  
)
```

Epoch 1/25

```
143/143 _____ 79s 365ms/step - accuracy: 0.2629 - loss:  
1.3848 - val_accuracy: 0.2796 - val_loss: 1.3840
```

Epoch 2/25

```
143/143 _____ 28s 191ms/step - accuracy: 0.2755 - loss:  
1.3844 - val_accuracy: 0.2752 - val_loss: 1.3841
```

Epoch 3/25

```
143/143 _____ 30s 199ms/step - accuracy: 0.2693 - loss:  
1.3855 - val_accuracy: 0.2796 - val_loss: 1.3833
```

Epoch 4/25

```
143/143 _____ 29s 195ms/step - accuracy: 0.2770 - loss:  
1.3858 - val_accuracy: 0.2796 - val_loss: 1.3836
```

Epoch 5/25

```
143/143 _____ 30s 203ms/step - accuracy: 0.2639 - loss:  
1.3875 - val_accuracy: 0.2796 - val_loss: 1.3832
```

Epoch 6/25

```
143/143 _____ 29s 194ms/step - accuracy: 0.2628 - loss:  
1.3844 - val_accuracy: 0.2796 - val_loss: 1.3836
```

Epoch 7/25

```
143/143 _____ 30s 200ms/step - accuracy: 0.2748 - loss:  
1.3850 - val_accuracy: 0.2796 - val_loss: 1.3837
```

Epoch 8/25

```
143/143 _____ 29s 197ms/step - accuracy: 0.2615 - loss:
```

1.3859 - val_accuracy: 0.2796 - val_loss: 1.3831
Epoch 9/25
143/143 ————— 29s 196ms/step - accuracy: 0.2897 - loss: 1.3828 - val_accuracy: 0.2796 - val_loss: 1.3832
Epoch 10/25
143/143 ————— 29s 197ms/step - accuracy: 0.2689 - loss: 1.3847 - val_accuracy: 0.2796 - val_loss: 1.3830
Epoch 11/25
143/143 ————— 29s 197ms/step - accuracy: 0.2841 - loss: 1.3824 - val_accuracy: 0.2796 - val_loss: 1.3829
Epoch 12/25
143/143 ————— 29s 193ms/step - accuracy: 0.2749 - loss: 1.3835 - val_accuracy: 0.2796 - val_loss: 1.3831
Epoch 13/25
143/143 ————— 29s 199ms/step - accuracy: 0.2742 - loss: 1.3849 - val_accuracy: 0.2796 - val_loss: 1.3830
Epoch 14/25
143/143 ————— 28s 191ms/step - accuracy: 0.2749 - loss: 1.3835 - val_accuracy: 0.2796 - val_loss: 1.3831
Epoch 15/25
143/143 ————— 29s 194ms/step - accuracy: 0.2820 - loss: 1.3831 - val_accuracy: 0.2796 - val_loss: 1.3830
Epoch 16/25
143/143 ————— 30s 200ms/step - accuracy: 0.2779 - loss: 1.3837 - val_accuracy: 0.2796 - val_loss: 1.3829
Epoch 17/25
143/143 ————— 29s 196ms/step - accuracy: 0.2739 - loss: 1.3841 - val_accuracy: 0.2796 - val_loss: 1.3829
Epoch 18/25
143/143 ————— 30s 200ms/step - accuracy: 0.2807 - loss: 1.3824 - val_accuracy: 0.2796 - val_loss: 1.3827
Epoch 19/25
143/143 ————— 29s 192ms/step - accuracy: 0.2874 - loss: 1.3813 - val_accuracy: 0.2796 - val_loss: 1.3828
Epoch 20/25
143/143 ————— 30s 201ms/step - accuracy: 0.2704 - loss: 1.3849 - val_accuracy: 0.2796 - val_loss: 1.3827
Epoch 21/25
143/143 ————— 29s 193ms/step - accuracy: 0.2744 - loss: 1.3834 - val_accuracy: 0.2796 - val_loss: 1.3827
Epoch 22/25
143/143 ————— 29s 196ms/step - accuracy: 0.2990 - loss: 1.3803 - val_accuracy: 0.2796 - val_loss: 1.3829
Epoch 23/25
143/143 ————— 29s 199ms/step - accuracy: 0.2840 - loss: 1.3823 - val_accuracy: 0.2796 - val_loss: 1.3832
Epoch 24/25
143/143 ————— 32s 215ms/step - accuracy: 0.2911 - loss: 1.3818 - val_accuracy: 0.2796 - val_loss: 1.3828

```
Epoch 25/25
143/143 ————— 29s 196ms/step - accuracy: 0.3061 - loss:
1.3793 - val_accuracy: 0.2796 - val_loss: 1.3828
```

Test Loss and accuracy

```
test_loss, test_acc = model.evaluate(test_generator)
print(f"Test Accuracy: {test_acc:.4f}")
print(f"Test Loss: {test_loss:.4f}")

41/41 ————— 8s 194ms/step - accuracy: 0.1600 - loss:
1.4247
Test Accuracy: 0.3089
Test Loss: 1.3785
```

Train Test Val Loss

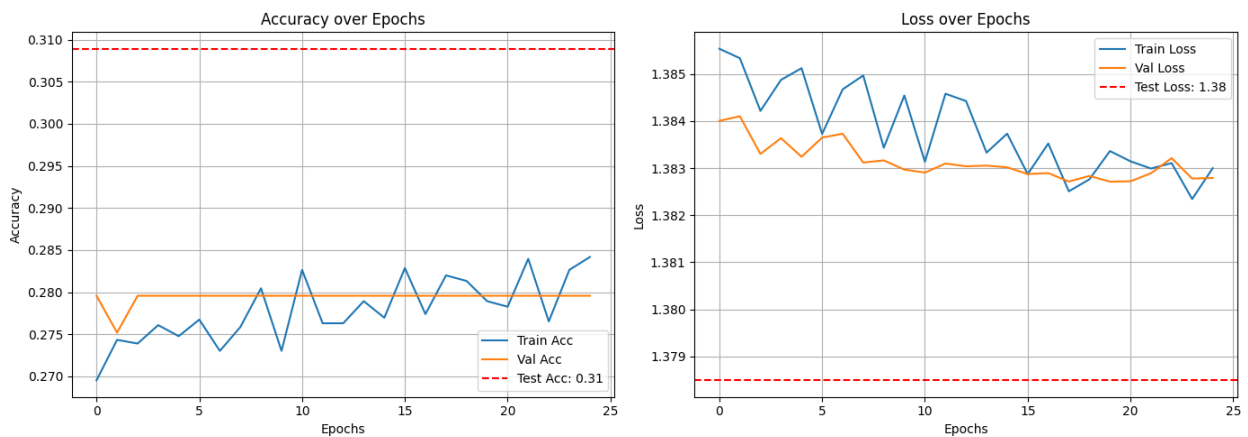
```
def plot_all_curves(history, test_acc, test_loss):
    plt.figure(figsize=(14, 5))

    # Accuracy plot
    plt.subplot(1, 2, 1)
    plt.plot(history.history['accuracy'], label='Train Acc')
    plt.plot(history.history['val_accuracy'], label='Val Acc')
    plt.axhline(y=test_acc, color='r', linestyle='--', label=f'Test
Acc: {test_acc:.2f}')
    plt.title('Accuracy over Epochs')
    plt.xlabel('Epochs')
    plt.ylabel('Accuracy')
    plt.legend()
    plt.grid(True)

    # Loss plot
    plt.subplot(1, 2, 2)
    plt.plot(history.history['loss'], label='Train Loss')
    plt.plot(history.history['val_loss'], label='Val Loss')
    plt.axhline(y=test_loss, color='r', linestyle='--', label=f'Test
Loss: {test_loss:.2f}')
    plt.title('Loss over Epochs')
    plt.xlabel('Epochs')
    plt.ylabel('Loss')
    plt.legend()
    plt.grid(True)

    plt.tight_layout()
```

```
plt.show()
plot_all_curves(history, test_acc, test_loss)
```



AUC - ROC Curve

```
from sklearn.metrics import roc_curve, auc
from sklearn.preprocessing import label_binarize

# Predictions
y_pred = model.predict(test_generator)
y_true = test_generator.classes
n_classes = y_pred.shape[1]

# One-hot encode the true labels
y_true_onehot = label_binarize(y_true, classes=np.arange(n_classes))

# Compute ROC curve and AUC for each class
fpr = {}
tpr = {}
roc_auc = {}

plt.figure(figsize=(10, 8))

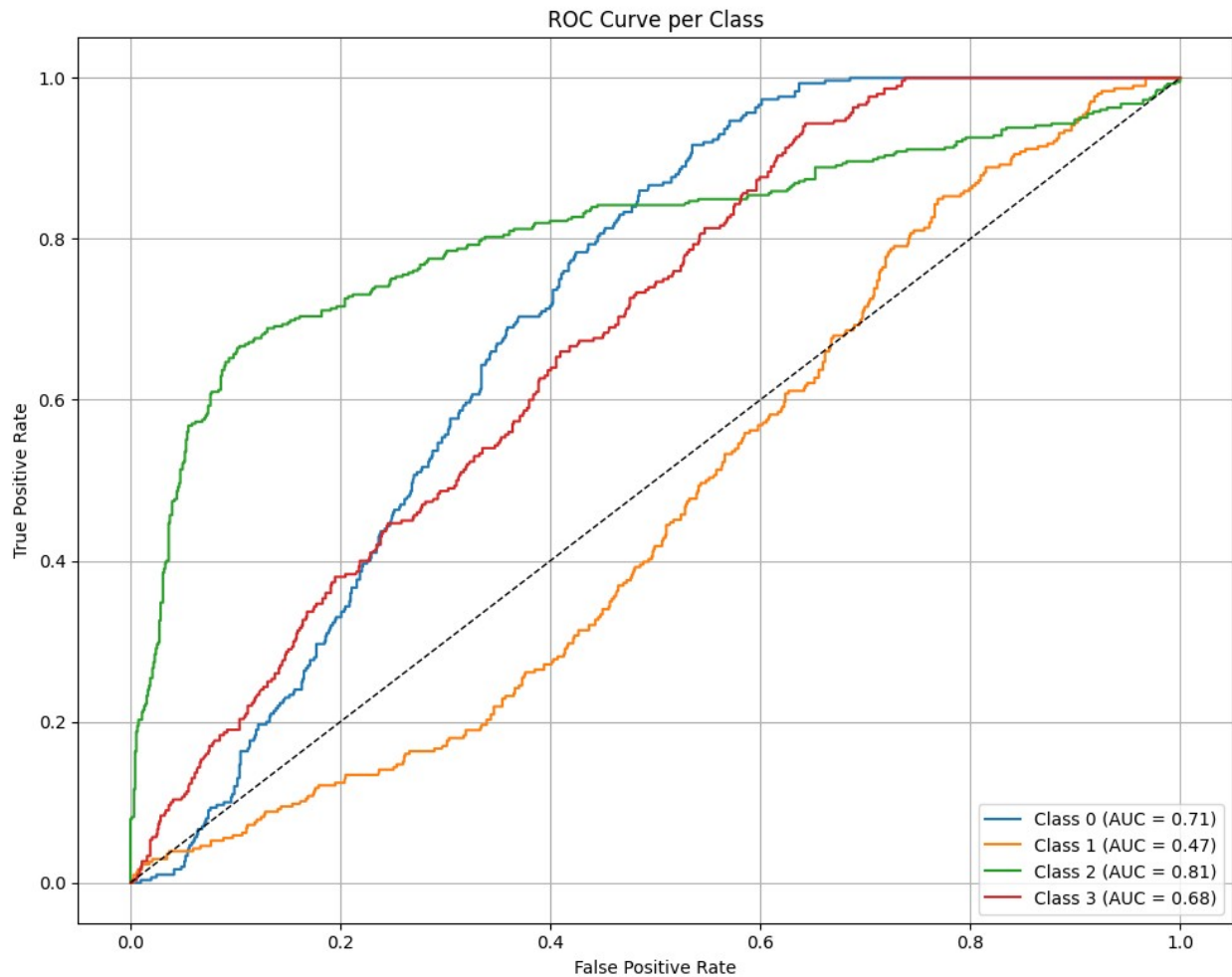
for i in range(n_classes):
    fpr[i], tpr[i], _ = roc_curve(y_true_onehot[:, i], y_pred[:, i])
    roc_auc[i] = auc(fpr[i], tpr[i])
    plt.plot(fpr[i], tpr[i], label=f'Class {i} (AUC = {roc_auc[i]:.2f})')

# Plot diagonal reference line
plt.plot([0, 1], [0, 1], 'k--', linewidth=1)

# Labels and title
```

```
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve per Class")
plt.legend(loc="lower right")
plt.grid(True)
plt.tight_layout()
plt.show()
```

41/41 ————— 15s 217ms/step



Confusion Matrix

```
from sklearn.metrics import confusion_matrix, classification_report,
precision_score, recall_score, f1_score
import seaborn as sns
import pandas as pd
```



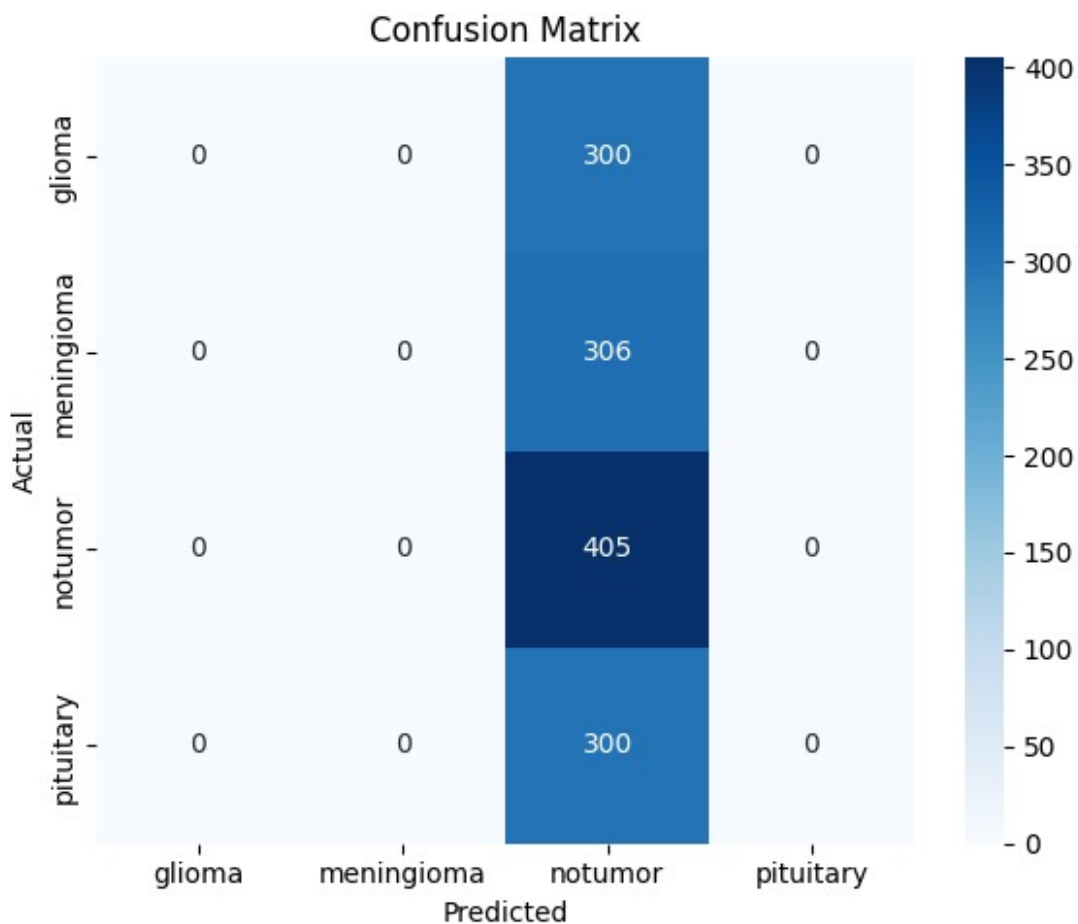
```
# Get predicted class indices
y_pred_class = np.argmax(y_pred, axis=1)
y_true_class = test_generator.classes
class_labels = list(test_generator.class_indices.keys())

# Confusion matrix
cm = confusion_matrix(y_true_class, y_pred_class)
df_cm = pd.DataFrame(cm, index=class_labels, columns=class_labels)

# Plot the confusion matrix
plt.figure(figsize=(6, 5))
sns.heatmap(df_cm, annot=True, fmt='d', cmap='Blues')
plt.title("Confusion Matrix")
plt.ylabel("Actual")
plt.xlabel("Predicted")
plt.tight_layout()
plt.show()

# Classification report (precision, recall, F1)
report = classification_report(y_true_class, y_pred_class,
target_names=class_labels, output_dict=True)
report_df = pd.DataFrame(report).transpose()

# Round values and print neatly
report_df = report_df.round(3)
display(report_df)
```



	precision	recall	f1-score	support
glioma	0.000	0.000	0.000	300.000
meningioma	0.000	0.000	0.000	306.000
notumor	0.309	1.000	0.472	405.000
pituitary	0.000	0.000	0.000	300.000
accuracy	0.309	0.309	0.309	0.309
macro avg	0.077	0.250	0.118	1311.000
weighted avg	0.095	0.309	0.146	1311.000

One-hot encode true labels for AUC

```
y_true_binarized = label_binarize(y_true_class,
classes=np.arange(n_classes))
```

Compute macro average metrics

```
macro_precision = precision_score(y_true_class, y_pred_class,
average='macro')
```

```
macro_recall = recall_score(y_true_class, y_pred_class,
average='macro')
```

```
macro_f1 = f1_score(y_true_class, y_pred_class, average='macro')
```

```
macro_auc = roc_auc_score(y_true_binarized, y_pred, average='macro',
multi_class='ovr')
```

```
# Append EfficientNetB0 results
results_df.loc[len(results_df)] = [
    "EfficientNetB0",
    test_acc,
    test_loss,
    round(macro_precision, 4),
    round(macro_recall, 4),
    round(macro_f1, 4),
    round(macro_auc, 4)
]
```

```
# Display the updated results
display(results_df)
```

	Model	Test Accuracy	Test Loss	Precision	Recall	F1-
Score \						
0	VGG16	0.791762	0.622208	0.7854	0.7815	
0.7778						
1	ResNet50	0.649123	0.968826	0.6300	0.6364	
0.6290						
2	DenseNet121	0.865751	0.351833	0.8606	0.8585	
0.8554						
3	EfficientNetB0	0.308924	1.378495	0.0772	0.2500	
0.1180						

	AUC
0	0.9282
1	0.8358
2	0.9781
3	0.6661

```
results_df = results_df[results_df['Model'] !=
"EfficientNetB0="].reset_index(drop=True)
```

```
# Display the cleaned DataFrame
display(results_df)
```

	Model	Test Accuracy	Test Loss	Precision	Recall	F1-
Score \						
0	VGG16	0.791762	0.622208	0.7854	0.7815	
0.7778						
1	ResNet50	0.649123	0.968826	0.6300	0.6364	
0.6290						
2	DenseNet121	0.865751	0.351833	0.8606	0.8585	
0.8554						
3	EfficientNetB0	0.308924	1.378495	0.0772	0.2500	
0.1180						

	AUC

```
0 0.9282
1 0.8358
2 0.9781
3 0.6661
```

results_df

Score \	Model	Test Accuracy	Test Loss	Precision	Recall	F1-
0	VGG16	0.791762	0.622208	0.7854	0.7815	
1	ResNet50	0.649123	0.968826	0.6300	0.6364	
2	DenseNet121	0.865751	0.351833	0.8606	0.8585	
3	EfficientNetB0	0.308924	1.378495	0.0772	0.2500	

```
AUC
0 0.9282
1 0.8358
2 0.9781
3 0.6661
```

MobileNetV2

Build and Compile MobileNetV2

```
from tensorflow.keras.applications import MobileNetV2

def build_mobilenetv2_model():
    input_tensor = Input(shape=(224, 224, 1))
    rgb_tensor = Lambda(lambda x: tf.image.grayscale_to_rgb(x))
    (input_tensor)

    base_model = MobileNetV2(weights='imagenet', include_top=False,
input_tensor=rgb_tensor)
    for layer in base_model.layers:
        layer.trainable = False

    x = base_model.output
    x = GlobalAveragePooling2D()(x)
    x = Dense(256, activation='relu')(x)
    x = Dropout(0.5)(x)
    output = Dense(4, activation='softmax')(x)

    model = Model(inputs=input_tensor, outputs=output)
    return model
```

```
# Compile the model
model = build_mobilenetv2_model()
model.compile(optimizer=Adam(learning_rate=1e-4),
loss='categorical_crossentropy', metrics=['accuracy'])

Downloading data from https://storage.googleapis.com/tensorflow/keras-
applications/mobilenet_v2/
mobilenet_v2_weights_tf_dim_ordering_tf_kernels_1.0_224_no_top.h5
9406464/9406464 _____ 1s 0us/step
```

Callback and train the model

```
early_stop = EarlyStopping(patience=5, restore_best_weights=True)
checkpoint = ModelCheckpoint("mobilenetv2_brain_tumor.keras",
save_best_only=True)

history = model.fit(
    train_generator,
    epochs=25,
    validation_data=val_generator,
    callbacks=[early_stop, checkpoint]
)

Epoch 1/25
143/143 _____ 51s 267ms/step - accuracy: 0.4967 - loss:
1.2030 - val_accuracy: 0.6784 - val_loss: 0.8604
Epoch 2/25
143/143 _____ 29s 196ms/step - accuracy: 0.7977 - loss:
0.5465 - val_accuracy: 0.7388 - val_loss: 0.7795
Epoch 3/25
143/143 _____ 30s 201ms/step - accuracy: 0.8347 - loss:
0.4417 - val_accuracy: 0.7572 - val_loss: 0.6535
Epoch 4/25
143/143 _____ 29s 194ms/step - accuracy: 0.8723 - loss:
0.3710 - val_accuracy: 0.7713 - val_loss: 0.6321
Epoch 5/25
143/143 _____ 30s 201ms/step - accuracy: 0.8872 - loss:
0.3368 - val_accuracy: 0.7809 - val_loss: 0.5822
Epoch 6/25
143/143 _____ 29s 195ms/step - accuracy: 0.8913 - loss:
0.3266 - val_accuracy: 0.7862 - val_loss: 0.5645
Epoch 7/25
143/143 _____ 29s 198ms/step - accuracy: 0.9052 - loss:
0.2930 - val_accuracy: 0.7888 - val_loss: 0.5584
Epoch 8/25
143/143 _____ 29s 197ms/step - accuracy: 0.8894 - loss:
0.3027 - val_accuracy: 0.7897 - val_loss: 0.5377
```

Epoch 9/25
143/143 _____ 31s 207ms/step - accuracy: 0.8878 - loss: 0.2937 - val_accuracy: 0.8098 - val_loss: 0.5278
Epoch 10/25
143/143 _____ 29s 197ms/step - accuracy: 0.9110 - loss: 0.2581 - val_accuracy: 0.8195 - val_loss: 0.4886
Epoch 11/25
143/143 _____ 30s 200ms/step - accuracy: 0.9029 - loss: 0.2765 - val_accuracy: 0.8168 - val_loss: 0.4747
Epoch 12/25
143/143 _____ 29s 199ms/step - accuracy: 0.9159 - loss: 0.2334 - val_accuracy: 0.8230 - val_loss: 0.4510
Epoch 13/25
143/143 _____ 30s 202ms/step - accuracy: 0.9150 - loss: 0.2325 - val_accuracy: 0.8133 - val_loss: 0.4920
Epoch 14/25
143/143 _____ 29s 197ms/step - accuracy: 0.9207 - loss: 0.2278 - val_accuracy: 0.8282 - val_loss: 0.4446
Epoch 15/25
143/143 _____ 30s 201ms/step - accuracy: 0.9272 - loss: 0.2114 - val_accuracy: 0.8230 - val_loss: 0.4752
Epoch 16/25
143/143 _____ 30s 205ms/step - accuracy: 0.9216 - loss: 0.2085 - val_accuracy: 0.8344 - val_loss: 0.4441
Epoch 17/25
143/143 _____ 29s 197ms/step - accuracy: 0.9232 - loss: 0.2145 - val_accuracy: 0.8309 - val_loss: 0.4522
Epoch 18/25
143/143 _____ 29s 198ms/step - accuracy: 0.9220 - loss: 0.2003 - val_accuracy: 0.8493 - val_loss: 0.4218
Epoch 19/25
143/143 _____ 29s 197ms/step - accuracy: 0.9271 - loss: 0.1925 - val_accuracy: 0.8457 - val_loss: 0.4251
Epoch 20/25
143/143 _____ 30s 199ms/step - accuracy: 0.9332 - loss: 0.1952 - val_accuracy: 0.8493 - val_loss: 0.4012
Epoch 21/25
143/143 _____ 29s 196ms/step - accuracy: 0.9259 - loss: 0.2021 - val_accuracy: 0.8457 - val_loss: 0.4119
Epoch 22/25
143/143 _____ 28s 191ms/step - accuracy: 0.9278 - loss: 0.1898 - val_accuracy: 0.8475 - val_loss: 0.4210
Epoch 23/25
143/143 _____ 30s 201ms/step - accuracy: 0.9372 - loss: 0.1748 - val_accuracy: 0.8449 - val_loss: 0.4089
Epoch 24/25
143/143 _____ 29s 195ms/step - accuracy: 0.9429 - loss: 0.1655 - val_accuracy: 0.8414 - val_loss: 0.4411
Epoch 25/25

```
143/143 ————— 29s 195ms/step - accuracy: 0.9405 - loss: 0.1687 - val_accuracy: 0.8493 - val_loss: 0.4289
```

Test and Loss Accuracy

```
test_loss, test_acc = model.evaluate(test_generator)
print(f"Test Accuracy: {test_acc:.4f}")
print(f"Test Loss: {test_loss:.4f}")
```

```
41/41 ————— 5s 130ms/step - accuracy: 0.7944 - loss: 0.5088
Test Accuracy: 0.8619
Test Loss: 0.3626
```

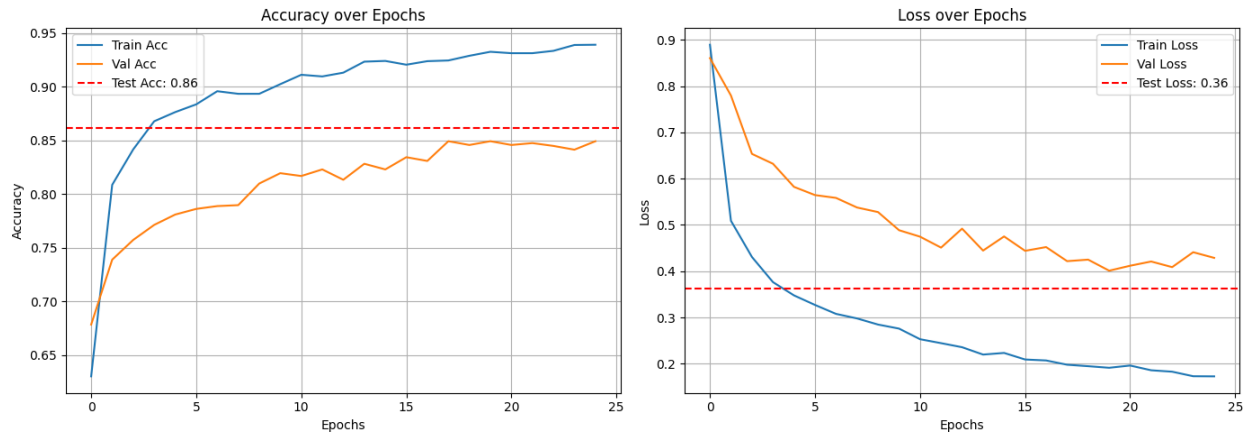
Train Test Val Loss

```
def plot_all_curves(history, test_acc, test_loss):
    plt.figure(figsize=(14, 5))

    # Accuracy plot
    plt.subplot(1, 2, 1)
    plt.plot(history.history['accuracy'], label='Train Acc')
    plt.plot(history.history['val_accuracy'], label='Val Acc')
    plt.axhline(y=test_acc, color='r', linestyle='--', label=f'Test
Acc: {test_acc:.2f}')
    plt.title('Accuracy over Epochs')
    plt.xlabel('Epochs')
    plt.ylabel('Accuracy')
    plt.legend()
    plt.grid(True)

    # Loss plot
    plt.subplot(1, 2, 2)
    plt.plot(history.history['loss'], label='Train Loss')
    plt.plot(history.history['val_loss'], label='Val Loss')
    plt.axhline(y=test_loss, color='r', linestyle='--', label=f'Test
Loss: {test_loss:.2f}')
    plt.title('Loss over Epochs')
    plt.xlabel('Epochs')
    plt.ylabel('Loss')
    plt.legend()
    plt.grid(True)

    plt.tight_layout()
    plt.show()
plot_all_curves(history, test_acc, test_loss)
```



AUC - ROC Curve

```
from sklearn.metrics import roc_curve, auc
from sklearn.preprocessing import label_binarize

# Predictions
y_pred = model.predict(test_generator)
y_true = test_generator.classes
n_classes = y_pred.shape[1]

# One-hot encode the true labels
y_true_onehot = label_binarize(y_true, classes=np.arange(n_classes))

# Compute ROC curve and AUC for each class
fpr = {}
tpr = {}
roc_auc = {}

plt.figure(figsize=(10, 8))

for i in range(n_classes):
    fpr[i], tpr[i], _ = roc_curve(y_true_onehot[:, i], y_pred[:, i])
    roc_auc[i] = auc(fpr[i], tpr[i])
    plt.plot(fpr[i], tpr[i], label=f'Class {i} (AUC = {roc_auc[i]:.2f})')

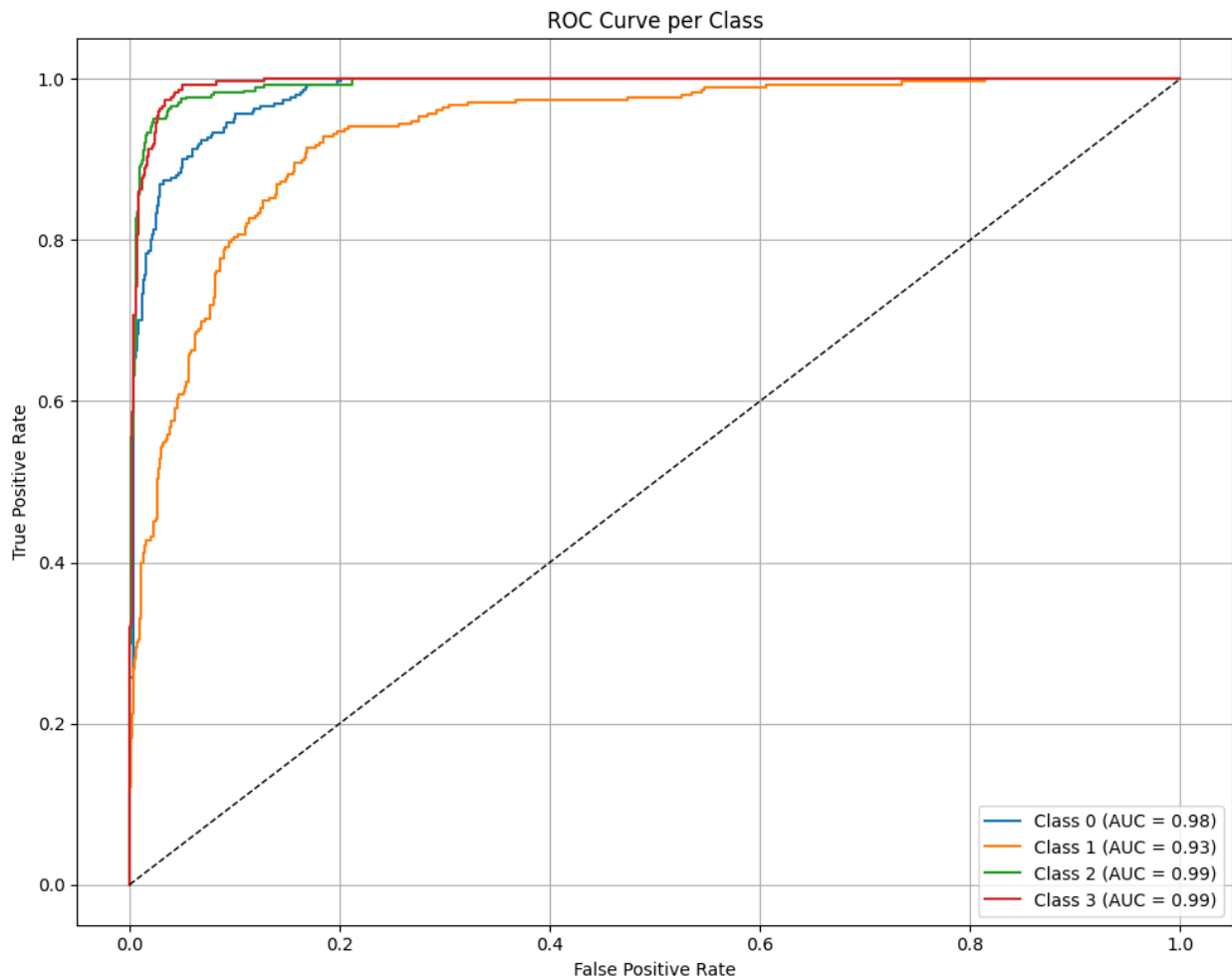
# Plot diagonal reference line
plt.plot([0, 1], [0, 1], 'k--', linewidth=1)

# Labels and title
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve per Class")
plt.legend(loc="lower right")
```



```
plt.grid(True)
plt.tight_layout()
plt.show()
```

41/41 ————— 9s 140ms/step



Confusion Matrix

```
from sklearn.metrics import confusion_matrix, classification_report,
precision_score, recall_score, f1_score
import seaborn as sns
import pandas as pd

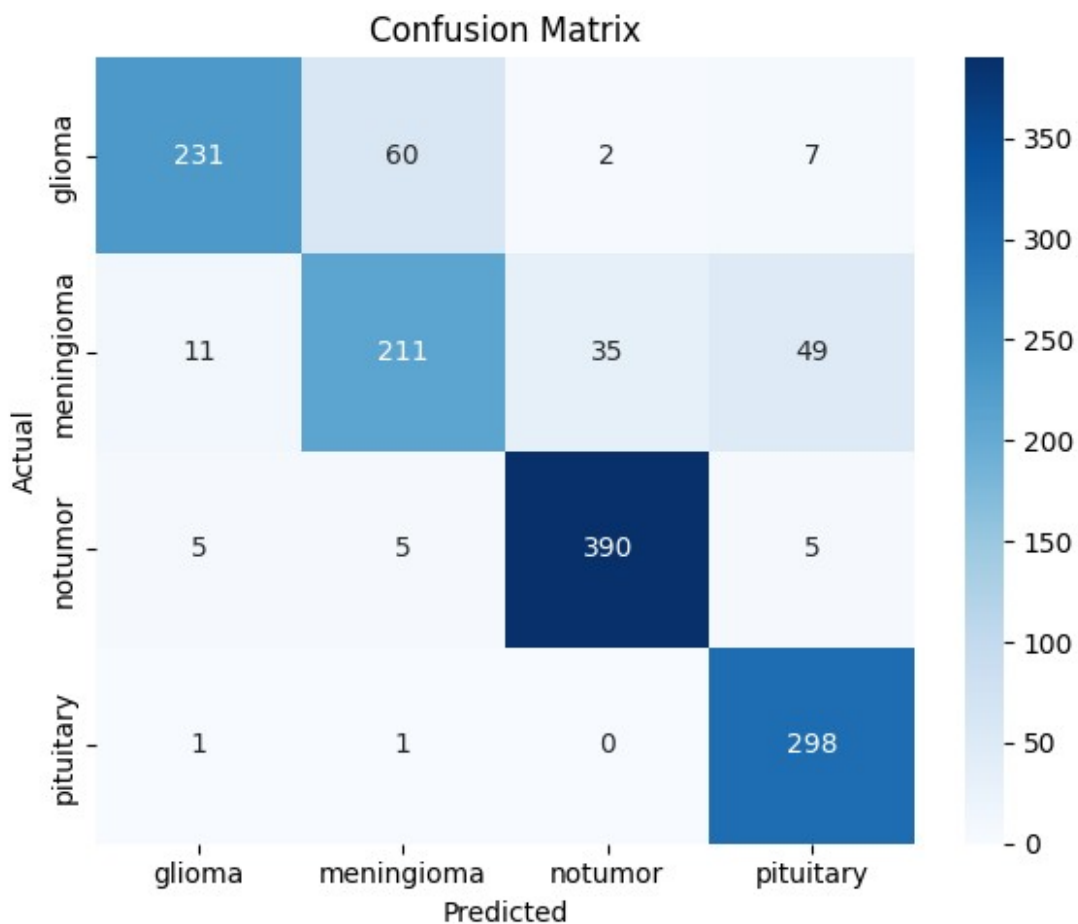
# Get predicted class indices
y_pred_class = np.argmax(y_pred, axis=1)
y_true_class = test_generator.classes
class_labels = list(test_generator.class_indices.keys())
```

```
# Confusion matrix
cm = confusion_matrix(y_true_class, y_pred_class)
df_cm = pd.DataFrame(cm, index=class_labels, columns=class_labels)

# Plot the confusion matrix
plt.figure(figsize=(6, 5))
sns.heatmap(df_cm, annot=True, fmt='d', cmap='Blues')
plt.title("Confusion Matrix")
plt.ylabel("Actual")
plt.xlabel("Predicted")
plt.tight_layout()
plt.show()

# Classification report (precision, recall, F1)
report = classification_report(y_true_class, y_pred_class,
target_names=class_labels, output_dict=True)
report_df = pd.DataFrame(report).transpose()

# Round values and print neatly
report_df = report_df.round(3)
display(report_df)
```



	precision	recall	f1-score	support
glioma	0.931	0.770	0.843	300.000
meningioma	0.762	0.690	0.724	306.000
notumor	0.913	0.963	0.938	405.000
pituitary	0.830	0.993	0.904	300.000
accuracy	0.862	0.862	0.862	0.862
macro avg	0.859	0.854	0.852	1311.000
weighted avg	0.863	0.862	0.858	1311.000

One-hot encode true labels for AUC

```
y_true_binarized = label_binarize(y_true_class,
classes=np.arange(n_classes))
```

Compute macro average metrics

```
macro_precision = precision_score(y_true_class, y_pred_class,
average='macro')
```

```
macro_recall = recall_score(y_true_class, y_pred_class,
average='macro')
```

```
macro_f1 = f1_score(y_true_class, y_pred_class, average='macro')
```

```
macro_auc = roc_auc_score(y_true_binarized, y_pred, average='macro',
multi_class='ovr')
```

```
# Append MobileNetV2 results
```

```
results_df.loc[len(results_df)] = [
    "MobileNetV2",
    test_acc,
    test_loss,
    round(macro_precision, 4),
    round(macro_recall, 4),
    round(macro_f1, 4),
    round(macro_auc, 4)
]
```

```
# Display the updated results
```

```
display(results_df)
```

	Model	Test Accuracy	Test Loss	Precision	Recall	F1-Score
0	VGG16	0.791762	0.622208	0.7854	0.7815	0.7778
1	ResNet50	0.649123	0.968826	0.6300	0.6364	0.6290
2	DenseNet121	0.865751	0.351833	0.8606	0.8585	0.8554
3	EfficientNetB0	0.308924	1.378495	0.0772	0.2500	0.1180
4	MobileNetV2	0.861937	0.362561	0.8592	0.8540	0.8522

	AUC
0	0.9282
1	0.8358
2	0.9781
3	0.6661
4	0.9755

```
# Drop the duplicate MobileNetV2 row (if present)
```

```
results_df = results_df.drop_duplicates(subset=["Model"],
keep="first").reset_index(drop=True)
```

```
# Display cleaned DataFrame
```

```
display(results_df)
```

	Model	Test Accuracy	Test Loss	Precision	Recall	F1-Score
0	VGG16	0.791762	0.622208	0.7854	0.7815	0.7778
1	ResNet50	0.649123	0.968826	0.6300	0.6364	0.6290

2	DenseNet121	0.865751	0.351833	0.8606	0.8585
0.8554					
3	EfficientNetB0	0.308924	1.378495	0.0772	0.2500
0.1180					
4	MobileNetV2	0.861937	0.362561	0.8592	0.8540
0.8522					

	AUC
0	0.9282
1	0.8358
2	0.9781
3	0.6661
4	0.9755

results_df

Score \	Model	Test Accuracy	Test Loss	Precision	Recall	F1-
0	VGG16	0.791762	0.622208	0.7854	0.7815	
0.7778						
1	ResNet50	0.649123	0.968826	0.6300	0.6364	
0.6290						
2	DenseNet121	0.865751	0.351833	0.8606	0.8585	
0.8554						
3	EfficientNetB0	0.308924	1.378495	0.0772	0.2500	
0.1180						
4	MobileNetV2	0.861937	0.362561	0.8592	0.8540	
0.8522						

	AUC
0	0.9282
1	0.8358
2	0.9781
3	0.6661
4	0.9755

InceptionV3

Build the Model

```
from tensorflow.keras.applications import InceptionV3

def build_inceptionv3_model():
    input_tensor = Input(shape=(224, 224, 1))
    rgb_tensor = Lambda(lambda x: tf.image.grayscale_to_rgb(x))
    (input_tensor)

    base_model = InceptionV3(weights='imagenet', include_top=False,
```

```

input_tensor=rgb_tensor)
    for layer in base_model.layers:
        layer.trainable = False

    x = base_model.output
    x = GlobalAveragePooling2D()(x)
    x = Dense(256, activation='relu')(x)
    x = Dropout(0.5)(x)
    output = Dense(4, activation='softmax')(x)

    model = Model(inputs=input_tensor, outputs=output)
    return model

# Compile the model
model = build_inceptionv3_model()
model.compile(optimizer=Adam(learning_rate=1e-4),
              loss='categorical_crossentropy', metrics=['accuracy'])

Downloading data from https://storage.googleapis.com/tensorflow/keras-
applications/inception_v3/
inception_v3_weights_tf_dim_ordering_tf_kernels_notop.h5
87910968/87910968 _____ 3s 0us/step

```

Callback and Train Model

```

early_stop = EarlyStopping(patience=5, restore_best_weights=True)
checkpoint = ModelCheckpoint("inceptionv3_brain_tumor.keras",
                             save_best_only=True)

history = model.fit(
    train_generator,
    validation_data=val_generator,
    epochs=25,
    callbacks=[early_stop, checkpoint]
)

Epoch 1/25
143/143 _____ 72s 357ms/step - accuracy: 0.5475 - loss:
1.1060 - val_accuracy: 0.6740 - val_loss: 0.8506
Epoch 2/25
143/143 _____ 31s 213ms/step - accuracy: 0.7856 - loss:
0.5785 - val_accuracy: 0.7283 - val_loss: 0.7282
Epoch 3/25
143/143 _____ 31s 211ms/step - accuracy: 0.8237 - loss:
0.4781 - val_accuracy: 0.7388 - val_loss: 0.6855
Epoch 4/25
143/143 _____ 31s 210ms/step - accuracy: 0.8375 - loss:

```

0.4259 - val_accuracy: 0.7441 - val_loss: 0.6947
Epoch 5/25
143/143 _____ 31s 213ms/step - accuracy: 0.8519 - loss:
0.4022 - val_accuracy: 0.7616 - val_loss: 0.5874
Epoch 6/25
143/143 _____ 31s 210ms/step - accuracy: 0.8673 - loss:
0.3680 - val_accuracy: 0.7634 - val_loss: 0.6220
Epoch 7/25
143/143 _____ 30s 203ms/step - accuracy: 0.8688 - loss:
0.3716 - val_accuracy: 0.7791 - val_loss: 0.6135
Epoch 8/25
143/143 _____ 31s 213ms/step - accuracy: 0.8633 - loss:
0.3366 - val_accuracy: 0.7669 - val_loss: 0.6189
Epoch 9/25
143/143 _____ 32s 215ms/step - accuracy: 0.8814 - loss:
0.3365 - val_accuracy: 0.8046 - val_loss: 0.5267
Epoch 10/25
143/143 _____ 31s 211ms/step - accuracy: 0.8751 - loss:
0.3283 - val_accuracy: 0.7914 - val_loss: 0.5594
Epoch 11/25
143/143 _____ 32s 214ms/step - accuracy: 0.8808 - loss:
0.3247 - val_accuracy: 0.8063 - val_loss: 0.5167
Epoch 12/25
143/143 _____ 31s 211ms/step - accuracy: 0.8852 - loss:
0.3041 - val_accuracy: 0.7862 - val_loss: 0.5491
Epoch 13/25
143/143 _____ 30s 205ms/step - accuracy: 0.8981 - loss:
0.2722 - val_accuracy: 0.7765 - val_loss: 0.6117
Epoch 14/25
143/143 _____ 32s 215ms/step - accuracy: 0.8957 - loss:
0.2865 - val_accuracy: 0.8133 - val_loss: 0.4964
Epoch 15/25
143/143 _____ 31s 213ms/step - accuracy: 0.9011 - loss:
0.2760 - val_accuracy: 0.8256 - val_loss: 0.4915
Epoch 16/25
143/143 _____ 31s 208ms/step - accuracy: 0.9013 - loss:
0.2619 - val_accuracy: 0.8063 - val_loss: 0.5173
Epoch 17/25
143/143 _____ 31s 208ms/step - accuracy: 0.9141 - loss:
0.2548 - val_accuracy: 0.8054 - val_loss: 0.5310
Epoch 18/25
143/143 _____ 31s 209ms/step - accuracy: 0.9047 - loss:
0.2499 - val_accuracy: 0.8221 - val_loss: 0.5007
Epoch 19/25
143/143 _____ 31s 212ms/step - accuracy: 0.9189 - loss:
0.2273 - val_accuracy: 0.8405 - val_loss: 0.4575
Epoch 20/25
143/143 _____ 31s 209ms/step - accuracy: 0.9213 - loss:
0.2208 - val_accuracy: 0.8256 - val_loss: 0.4826

```

Epoch 21/25
143/143 _____ 32s 216ms/step - accuracy: 0.9197 - loss:
0.2273 - val_accuracy: 0.8344 - val_loss: 0.4371
Epoch 22/25
143/143 _____ 31s 212ms/step - accuracy: 0.9197 - loss:
0.2220 - val_accuracy: 0.8422 - val_loss: 0.4301
Epoch 23/25
143/143 _____ 31s 210ms/step - accuracy: 0.9171 - loss:
0.2282 - val_accuracy: 0.8352 - val_loss: 0.4591
Epoch 24/25
143/143 _____ 34s 228ms/step - accuracy: 0.9248 - loss:
0.2108 - val_accuracy: 0.8545 - val_loss: 0.4264
Epoch 25/25
143/143 _____ 33s 225ms/step - accuracy: 0.9234 - loss:
0.2063 - val_accuracy: 0.8326 - val_loss: 0.4646

```

Test And Loss Accuracy

```

test_loss, test_acc = model.evaluate(test_generator)
print(f"Test Accuracy: {test_acc:.4f}")
print(f"Test Loss: {test_loss:.4f}")

41/41 _____ 10s 241ms/step - accuracy: 0.8371 - loss:
0.3916
Test Accuracy: 0.8810
Test Loss: 0.3066

```

Train Test Val Plot

```

def plot_all_curves(history, test_acc, test_loss):
    plt.figure(figsize=(14, 5))

    # Accuracy plot
    plt.subplot(1, 2, 1)
    plt.plot(history.history['accuracy'], label='Train Acc')
    plt.plot(history.history['val_accuracy'], label='Val Acc')
    plt.axhline(y=test_acc, color='r', linestyle='--', label=f'Test
Acc: {test_acc:.2f}')
    plt.title('Accuracy over Epochs')
    plt.xlabel('Epochs')
    plt.ylabel('Accuracy')
    plt.legend()
    plt.grid(True)

    # Loss plot
    plt.subplot(1, 2, 2)

```

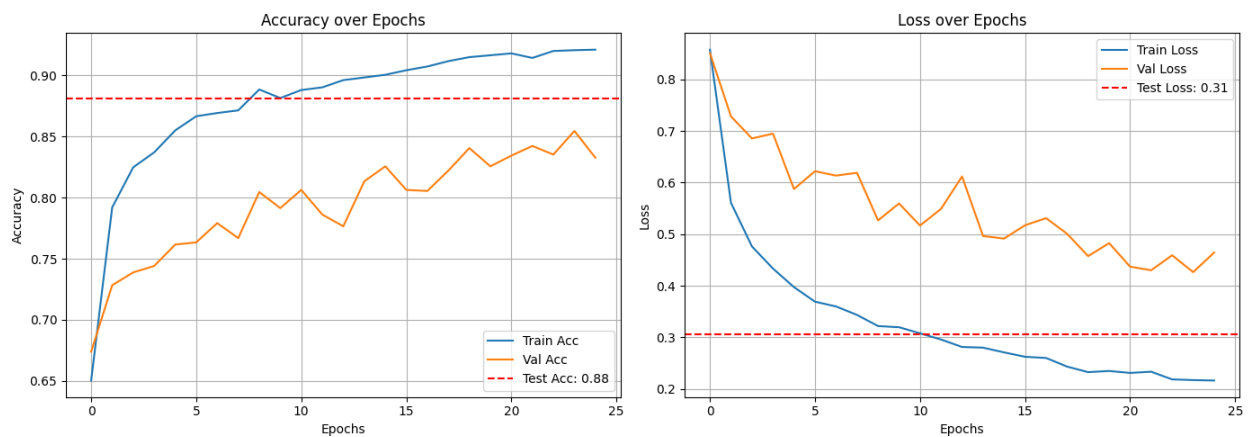


```

plt.plot(history.history['loss'], label='Train Loss')
plt.plot(history.history['val_loss'], label='Val Loss')
plt.axhline(y=test_loss, color='r', linestyle='--', label=f'Test
Loss: {test_loss:.2f}')
plt.title('Loss over Epochs')
plt.xlabel('Epochs')
plt.ylabel('Loss')
plt.legend()
plt.grid(True)

plt.tight_layout()
plt.show()
plot_all_curves(history, test_acc, test_loss)

```



AUC - ROC Curve

```

from sklearn.metrics import roc_curve, auc
from sklearn.preprocessing import label_binarize

# Predictions
y_pred = model.predict(test_generator)
y_true = test_generator.classes
n_classes = y_pred.shape[1]

# One-hot encode the true labels
y_true_onehot = label_binarize(y_true, classes=np.arange(n_classes))

# Compute ROC curve and AUC for each class
fpr = {}
tpr = {}
roc_auc = {}

```

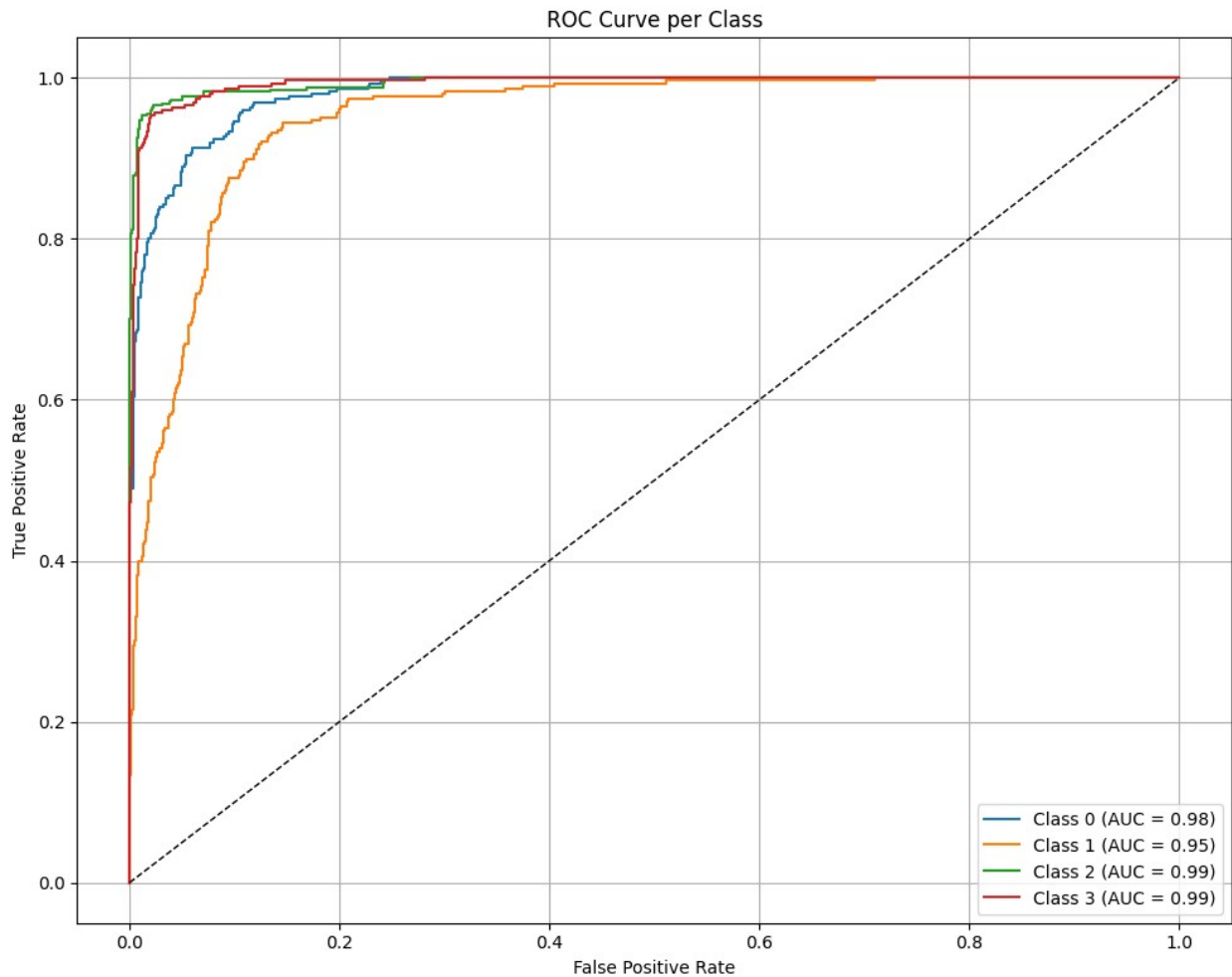
```
plt.figure(figsize=(10, 8))

for i in range(n_classes):
    fpr[i], tpr[i], _ = roc_curve(y_true_onehot[:, i], y_pred[:, i])
    roc_auc[i] = auc(fpr[i], tpr[i])
    plt.plot(fpr[i], tpr[i], label=f'Class {i} (AUC =
{roc_auc[i]:.2f})')

# Plot diagonal reference line
plt.plot([0, 1], [0, 1], 'k--', linewidth=1)

# Labels and title
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve per Class")
plt.legend(loc="lower right")
plt.grid(True)
plt.tight_layout()
plt.show()
```

41/41 ————— 15s 223ms/step



Confusion Matrix

```
from sklearn.metrics import confusion_matrix, classification_report,
precision_score, recall_score, f1_score
import seaborn as sns
import pandas as pd

# Get predicted class indices
y_pred_class = np.argmax(y_pred, axis=1)
y_true_class = test_generator.classes
class_labels = list(test_generator.class_indices.keys())

# Confusion matrix
cm = confusion_matrix(y_true_class, y_pred_class)
df_cm = pd.DataFrame(cm, index=class_labels, columns=class_labels)

# Plot the confusion matrix
plt.figure(figsize=(6, 5))
```

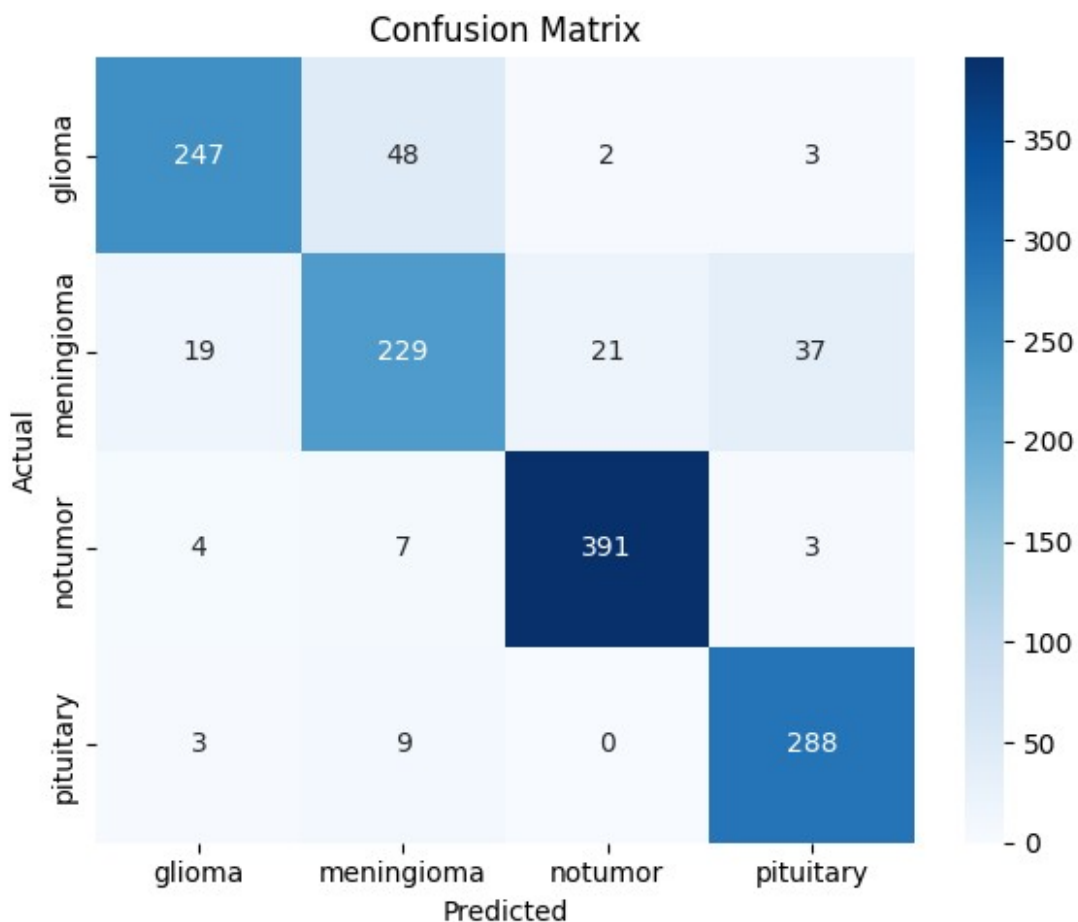
```

sns.heatmap(df_cm, annot=True, fmt='d', cmap='Blues')
plt.title("Confusion Matrix")
plt.ylabel("Actual")
plt.xlabel("Predicted")
plt.tight_layout()
plt.show()

# Classification report (precision, recall, F1)
report = classification_report(y_true_class, y_pred_class,
target_names=class_labels, output_dict=True)
report_df = pd.DataFrame(report).transpose()

# Round values and print neatly
report_df = report_df.round(3)
display(report_df)

```



	precision	recall	f1-score	support
glioma	0.905	0.823	0.862	300.000
meningioma	0.782	0.748	0.765	306.000
notumor	0.944	0.965	0.955	405.000

pituitary	0.870	0.960	0.913	300.000
accuracy	0.881	0.881	0.881	0.881
macro avg	0.875	0.874	0.874	1311.000
weighted avg	0.880	0.881	0.880	1311.000

```
# # Ensure DataFrame has correct structure
# try:
#     results_df
# except NameError:
#     results_df = pd.DataFrame(columns=[
#         "Model", "Test Accuracy", "Test Loss", "F1-Score", "AUC"
#     ])

# # One-hot encode true labels for AUC
# y_true_binarized = label_binarize(y_true_class,
#     classes=np.arange(n_classes))

# # Compute F1 and AUC
# macro_f1 = f1_score(y_true_class, y_pred_class, average='macro')
# macro_auc = roc_auc_score(y_true_binarized, y_pred, average='macro',
#     multi_class='ovr')

# # Append InceptionV3 results
# results_df.loc[len(results_df)] = [
#     "InceptionV3",
#     test_acc,
#     test_loss,
#     round(macro_f1, 4),
#     round(macro_auc, 4)
# ]

# # Display the updated DataFrame
# display(results_df)
# One-hot encode true labels for AUC
y_true_binarized = label_binarize(y_true_class,
    classes=np.arange(n_classes))

# Compute macro average metrics
macro_precision = precision_score(y_true_class, y_pred_class,
    average='macro')
macro_recall = recall_score(y_true_class, y_pred_class,
    average='macro')
macro_f1 = f1_score(y_true_class, y_pred_class, average='macro')
macro_auc = roc_auc_score(y_true_binarized, y_pred, average='macro',
    multi_class='ovr')

# Append MobileNetV2 results
results_df.loc[len(results_df)] = [
    "InceptionV3",
    test_acc,
```

```

    test_loss,
    round(macro_precision, 4),
    round(macro_recall, 4),
    round(macro_f1, 4),
    round(macro_auc, 4)
]

```

```

# Display the updated results
display(results_df)

```

	Model	Test Accuracy	Test Loss	Precision	Recall	F1-
Score \						
0	VGG16	0.791762	0.622208	0.7854	0.7815	
0.7778						
1	ResNet50	0.649123	0.968826	0.6300	0.6364	
0.6290						
2	DenseNet121	0.865751	0.351833	0.8606	0.8585	
0.8554						
3	EfficientNetB0	0.308924	1.378495	0.0772	0.2500	
0.1180						
4	MobileNetV2	0.861937	0.362561	0.8592	0.8540	
0.8522						
5	InceptionV3	0.881007	0.306638	0.8752	0.8743	
0.8736						
	AUC					
0	0.9282					
1	0.8358					
2	0.9781					
3	0.6661					
4	0.9755					
5	0.9800					

Xception

Build the Model

```

from tensorflow.keras.applications import Xception
def build_xception_model():
    input_tensor = Input(shape=(224, 224, 1))
    rgb_tensor = Lambda(lambda x: tf.image.grayscale_to_rgb(x))
    (input_tensor)

    base_model = Xception(weights='imagenet', include_top=False,
input_tensor=rgb_tensor)
    for layer in base_model.layers:

```

```

        layer.trainable = False

    x = base_model.output
    x = GlobalAveragePooling2D()(x)
    x = Dense(256, activation='relu')(x)
    x = Dropout(0.5)(x)
    output = Dense(4, activation='softmax')(x)

    model = Model(inputs=input_tensor, outputs=output)
    return model

# Compile the model
model = build_xception_model()
model.compile(optimizer=Adam(learning_rate=1e-4),
              loss='categorical_crossentropy', metrics=['accuracy'])

Downloading data from https://storage.googleapis.com/tensorflow/keras-
applications/xception/
xception_weights_tf_dim_ordering_tf_kernels_notop.h5
83683744/83683744 _____ 3s 0us/step

```

Callbacks and Train

```

early_stop = EarlyStopping(patience=5, restore_best_weights=True)
checkpoint = ModelCheckpoint("xception_brain_tumor.keras",
                             save_best_only=True)

history = model.fit(
    train_generator,
    validation_data=val_generator,
    epochs=25,
    callbacks=[early_stop, checkpoint]
)

Epoch 1/25
143/143 _____ 61s 331ms/step - accuracy: 0.5499 - loss:
1.0428 - val_accuracy: 0.7064 - val_loss: 0.8332
Epoch 2/25
143/143 _____ 35s 236ms/step - accuracy: 0.8107 - loss:
0.5205 - val_accuracy: 0.7485 - val_loss: 0.7240
Epoch 3/25
143/143 _____ 32s 218ms/step - accuracy: 0.8440 - loss:
0.4434 - val_accuracy: 0.7826 - val_loss: 0.6140
Epoch 4/25
143/143 _____ 33s 225ms/step - accuracy: 0.8548 - loss:
0.3995 - val_accuracy: 0.7695 - val_loss: 0.5976
Epoch 5/25
143/143 _____ 33s 223ms/step - accuracy: 0.8654 - loss:

```

0.3667 - val_accuracy: 0.7862 - val_loss: 0.5614
Epoch 6/25
143/143 _____ 33s 222ms/step - accuracy: 0.8791 - loss:
0.3419 - val_accuracy: 0.8011 - val_loss: 0.5425
Epoch 7/25
143/143 _____ 32s 217ms/step - accuracy: 0.8769 - loss:
0.3553 - val_accuracy: 0.8089 - val_loss: 0.5326
Epoch 8/25
143/143 _____ 41s 214ms/step - accuracy: 0.8979 - loss:
0.3026 - val_accuracy: 0.7800 - val_loss: 0.5971
Epoch 9/25
143/143 _____ 33s 223ms/step - accuracy: 0.8875 - loss:
0.2975 - val_accuracy: 0.8072 - val_loss: 0.4970
Epoch 10/25
143/143 _____ 32s 218ms/step - accuracy: 0.8983 - loss:
0.2849 - val_accuracy: 0.8203 - val_loss: 0.4753
Epoch 11/25
143/143 _____ 32s 214ms/step - accuracy: 0.8970 - loss:
0.2808 - val_accuracy: 0.8221 - val_loss: 0.4818
Epoch 12/25
143/143 _____ 32s 215ms/step - accuracy: 0.9057 - loss:
0.2589 - val_accuracy: 0.8335 - val_loss: 0.4463
Epoch 13/25
143/143 _____ 32s 214ms/step - accuracy: 0.8966 - loss:
0.2836 - val_accuracy: 0.8309 - val_loss: 0.4471
Epoch 14/25
143/143 _____ 31s 209ms/step - accuracy: 0.9155 - loss:
0.2420 - val_accuracy: 0.8265 - val_loss: 0.4535
Epoch 15/25
143/143 _____ 32s 218ms/step - accuracy: 0.9167 - loss:
0.2435 - val_accuracy: 0.8344 - val_loss: 0.4447
Epoch 16/25
143/143 _____ 31s 211ms/step - accuracy: 0.9162 - loss:
0.2431 - val_accuracy: 0.8344 - val_loss: 0.4587
Epoch 17/25
143/143 _____ 32s 218ms/step - accuracy: 0.9191 - loss:
0.2312 - val_accuracy: 0.8387 - val_loss: 0.4317
Epoch 18/25
143/143 _____ 32s 217ms/step - accuracy: 0.9195 - loss:
0.2224 - val_accuracy: 0.8405 - val_loss: 0.4300
Epoch 19/25
143/143 _____ 34s 228ms/step - accuracy: 0.9138 - loss:
0.2332 - val_accuracy: 0.8422 - val_loss: 0.4208
Epoch 20/25
143/143 _____ 35s 234ms/step - accuracy: 0.9241 - loss:
0.2266 - val_accuracy: 0.8501 - val_loss: 0.4189
Epoch 21/25
143/143 _____ 33s 223ms/step - accuracy: 0.9189 - loss:
0.2254 - val_accuracy: 0.8457 - val_loss: 0.4111


```

Epoch 22/25
143/143 _____ 33s 222ms/step - accuracy: 0.9215 - loss:
0.2177 - val_accuracy: 0.8659 - val_loss: 0.4025
Epoch 23/25
143/143 _____ 31s 212ms/step - accuracy: 0.9246 - loss:
0.2114 - val_accuracy: 0.8422 - val_loss: 0.4234
Epoch 24/25
143/143 _____ 32s 216ms/step - accuracy: 0.9293 - loss:
0.2017 - val_accuracy: 0.8528 - val_loss: 0.4075
Epoch 25/25
143/143 _____ 31s 211ms/step - accuracy: 0.9309 - loss:
0.1935 - val_accuracy: 0.8528 - val_loss: 0.4040

```

Test and Loss Accuracy

```

test_loss, test_acc = model.evaluate(test_generator)
print(f"Test Accuracy: {test_acc:.4f}")
print(f"Test Loss: {test_loss:.4f}")

41/41 _____ 7s 180ms/step - accuracy: 0.8448 - loss:
0.3858
Test Accuracy: 0.8833
Test Loss: 0.3102

```

Train Test Loss Curve

```

def plot_all_curves(history, test_acc, test_loss):
    plt.figure(figsize=(14, 5))

    # Accuracy plot
    plt.subplot(1, 2, 1)
    plt.plot(history.history['accuracy'], label='Train Acc')
    plt.plot(history.history['val_accuracy'], label='Val Acc')
    plt.axhline(y=test_acc, color='r', linestyle='--', label=f'Test
Acc: {test_acc:.2f}')
    plt.title('Accuracy over Epochs')
    plt.xlabel('Epochs')
    plt.ylabel('Accuracy')
    plt.legend()
    plt.grid(True)

    # Loss plot
    plt.subplot(1, 2, 2)
    plt.plot(history.history['loss'], label='Train Loss')
    plt.plot(history.history['val_loss'], label='Val Loss')

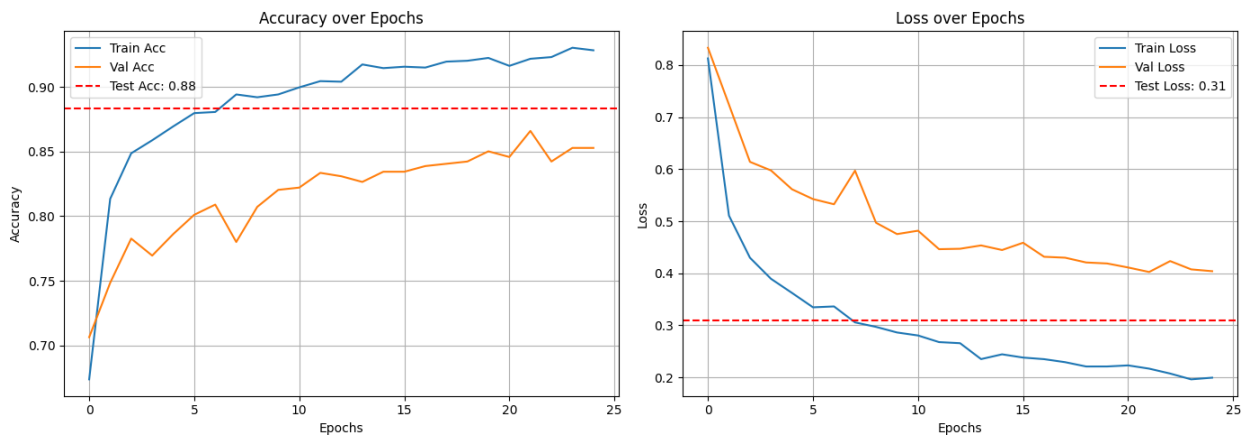
```

```

plt.axhline(y=test_loss, color='r', linestyle='--', label=f'Test
Loss: {test_loss:.2f}')
plt.title('Loss over Epochs')
plt.xlabel('Epochs')
plt.ylabel('Loss')
plt.legend()
plt.grid(True)

plt.tight_layout()
plt.show()
plot_all_curves(history, test_acc, test_loss)

```



AUC - ROC Curve

```

from sklearn.metrics import roc_curve, auc
from sklearn.preprocessing import label_binarize

# Predictions
y_pred = model.predict(test_generator)
y_true = test_generator.classes
n_classes = y_pred.shape[1]

# One-hot encode the true labels
y_true_onehot = label_binarize(y_true, classes=np.arange(n_classes))

# Compute ROC curve and AUC for each class
fpr = {}
tpr = {}
roc_auc = {}

plt.figure(figsize=(10, 8))

```

```

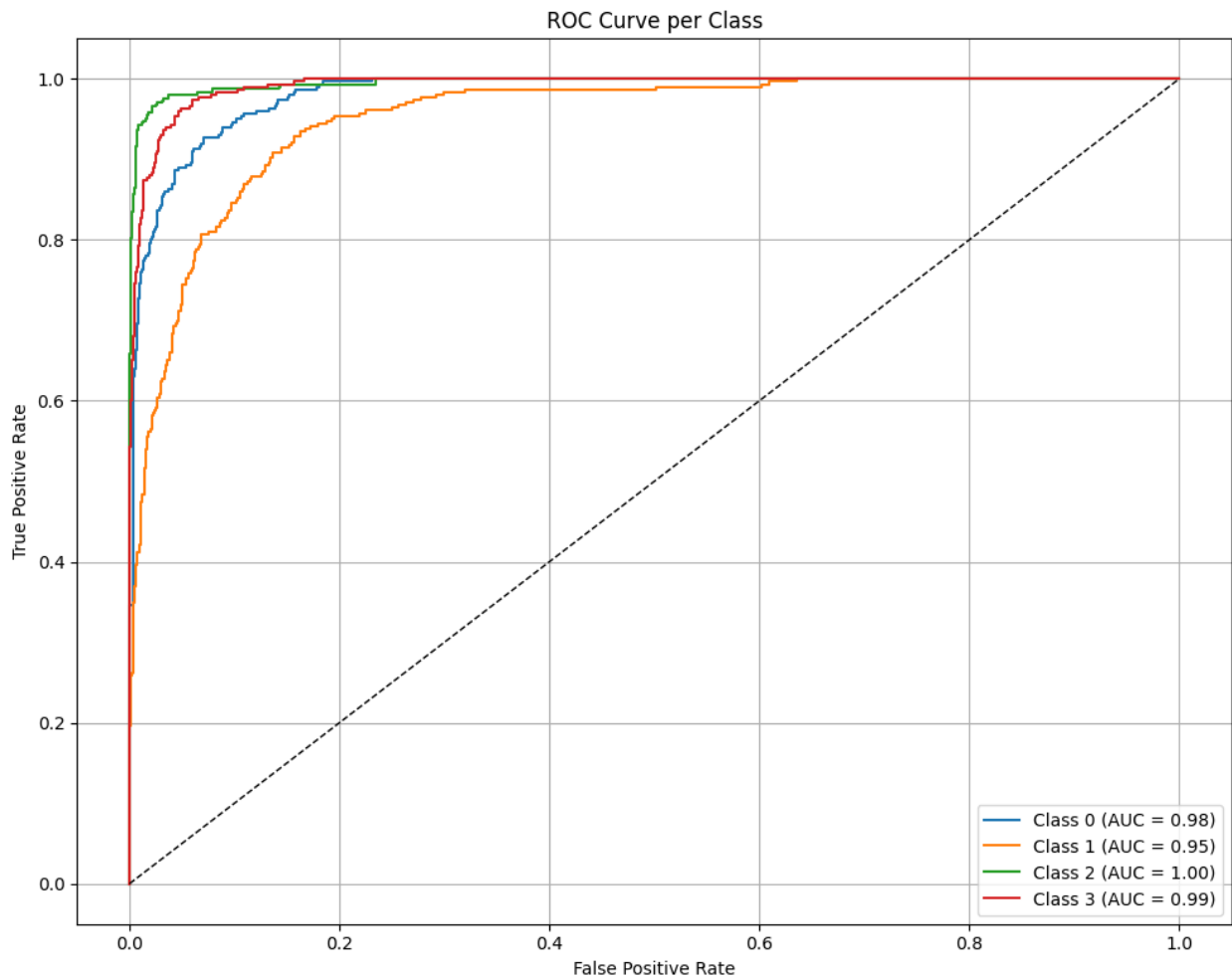
for i in range(n_classes):
    fpr[i], tpr[i], _ = roc_curve(y_true_onehot[:, i], y_pred[:, i])
    roc_auc[i] = auc(fpr[i], tpr[i])
    plt.plot(fpr[i], tpr[i], label=f'Class {i} (AUC =
{roc_auc[i]:.2f})')

# Plot diagonal reference line
plt.plot([0, 1], [0, 1], 'k--', linewidth=1)

# Labels and title
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve per Class")
plt.legend(loc="lower right")
plt.grid(True)
plt.tight_layout()
plt.show()

```

41/41 ————— 9s 137ms/step



Confusion Matrix

```
from sklearn.metrics import confusion_matrix, classification_report,
precision_score, recall_score, f1_score
import seaborn as sns
import pandas as pd

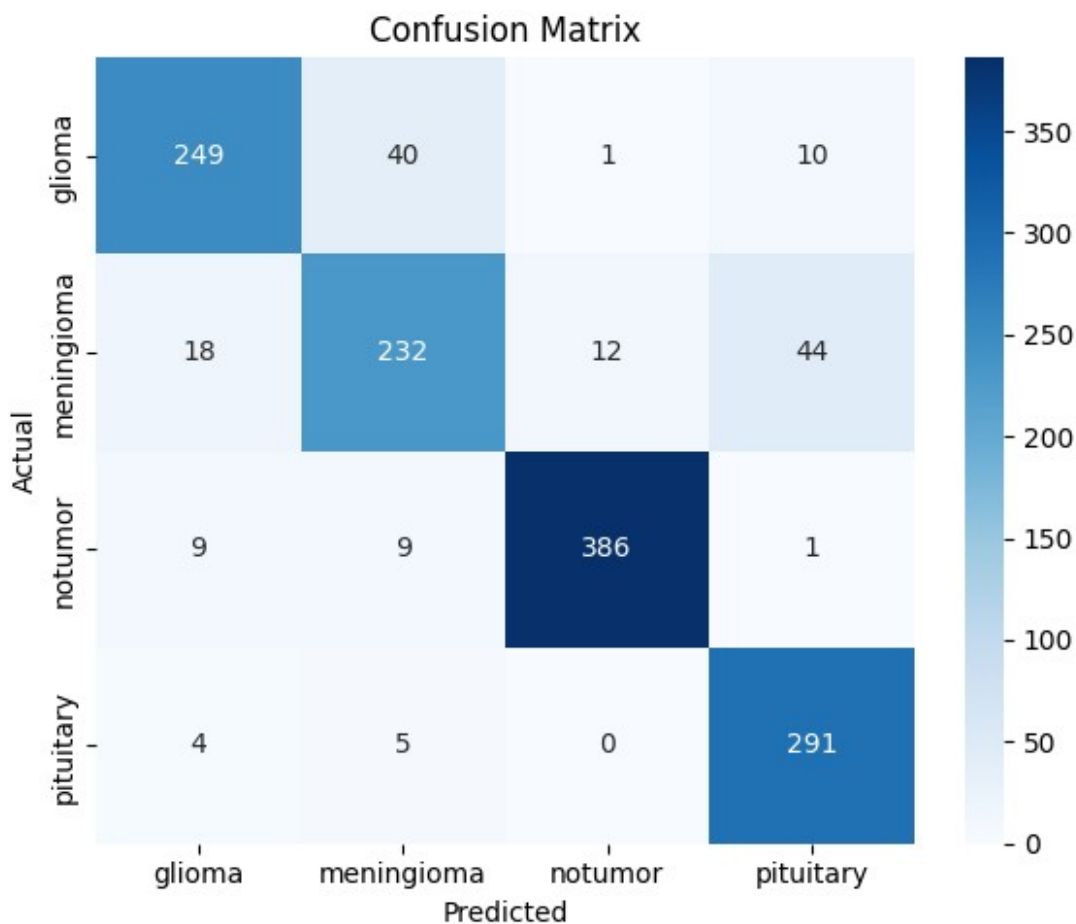
# Get predicted class indices
y_pred_class = np.argmax(y_pred, axis=1)
y_true_class = test_generator.classes
class_labels = list(test_generator.class_indices.keys())

# Confusion matrix
cm = confusion_matrix(y_true_class, y_pred_class)
df_cm = pd.DataFrame(cm, index=class_labels, columns=class_labels)

# Plot the confusion matrix
plt.figure(figsize=(6, 5))
sns.heatmap(df_cm, annot=True, fmt='d', cmap='Blues')
plt.title("Confusion Matrix")
plt.ylabel("Actual")
plt.xlabel("Predicted")
plt.tight_layout()
plt.show()

# Classification report (precision, recall, F1)
report = classification_report(y_true_class, y_pred_class,
target_names=class_labels, output_dict=True)
report_df = pd.DataFrame(report).transpose()

# Round values and print neatly
report_df = report_df.round(3)
display(report_df)
```



	precision	recall	f1-score	support
glioma	0.889	0.830	0.859	300.000
meningioma	0.811	0.758	0.784	306.000
notumor	0.967	0.953	0.960	405.000
pituitary	0.841	0.970	0.901	300.000
accuracy	0.883	0.883	0.883	0.883
macro avg	0.877	0.878	0.876	1311.000
weighted avg	0.884	0.883	0.882	1311.000

```
y_true_binarized = label_binarize(y_true_class,
classes=np.arange(n_classes))
```

```
# Compute macro average metrics
```

```
macro_precision = precision_score(y_true_class, y_pred_class,
average='macro')
```

```
macro_recall = recall_score(y_true_class, y_pred_class,
average='macro')
```

```
macro_f1 = f1_score(y_true_class, y_pred_class, average='macro')
```

```
macro_auc = roc_auc_score(y_true_binarized, y_pred, average='macro',
multi_class='ovr')
```

```
# Append MobileNetV2 results
results_df.loc[len(results_df)] = [
    "Xception",
    test_acc,
    test_loss,
    round(macro_precision, 4),
    round(macro_recall, 4),
    round(macro_f1, 4),
    round(macro_auc, 4)
]
```

```
# Display the updated results
display(results_df)
```

	Model	Test Accuracy	Test Loss	Precision	Recall	F1-
Score \						
0	VGG16	0.791762	0.622208	0.7854	0.7815	
0.7778						
1	ResNet50	0.649123	0.968826	0.6300	0.6364	
0.6290						
2	DenseNet121	0.865751	0.351833	0.8606	0.8585	
0.8554						
3	EfficientNetB0	0.308924	1.378495	0.0772	0.2500	
0.1180						
4	MobileNetV2	0.861937	0.362561	0.8592	0.8540	
0.8522						
5	InceptionV3	0.881007	0.306638	0.8752	0.8743	
0.8736						
6	Xception	0.883295	0.310204	0.8772	0.8778	
0.8759						
	AUC					
0	0.9282					
1	0.8358					
2	0.9781					
3	0.6661					
4	0.9755					
5	0.9800					
6	0.9803					

NASNetMobile

```
from tensorflow.keras.applications import NASNetMobile

def build_nasnetmobile_model():
    input_tensor = Input(shape=(224, 224, 1))
```

```

    rgb_tensor = Lambda(lambda x: tf.image.grayscale_to_rgb(x))
(input_tensor)

    base_model = NASNetMobile(weights='imagenet', include_top=False,
input_tensor=rgb_tensor)
    for layer in base_model.layers:
        layer.trainable = False

    x = base_model.output
    x = GlobalAveragePooling2D()(x)
    x = Dense(256, activation='relu')(x)
    x = Dropout(0.5)(x)
    output = Dense(4, activation='softmax')(x)

    model = Model(inputs=input_tensor, outputs=output)
    return model

# Compile the model
model = build_nasnetmobile_model()
model.compile(optimizer=Adam(learning_rate=1e-4),
loss='categorical_crossentropy', metrics=['accuracy'])

Downloading data from https://storage.googleapis.com/tensorflow/keras-
applications/nasnet/NASNet-mobile-no-top.h5
19993432/19993432 _____ 2s 0us/step

```

Train with Callbacks

```

early_stop = EarlyStopping(patience=5, restore_best_weights=True)
checkpoint = ModelCheckpoint("nasnetmobile_brain_tumor.keras",
save_best_only=True)

history = model.fit(
    train_generator,
    validation_data=val_generator,
    epochs=25,
    callbacks=[early_stop, checkpoint]
)

Epoch 1/25
143/143 _____ 108s 450ms/step - accuracy: 0.4507 -
loss: 1.2651 - val_accuracy: 0.6731 - val_loss: 0.8887
Epoch 2/25
143/143 _____ 32s 216ms/step - accuracy: 0.7289 - loss:
0.7050 - val_accuracy: 0.6933 - val_loss: 0.8174
Epoch 3/25
143/143 _____ 32s 214ms/step - accuracy: 0.7798 - loss:

```

0.5807 - val_accuracy: 0.7143 - val_loss: 0.7881
Epoch 4/25
143/143 _____ 32s 217ms/step - accuracy: 0.8196 - loss:
0.5094 - val_accuracy: 0.7160 - val_loss: 0.7734
Epoch 5/25
143/143 _____ 33s 223ms/step - accuracy: 0.8180 - loss:
0.4840 - val_accuracy: 0.7239 - val_loss: 0.7289
Epoch 6/25
143/143 _____ 32s 216ms/step - accuracy: 0.8312 - loss:
0.4548 - val_accuracy: 0.7415 - val_loss: 0.6692
Epoch 7/25
143/143 _____ 32s 218ms/step - accuracy: 0.8424 - loss:
0.4203 - val_accuracy: 0.7388 - val_loss: 0.6626
Epoch 8/25
143/143 _____ 32s 217ms/step - accuracy: 0.8490 - loss:
0.4117 - val_accuracy: 0.7511 - val_loss: 0.6519
Epoch 9/25
143/143 _____ 32s 218ms/step - accuracy: 0.8555 - loss:
0.3852 - val_accuracy: 0.7695 - val_loss: 0.6208
Epoch 10/25
143/143 _____ 32s 213ms/step - accuracy: 0.8638 - loss:
0.3640 - val_accuracy: 0.7669 - val_loss: 0.6170
Epoch 11/25
143/143 _____ 32s 217ms/step - accuracy: 0.8634 - loss:
0.3650 - val_accuracy: 0.7818 - val_loss: 0.5734
Epoch 12/25
143/143 _____ 30s 203ms/step - accuracy: 0.8697 - loss:
0.3498 - val_accuracy: 0.7879 - val_loss: 0.5838
Epoch 13/25
143/143 _____ 31s 207ms/step - accuracy: 0.8771 - loss:
0.3398 - val_accuracy: 0.7721 - val_loss: 0.6036
Epoch 14/25
143/143 _____ 30s 203ms/step - accuracy: 0.8661 - loss:
0.3713 - val_accuracy: 0.7818 - val_loss: 0.5890
Epoch 15/25
143/143 _____ 32s 219ms/step - accuracy: 0.8792 - loss:
0.3247 - val_accuracy: 0.7888 - val_loss: 0.5550
Epoch 16/25
143/143 _____ 30s 206ms/step - accuracy: 0.8926 - loss:
0.3061 - val_accuracy: 0.7783 - val_loss: 0.5741
Epoch 17/25
143/143 _____ 30s 204ms/step - accuracy: 0.8838 - loss:
0.3135 - val_accuracy: 0.7914 - val_loss: 0.5603
Epoch 18/25
143/143 _____ 31s 213ms/step - accuracy: 0.8875 - loss:
0.3132 - val_accuracy: 0.7897 - val_loss: 0.5527
Epoch 19/25
143/143 _____ 32s 213ms/step - accuracy: 0.8887 - loss:
0.3089 - val_accuracy: 0.7914 - val_loss: 0.5438


```

Epoch 20/25
143/143 _____ 30s 203ms/step - accuracy: 0.8939 - loss:
0.3019 - val_accuracy: 0.7932 - val_loss: 0.5535
Epoch 21/25
143/143 _____ 32s 219ms/step - accuracy: 0.8984 - loss:
0.2959 - val_accuracy: 0.8011 - val_loss: 0.5337
Epoch 22/25
143/143 _____ 30s 204ms/step - accuracy: 0.8894 - loss:
0.2992 - val_accuracy: 0.7932 - val_loss: 0.5341
Epoch 23/25
143/143 _____ 32s 216ms/step - accuracy: 0.8942 - loss:
0.2875 - val_accuracy: 0.8142 - val_loss: 0.5066
Epoch 24/25
143/143 _____ 30s 203ms/step - accuracy: 0.9034 - loss:
0.2752 - val_accuracy: 0.8098 - val_loss: 0.5343
Epoch 25/25
143/143 _____ 30s 204ms/step - accuracy: 0.9041 - loss:
0.2796 - val_accuracy: 0.7844 - val_loss: 0.5687

```

Test Accuracy and Loss

```

test_loss, test_acc = model.evaluate(test_generator)
print(f"Test Accuracy: {test_acc:.4f}")
print(f"Test Loss: {test_loss:.4f}")

41/41 _____ 10s 255ms/step - accuracy: 0.7834 - loss:
0.5310
Test Accuracy: 0.8345
Test Loss: 0.4257

```

Train Test and val plot

```

def plot_all_curves(history, test_acc, test_loss):
    plt.figure(figsize=(14, 5))

    # Accuracy plot
    plt.subplot(1, 2, 1)
    plt.plot(history.history['accuracy'], label='Train Acc')
    plt.plot(history.history['val_accuracy'], label='Val Acc')
    plt.axhline(y=test_acc, color='r', linestyle='--', label=f'Test
Acc: {test_acc:.2f}')
    plt.title('Accuracy over Epochs')
    plt.xlabel('Epochs')
    plt.ylabel('Accuracy')
    plt.legend()
    plt.grid(True)

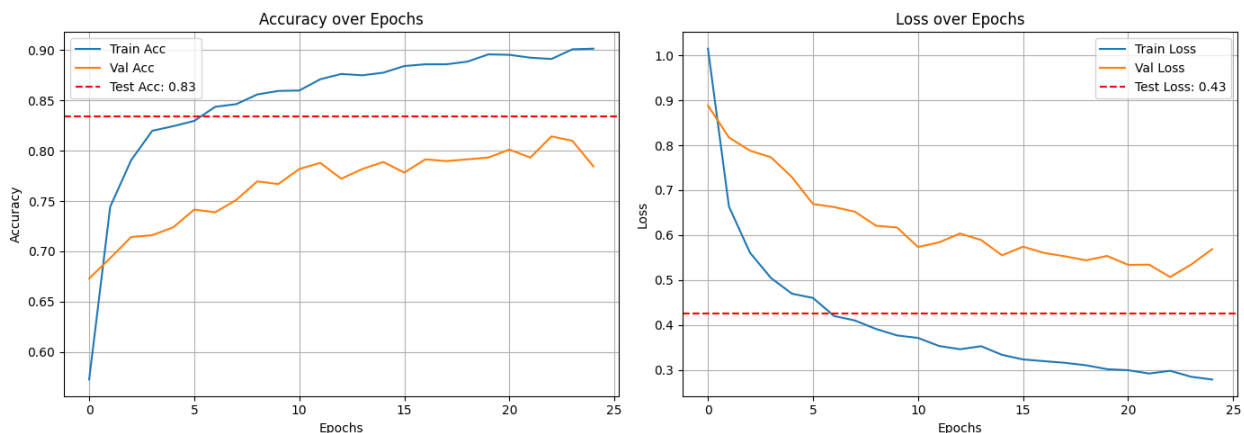
```

```

# Loss plot
plt.subplot(1, 2, 2)
plt.plot(history.history['loss'], label='Train Loss')
plt.plot(history.history['val_loss'], label='Val Loss')
plt.axhline(y=test_loss, color='r', linestyle='--', label=f'Test
Loss: {test_loss:.2f}')
plt.title('Loss over Epochs')
plt.xlabel('Epochs')
plt.ylabel('Loss')
plt.legend()
plt.grid(True)

plt.tight_layout()
plt.show()
plot_all_curves(history, test_acc, test_loss)

```



AUC - ROC Curve

```

from sklearn.metrics import roc_curve, auc
from sklearn.preprocessing import label_binarize

# Predictions
y_pred = model.predict(test_generator)
y_true = test_generator.classes
n_classes = y_pred.shape[1]

# One-hot encode the true labels
y_true_onehot = label_binarize(y_true, classes=np.arange(n_classes))

# Compute ROC curve and AUC for each class
fpr = {}
tpr = {}

```

```
roc_auc = {}

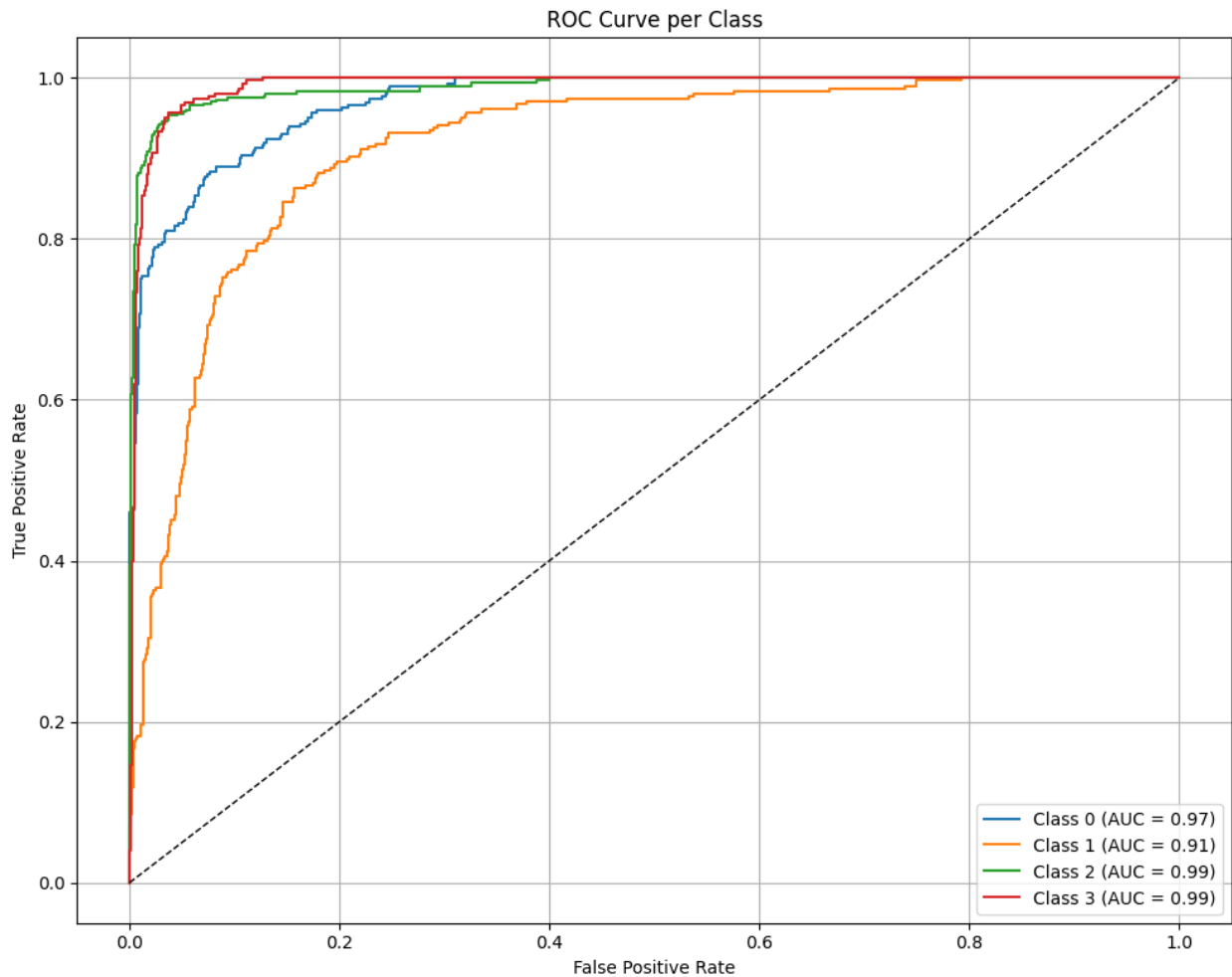
plt.figure(figsize=(10, 8))

for i in range(n_classes):
    fpr[i], tpr[i], _ = roc_curve(y_true_onehot[:, i], y_pred[:, i])
    roc_auc[i] = auc(fpr[i], tpr[i])
    plt.plot(fpr[i], tpr[i], label=f'Class {i} (AUC =
{roc_auc[i]:.2f})')

# Plot diagonal reference line
plt.plot([0, 1], [0, 1], 'k--', linewidth=1)

# Labels and title
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve per Class")
plt.legend(loc="lower right")
plt.grid(True)
plt.tight_layout()
plt.show()
```

41/41 ————— 32s 422ms/step



Confusion Matrix

```
from sklearn.metrics import confusion_matrix, classification_report,
precision_score, recall_score, f1_score
import seaborn as sns
import pandas as pd

# Get predicted class indices
y_pred_class = np.argmax(y_pred, axis=1)
y_true_class = test_generator.classes
class_labels = list(test_generator.class_indices.keys())

# Confusion matrix
cm = confusion_matrix(y_true_class, y_pred_class)
df_cm = pd.DataFrame(cm, index=class_labels, columns=class_labels)
```

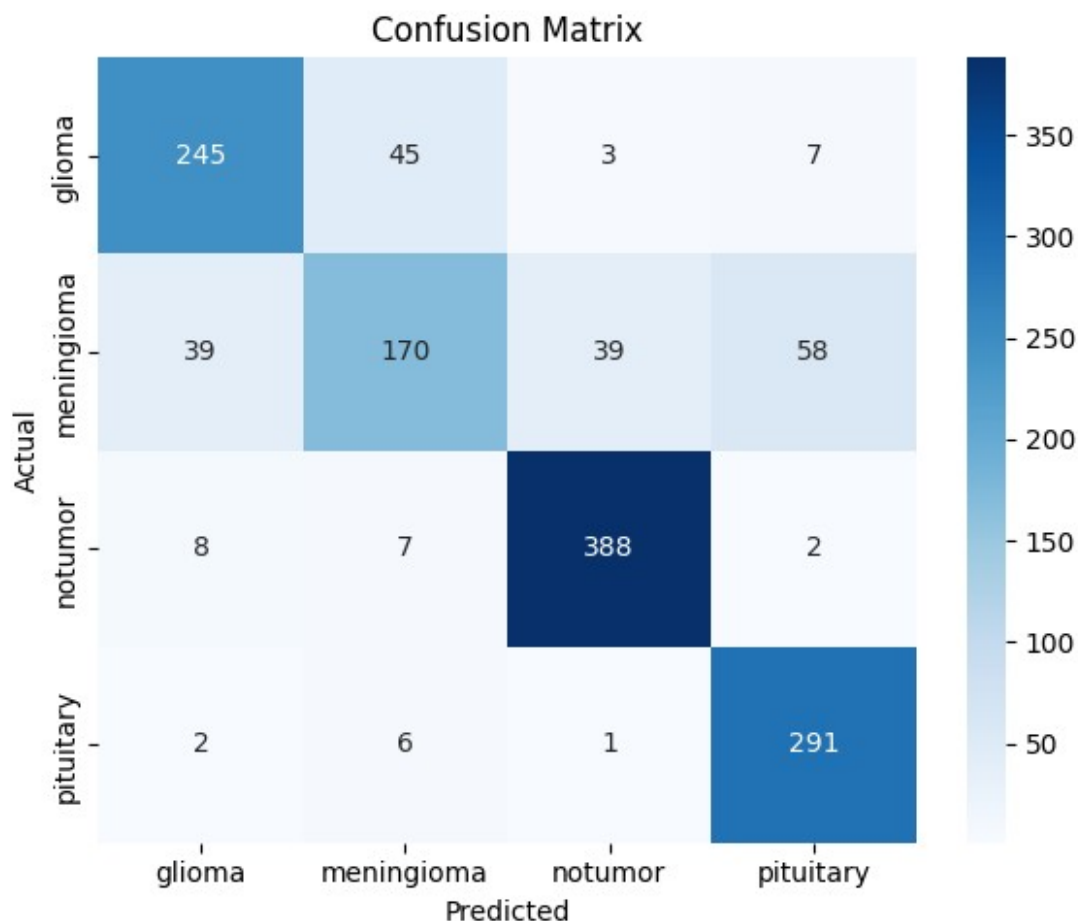
```

# Plot the confusion matrix
plt.figure(figsize=(6, 5))
sns.heatmap(df_cm, annot=True, fmt='d', cmap='Blues')
plt.title("Confusion Matrix")
plt.ylabel("Actual")
plt.xlabel("Predicted")
plt.tight_layout()
plt.show()

# Classification report (precision, recall, F1)
report = classification_report(y_true_class, y_pred_class,
target_names=class_labels, output_dict=True)
report_df = pd.DataFrame(report).transpose()

# Round values and print neatly
report_df = report_df.round(3)
display(report_df)

```



	precision	recall	f1-score	support
glioma	0.833	0.817	0.825	300.000

meningioma	0.746	0.556	0.637	306.000
notumor	0.900	0.958	0.928	405.000
pituitary	0.813	0.970	0.884	300.000
accuracy	0.834	0.834	0.834	0.834
macro avg	0.823	0.825	0.819	1311.000
weighted avg	0.829	0.834	0.827	1311.000

```
y_true_binarized = label_binarize(y_true_class,
classes=np.arange(n_classes))
```

```
# Compute macro average metrics
```

```
macro_precision = precision_score(y_true_class, y_pred_class,
average='macro')
macro_recall = recall_score(y_true_class, y_pred_class,
average='macro')
macro_f1 = f1_score(y_true_class, y_pred_class, average='macro')
macro_auc = roc_auc_score(y_true_binarized, y_pred, average='macro',
multi_class='ovr')
```

```
# Append MobileNetV2 results
```

```
results_df.loc[len(results_df)] = [
    "NASANetMobile",
    test_acc,
    test_loss,
    round(macro_precision, 4),
    round(macro_recall, 4),
    round(macro_f1, 4),
    round(macro_auc, 4)
]
```

```
# Display the updated results
```

```
display(results_df)
```

	Model	Test Accuracy	Test Loss	Precision	Recall	F1-Score
0	VGG16	0.791762	0.622208	0.7854	0.7815	0.7778
1	ResNet50	0.649123	0.968826	0.6300	0.6364	0.6290
2	DenseNet121	0.865751	0.351833	0.8606	0.8585	0.8554
3	EfficientNetB0	0.308924	1.378495	0.0772	0.2500	0.1180
4	MobileNetV2	0.861937	0.362561	0.8592	0.8540	0.8522
5	InceptionV3	0.881007	0.306638	0.8752	0.8743	0.8736
6	Xception	0.883295	0.310204	0.8772	0.8778	0.8759
7	NASANetMobile	0.834477	0.425686	0.8230	0.8251	

0.8186

	AUC
0	0.9282
1	0.8358
2	0.9781
3	0.6661
4	0.9755
5	0.9800
6	0.9803
7	0.9664

InceptionResNetV2

Build the Model

```
from tensorflow.keras.applications import InceptionResNetV2

def build_inceptionresnetv2_model():
    input_tensor = Input(shape=(224, 224, 1))
    rgb_tensor = Lambda(lambda x: tf.image.grayscale_to_rgb(x))
    (input_tensor)

    base_model = InceptionResNetV2(weights='imagenet',
    include_top=False, input_tensor=rgb_tensor)
    for layer in base_model.layers:
        layer.trainable = False

    x = base_model.output
    x = GlobalAveragePooling2D()(x)
    x = Dense(256, activation='relu')(x)
    x = Dropout(0.5)(x)
    output = Dense(4, activation='softmax')(x)

    model = Model(inputs=input_tensor, outputs=output)
    return model

# Compile the model
model = build_inceptionresnetv2_model()
model.compile(optimizer=Adam(learning_rate=1e-4),
loss='categorical_crossentropy', metrics=['accuracy'])

Downloading data from https://storage.googleapis.com/tensorflow/keras-
applications/inception_resnet_v2/
inception_resnet_v2_weights_tf_dim_ordering_tf_kernels_notop.h5
219055592/219055592 ————— 8s 0us/step
```

```
early_stop = EarlyStopping(patience=5, restore_best_weights=True)
checkpoint = ModelCheckpoint("inceptionresnetv2_brain_tumor.keras",
save_best_only=True)
```

```
history = model.fit(
    train_generator,
    validation_data=val_generator,
    epochs=25,
    callbacks=[early_stop, checkpoint]
)
```

Epoch 1/25

143/143 ————— 103s 463ms/step - accuracy: 0.5357 -
loss: 1.1690 - val_accuracy: 0.6626 - val_loss: 0.9661

Epoch 2/25

143/143 ————— 35s 239ms/step - accuracy: 0.7765 - loss:
0.5706 - val_accuracy: 0.6976 - val_loss: 0.8029

Epoch 3/25

143/143 ————— 33s 225ms/step - accuracy: 0.8166 - loss:
0.5041 - val_accuracy: 0.7152 - val_loss: 0.8168

Epoch 4/25

143/143 ————— 36s 242ms/step - accuracy: 0.8463 - loss:
0.4371 - val_accuracy: 0.7195 - val_loss: 0.7498

Epoch 5/25

143/143 ————— 35s 239ms/step - accuracy: 0.8468 - loss:
0.4143 - val_accuracy: 0.7362 - val_loss: 0.7290

Epoch 6/25

143/143 ————— 35s 238ms/step - accuracy: 0.8488 - loss:
0.4294 - val_accuracy: 0.7450 - val_loss: 0.6715

Epoch 7/25

143/143 ————— 33s 221ms/step - accuracy: 0.8639 - loss:
0.3807 - val_accuracy: 0.7432 - val_loss: 0.7113

Epoch 8/25

143/143 ————— 36s 244ms/step - accuracy: 0.8590 - loss:
0.3944 - val_accuracy: 0.7818 - val_loss: 0.6074

Epoch 9/25

143/143 ————— 35s 238ms/step - accuracy: 0.8748 - loss:
0.3517 - val_accuracy: 0.7844 - val_loss: 0.5957

Epoch 10/25

143/143 ————— 35s 236ms/step - accuracy: 0.8744 - loss:
0.3500 - val_accuracy: 0.7800 - val_loss: 0.5660

Epoch 11/25

143/143 ————— 33s 225ms/step - accuracy: 0.8881 - loss:
0.3233 - val_accuracy: 0.7695 - val_loss: 0.6585

Epoch 12/25

143/143 ————— 36s 242ms/step - accuracy: 0.8871 - loss:
0.3118 - val_accuracy: 0.7967 - val_loss: 0.5420

Epoch 13/25


```

143/143 _____ 33s 225ms/step - accuracy: 0.8968 - loss:
0.2987 - val_accuracy: 0.7949 - val_loss: 0.5425
Epoch 14/25
143/143 _____ 33s 225ms/step - accuracy: 0.9036 - loss:
0.2878 - val_accuracy: 0.7826 - val_loss: 0.5634
Epoch 15/25
143/143 _____ 33s 224ms/step - accuracy: 0.8904 - loss:
0.3023 - val_accuracy: 0.7914 - val_loss: 0.5675
Epoch 16/25
143/143 _____ 33s 221ms/step - accuracy: 0.8987 - loss:
0.2955 - val_accuracy: 0.8019 - val_loss: 0.5522
Epoch 17/25
143/143 _____ 35s 241ms/step - accuracy: 0.8953 - loss:
0.2823 - val_accuracy: 0.8256 - val_loss: 0.5001
Epoch 18/25
143/143 _____ 33s 224ms/step - accuracy: 0.9001 - loss:
0.2746 - val_accuracy: 0.7923 - val_loss: 0.5687
Epoch 19/25
143/143 _____ 33s 226ms/step - accuracy: 0.9102 - loss:
0.2562 - val_accuracy: 0.8089 - val_loss: 0.5414
Epoch 20/25
143/143 _____ 33s 226ms/step - accuracy: 0.9055 - loss:
0.2661 - val_accuracy: 0.8142 - val_loss: 0.5007
Epoch 21/25
143/143 _____ 33s 222ms/step - accuracy: 0.9114 - loss:
0.2623 - val_accuracy: 0.8265 - val_loss: 0.5138
Epoch 22/25
143/143 _____ 33s 220ms/step - accuracy: 0.9170 - loss:
0.2403 - val_accuracy: 0.8116 - val_loss: 0.5184

```

Accuracy and Loss of Test Dataset

```

test_loss, test_acc = model.evaluate(test_generator)
print(f"Test Accuracy: {test_acc:.4f}")
print(f"Test Loss: {test_loss:.4f}")

41/41 _____ 11s 268ms/step - accuracy: 0.8060 - loss:
0.4946
Test Accuracy: 0.8680
Test Loss: 0.3694

```

Train Test and Loss Plot

```

def plot_all_curves(history, test_acc, test_loss):
    plt.figure(figsize=(14, 5))

```

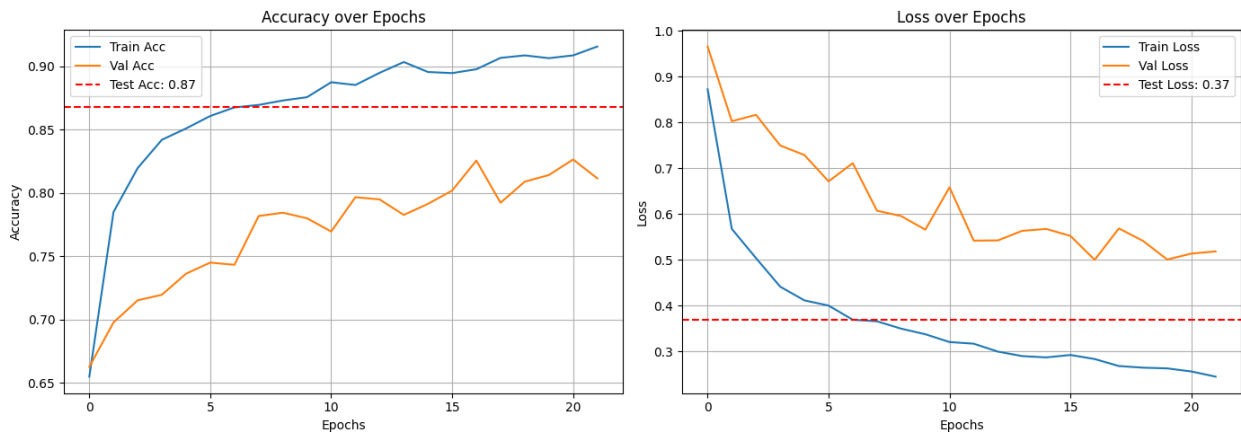
```

# Accuracy plot
plt.subplot(1, 2, 1)
plt.plot(history.history['accuracy'], label='Train Acc')
plt.plot(history.history['val_accuracy'], label='Val Acc')
plt.axhline(y=test_acc, color='r', linestyle='--', label=f'Test
Acc: {test_acc:.2f}')
plt.title('Accuracy over Epochs')
plt.xlabel('Epochs')
plt.ylabel('Accuracy')
plt.legend()
plt.grid(True)

# Loss plot
plt.subplot(1, 2, 2)
plt.plot(history.history['loss'], label='Train Loss')
plt.plot(history.history['val_loss'], label='Val Loss')
plt.axhline(y=test_loss, color='r', linestyle='--', label=f'Test
Loss: {test_loss:.2f}')
plt.title('Loss over Epochs')
plt.xlabel('Epochs')
plt.ylabel('Loss')
plt.legend()
plt.grid(True)

plt.tight_layout()
plt.show()
plot_all_curves(history, test_acc, test_loss)

```



AUC - ROC Curve

```

from sklearn.metrics import roc_curve, auc
from sklearn.preprocessing import label_binarize

```

```

# Predictions
y_pred = model.predict(test_generator)
y_true = test_generator.classes
n_classes = y_pred.shape[1]

# One-hot encode the true labels
y_true_onehot = label_binarize(y_true, classes=np.arange(n_classes))

# Compute ROC curve and AUC for each class
fpr = {}
tpr = {}
roc_auc = {}

plt.figure(figsize=(10, 8))

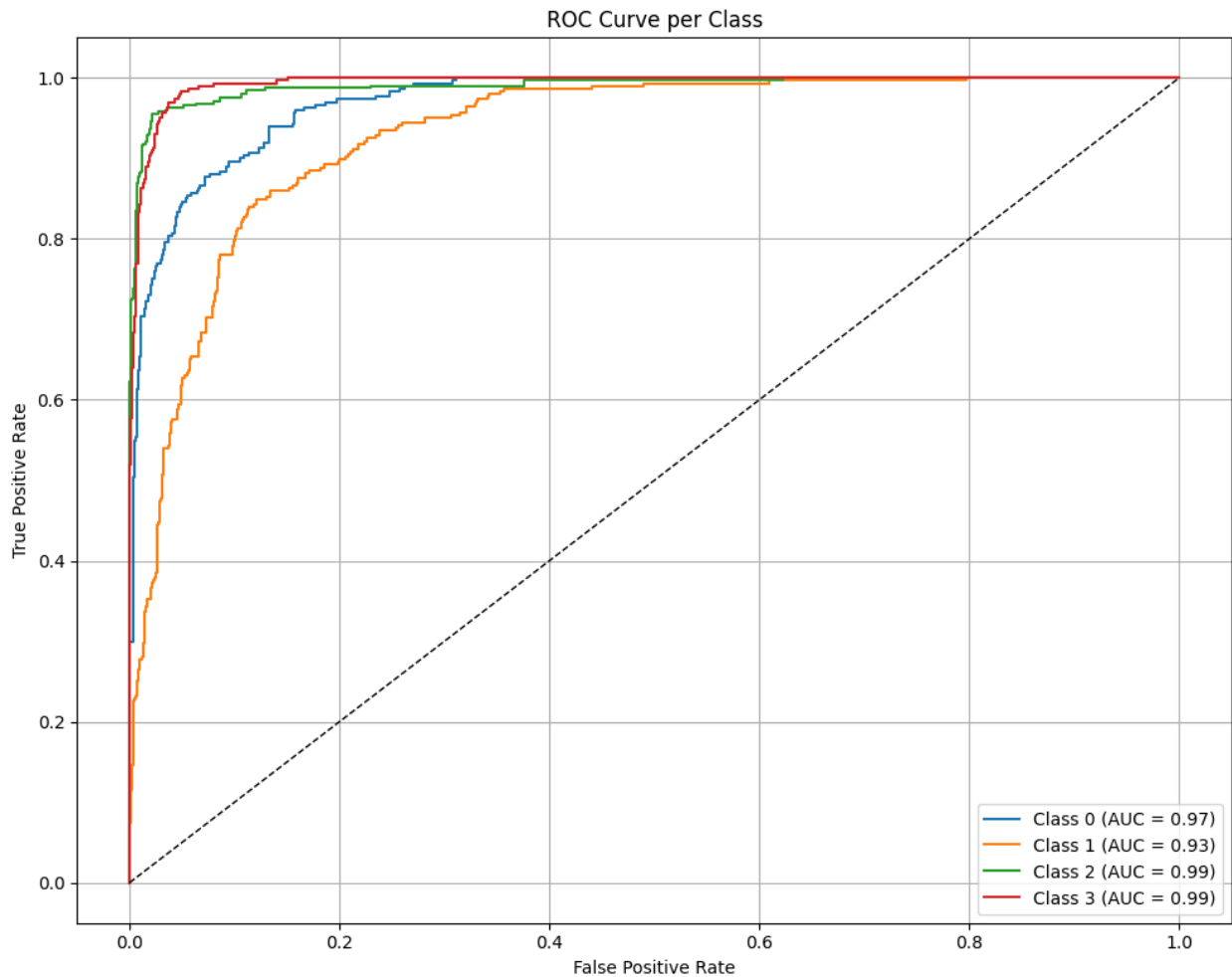
for i in range(n_classes):
    fpr[i], tpr[i], _ = roc_curve(y_true_onehot[:, i], y_pred[:, i])
    roc_auc[i] = auc(fpr[i], tpr[i])
    plt.plot(fpr[i], tpr[i], label=f'Class {i} (AUC =
{roc_auc[i]:.2f})')

# Plot diagonal reference line
plt.plot([0, 1], [0, 1], 'k--', linewidth=1)

# Labels and title
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve per Class")
plt.legend(loc="lower right")
plt.grid(True)
plt.tight_layout()
plt.show()

```

41/41 ————— 31s 416ms/step



Confusion Matrix

```
from sklearn.metrics import confusion_matrix, classification_report,
precision_score, recall_score, f1_score
import seaborn as sns
import pandas as pd

# Get predicted class indices
y_pred_class = np.argmax(y_pred, axis=1)
y_true_class = test_generator.classes
class_labels = list(test_generator.class_indices.keys())

# Confusion matrix
cm = confusion_matrix(y_true_class, y_pred_class)
df_cm = pd.DataFrame(cm, index=class_labels, columns=class_labels)

# Plot the confusion matrix
plt.figure(figsize=(6, 5))
```

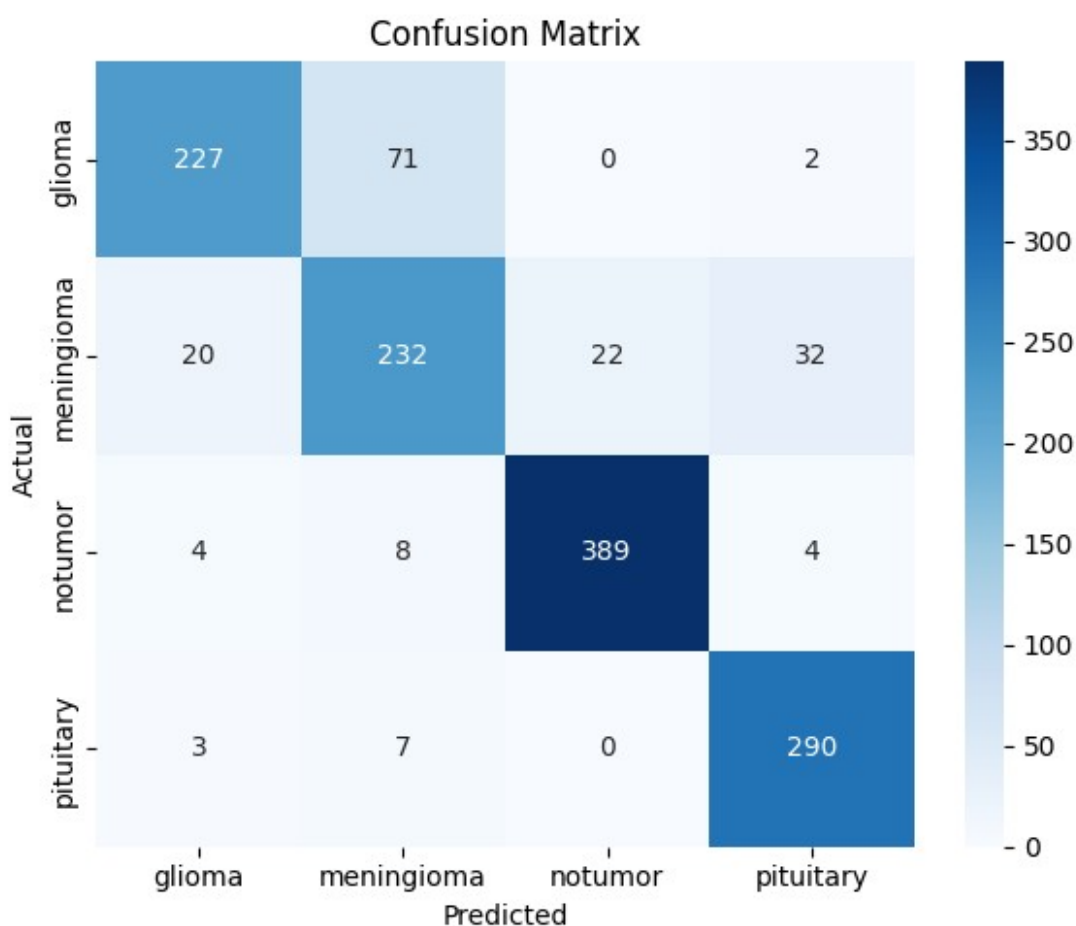
```

sns.heatmap(df_cm, annot=True, fmt='d', cmap='Blues')
plt.title("Confusion Matrix")
plt.ylabel("Actual")
plt.xlabel("Predicted")
plt.tight_layout()
plt.show()

# Classification report (precision, recall, F1)
report = classification_report(y_true_class, y_pred_class,
target_names=class_labels, output_dict=True)
report_df = pd.DataFrame(report).transpose()

# Round values and print neatly
report_df = report_df.round(3)
display(report_df)

```



	precision	recall	f1-score	support
glioma	0.894	0.757	0.819	300.000
meningioma	0.730	0.758	0.744	306.000
notumor	0.946	0.960	0.953	405.000

pituitary	0.884	0.967	0.924	300.000
accuracy	0.868	0.868	0.868	0.868
macro avg	0.863	0.860	0.860	1311.000
weighted avg	0.870	0.868	0.867	1311.000

```
y_true_binarized = label_binarize(y_true_class,
classes=np.arange(n_classes))
```

```
# Compute macro average metrics
```

```
macro_precision = precision_score(y_true_class, y_pred_class,
average='macro')
```

```
macro_recall = recall_score(y_true_class, y_pred_class,
average='macro')
```

```
macro_f1 = f1_score(y_true_class, y_pred_class, average='macro')
```

```
macro_auc = roc_auc_score(y_true_binarized, y_pred, average='macro',
multi_class='ovr')
```

```
# Append MobileNetV2 results
```

```
results_df.loc[len(results_df)] = [
    "InceptionResNetV2",
    test_acc,
    test_loss,
    round(macro_precision, 4),
    round(macro_recall, 4),
    round(macro_f1, 4),
    round(macro_auc, 4)
]
```

```
# Display the updated results
```

```
display(results_df)
```

	Model	Test Accuracy	Test Loss	Precision	Recall	F1-Score
0	VGG16	0.791762	0.622208	0.7854	0.7815	0.7778
1	ResNet50	0.649123	0.968826	0.6300	0.6364	0.6290
2	DenseNet121	0.865751	0.351833	0.8606	0.8585	0.8554
3	EfficientNetB0	0.308924	1.378495	0.0772	0.2500	0.1180
4	MobileNetV2	0.861937	0.362561	0.8592	0.8540	0.8522
5	InceptionV3	0.881007	0.306638	0.8752	0.8743	0.8736
6	Xception	0.883295	0.310204	0.8772	0.8778	0.8759
7	NASANetMobile	0.834477	0.425686	0.8230	0.8251	

```
0.8186
8 InceptionResNetV2      0.868040  0.369368    0.8635  0.8605
0.8600
```

```

      AUC
0  0.9282
1  0.8358
2  0.9781
3  0.6661
4  0.9755
5  0.9800
6  0.9803
7  0.9664
8  0.9715
```

Resnet101

Build The Model

```
from tensorflow.keras.applications import ResNet101
def build_resnet101_model():
    input_tensor = Input(shape=(224, 224, 1))
    rgb_tensor = Lambda(lambda x: tf.image.grayscale_to_rgb(x))
    (input_tensor)

    base_model = ResNet101(weights='imagenet', include_top=False,
input_tensor=rgb_tensor)
    for layer in base_model.layers:
        layer.trainable = False

    x = base_model.output
    x = GlobalAveragePooling2D()(x)
    x = Dense(256, activation='relu')(x)
    x = Dropout(0.5)(x)
    output = Dense(4, activation='softmax')(x)

    model = Model(inputs=input_tensor, outputs=output)
    return model
# Compile the model
model = build_resnet101_model()
model.compile(optimizer=Adam(learning_rate=1e-4),
loss='categorical_crossentropy', metrics=['accuracy'])
```

Downloading data from <https://storage.googleapis.com/tensorflow/keras-applications/resnet/>

```
resnet101_weights_tf_dim_ordering_tf_kernels_notop.h5  
171446536/171446536 _____ 6s 0us/step
```

Call Back and Call The Model

```
early_stop = EarlyStopping(patience=5, restore_best_weights=True)  
checkpoint = ModelCheckpoint("resnet101_brain_tumor.keras",  
save_best_only=True)
```

```
history = model.fit(  
    train_generator,  
    validation_data=val_generator,  
    epochs=25,  
    callbacks=[early_stop, checkpoint]  
)
```

Epoch 1/25

```
143/143 _____ 75s 358ms/step - accuracy: 0.3400 - loss:  
1.5314 - val_accuracy: 0.5329 - val_loss: 1.2316
```

Epoch 2/25

```
143/143 _____ 34s 229ms/step - accuracy: 0.4710 - loss:  
1.1648 - val_accuracy: 0.5048 - val_loss: 1.1828
```

Epoch 3/25

```
143/143 _____ 35s 234ms/step - accuracy: 0.5429 - loss:  
1.0749 - val_accuracy: 0.5916 - val_loss: 1.1449
```

Epoch 4/25

```
143/143 _____ 35s 235ms/step - accuracy: 0.5622 - loss:  
1.0517 - val_accuracy: 0.5706 - val_loss: 1.1180
```

Epoch 5/25

```
143/143 _____ 35s 237ms/step - accuracy: 0.5901 - loss:  
1.0154 - val_accuracy: 0.6214 - val_loss: 1.0918
```

Epoch 6/25

```
143/143 _____ 34s 233ms/step - accuracy: 0.6324 - loss:  
0.9791 - val_accuracy: 0.5951 - val_loss: 1.0627
```

Epoch 7/25

```
143/143 _____ 34s 233ms/step - accuracy: 0.6487 - loss:  
0.9553 - val_accuracy: 0.6310 - val_loss: 1.0402
```

Epoch 8/25

```
143/143 _____ 35s 237ms/step - accuracy: 0.6386 - loss:  
0.9350 - val_accuracy: 0.6284 - val_loss: 1.0174
```

Epoch 9/25

```
143/143 _____ 34s 229ms/step - accuracy: 0.6461 - loss:  
0.9194 - val_accuracy: 0.6459 - val_loss: 1.0040
```

Epoch 10/25

```
143/143 _____ 34s 232ms/step - accuracy: 0.6703 - loss:  
0.8943 - val_accuracy: 0.6635 - val_loss: 0.9837
```

Epoch 11/25

```
143/143 _____ 33s 225ms/step - accuracy: 0.6709 - loss:
```


0.8755 - val_accuracy: 0.6827 - val_loss: 0.9699
Epoch 12/25
143/143 ————— 32s 220ms/step - accuracy: 0.6824 - loss:
0.8545 - val_accuracy: 0.6678 - val_loss: 0.9706
Epoch 13/25
143/143 ————— 34s 229ms/step - accuracy: 0.6781 - loss:
0.8589 - val_accuracy: 0.6757 - val_loss: 0.9552
Epoch 14/25
143/143 ————— 34s 231ms/step - accuracy: 0.6857 - loss:
0.8287 - val_accuracy: 0.6582 - val_loss: 0.9487
Epoch 15/25
143/143 ————— 33s 225ms/step - accuracy: 0.6863 - loss:
0.8265 - val_accuracy: 0.6696 - val_loss: 0.9355
Epoch 16/25
143/143 ————— 34s 231ms/step - accuracy: 0.6965 - loss:
0.8065 - val_accuracy: 0.6643 - val_loss: 0.9226
Epoch 17/25
143/143 ————— 32s 218ms/step - accuracy: 0.6994 - loss:
0.7978 - val_accuracy: 0.6678 - val_loss: 0.9289
Epoch 18/25
143/143 ————— 34s 227ms/step - accuracy: 0.6993 - loss:
0.7909 - val_accuracy: 0.6775 - val_loss: 0.9028
Epoch 19/25
143/143 ————— 32s 219ms/step - accuracy: 0.6899 - loss:
0.7954 - val_accuracy: 0.6784 - val_loss: 0.9041
Epoch 20/25
143/143 ————— 32s 216ms/step - accuracy: 0.6935 - loss:
0.7904 - val_accuracy: 0.6661 - val_loss: 0.9032
Epoch 21/25
143/143 ————— 34s 230ms/step - accuracy: 0.7017 - loss:
0.7667 - val_accuracy: 0.6836 - val_loss: 0.8820
Epoch 22/25
143/143 ————— 32s 218ms/step - accuracy: 0.6939 - loss:
0.7604 - val_accuracy: 0.6687 - val_loss: 0.8914
Epoch 23/25
143/143 ————— 32s 219ms/step - accuracy: 0.7042 - loss:
0.7518 - val_accuracy: 0.6748 - val_loss: 0.8984
Epoch 24/25
143/143 ————— 33s 221ms/step - accuracy: 0.6927 - loss:
0.7611 - val_accuracy: 0.6845 - val_loss: 0.8925
Epoch 25/25
143/143 ————— 33s 223ms/step - accuracy: 0.7004 - loss:
0.7548 - val_accuracy: 0.6696 - val_loss: 0.8902

Accuracy and Loss Of The TEst Dataset

```
test_loss, test_acc = model.evaluate(test_generator)
print(f"Test Accuracy: {test_acc:.4f}")
print(f"Test Loss: {test_loss:.4f}")
```

41/41 ————— 6s 156ms/step - accuracy: 0.5608 - loss: 1.1176
Test Accuracy: 0.6621
Test Loss: 0.9177

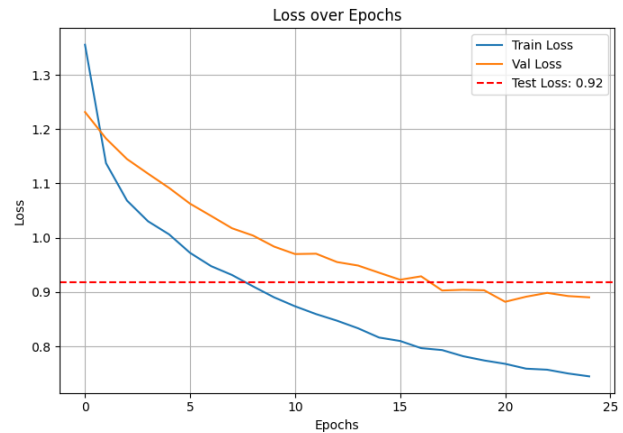
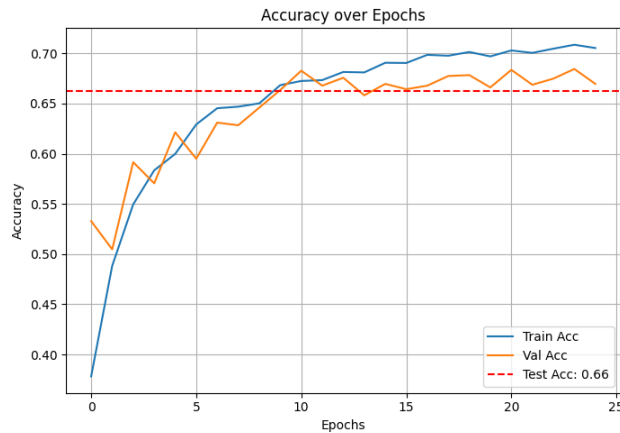
Test Train and Val Accuracy

```
def plot_all_curves(history, test_acc, test_loss):
    plt.figure(figsize=(14, 5))

    # Accuracy plot
    plt.subplot(1, 2, 1)
    plt.plot(history.history['accuracy'], label='Train Acc')
    plt.plot(history.history['val_accuracy'], label='Val Acc')
    plt.axhline(y=test_acc, color='r', linestyle='--', label=f'Test
Acc: {test_acc:.2f}')
    plt.title('Accuracy over Epochs')
    plt.xlabel('Epochs')
    plt.ylabel('Accuracy')
    plt.legend()
    plt.grid(True)

    # Loss plot
    plt.subplot(1, 2, 2)
    plt.plot(history.history['loss'], label='Train Loss')
    plt.plot(history.history['val_loss'], label='Val Loss')
    plt.axhline(y=test_loss, color='r', linestyle='--', label=f'Test
Loss: {test_loss:.2f}')
    plt.title('Loss over Epochs')
    plt.xlabel('Epochs')
    plt.ylabel('Loss')
    plt.legend()
    plt.grid(True)

    plt.tight_layout()
    plt.show()
plot_all_curves(history, test_acc, test_loss)
```



AUC - ROC Curve

```
from sklearn.metrics import roc_curve, auc
from sklearn.preprocessing import label_binarize

# Predictions
y_pred = model.predict(test_generator)
y_true = test_generator.classes
n_classes = y_pred.shape[1]

# One-hot encode the true labels
y_true_onehot = label_binarize(y_true, classes=np.arange(n_classes))

# Compute ROC curve and AUC for each class
fpr = {}
tpr = {}
roc_auc = {}

plt.figure(figsize=(10, 8))

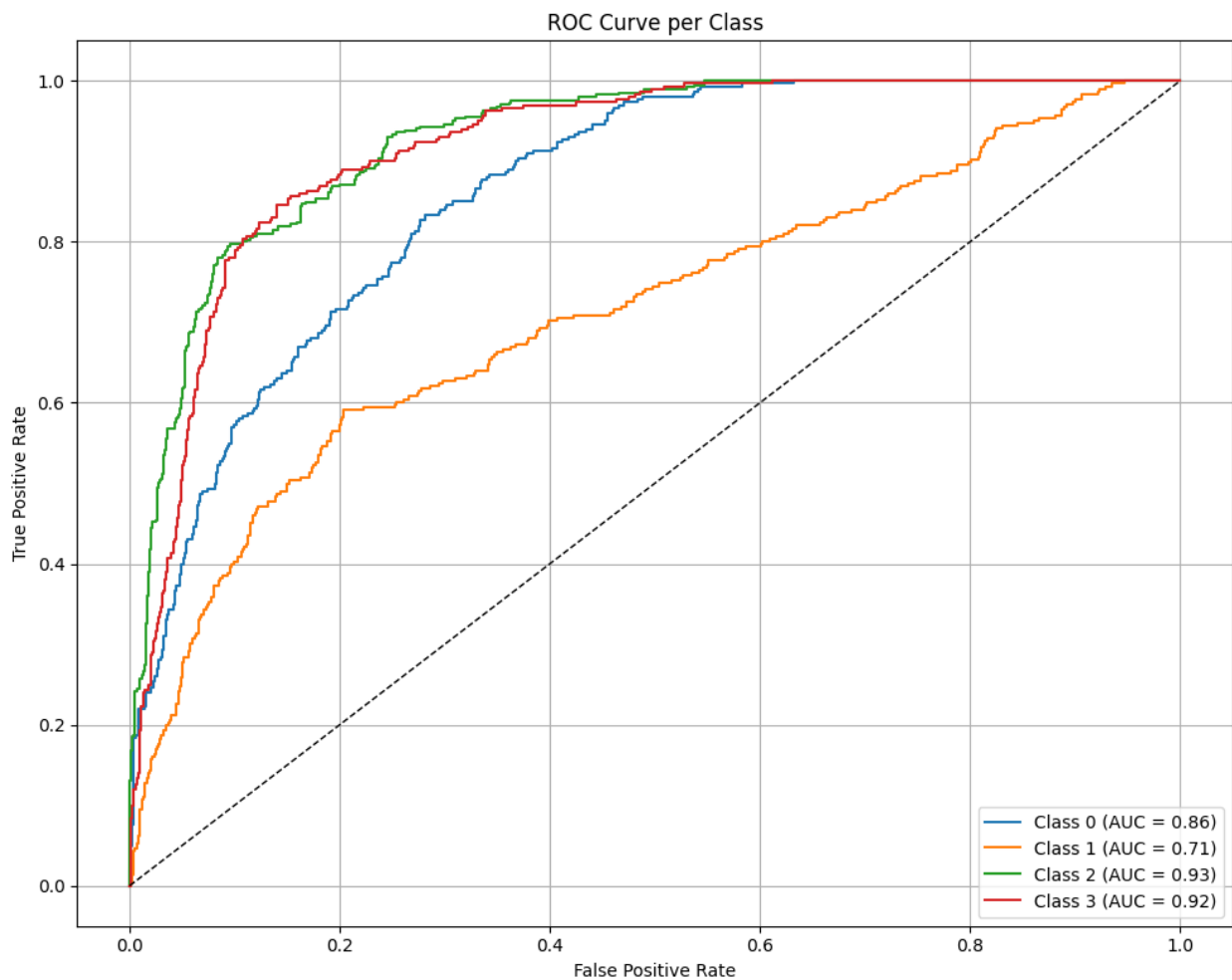
for i in range(n_classes):
    fpr[i], tpr[i], _ = roc_curve(y_true_onehot[:, i], y_pred[:, i])
    roc_auc[i] = auc(fpr[i], tpr[i])
    plt.plot(fpr[i], tpr[i], label=f'Class {i} (AUC = {roc_auc[i]:.2f})')

# Plot diagonal reference line
plt.plot([0, 1], [0, 1], 'k--', linewidth=1)

# Labels and title
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
```

```
plt.title("ROC Curve per Class")
plt.legend(loc="lower right")
plt.grid(True)
plt.tight_layout()
plt.show()
```

41/41 ————— 20s 325ms/step



Confusion Matrix

```
from sklearn.metrics import confusion_matrix, classification_report,
precision_score, recall_score, f1_score
import seaborn as sns
import pandas as pd
```

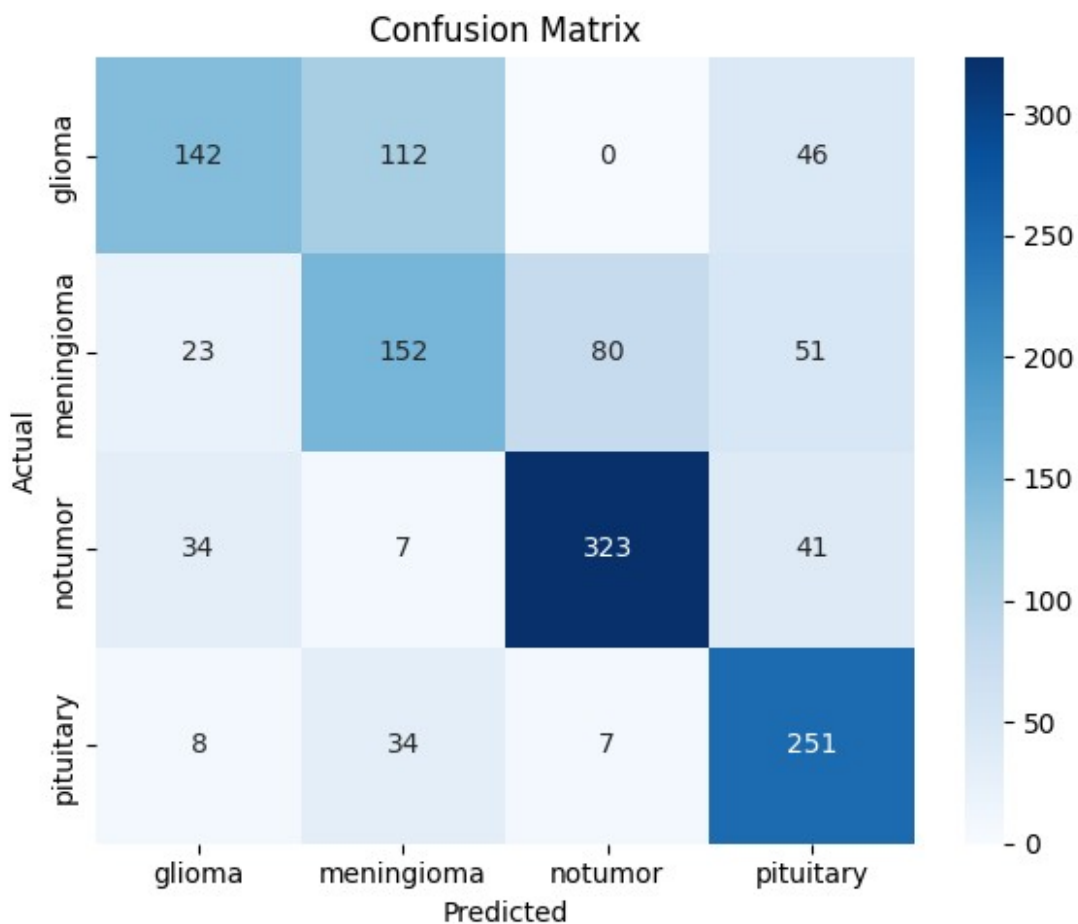
```
# Get predicted class indices
y_pred_class = np.argmax(y_pred, axis=1)
y_true_class = test_generator.classes
class_labels = list(test_generator.class_indices.keys())

# Confusion matrix
cm = confusion_matrix(y_true_class, y_pred_class)
df_cm = pd.DataFrame(cm, index=class_labels, columns=class_labels)

# Plot the confusion matrix
plt.figure(figsize=(6, 5))
sns.heatmap(df_cm, annot=True, fmt='d', cmap='Blues')
plt.title("Confusion Matrix")
plt.ylabel("Actual")
plt.xlabel("Predicted")
plt.tight_layout()
plt.show()

# Classification report (precision, recall, F1)
report = classification_report(y_true_class, y_pred_class,
target_names=class_labels, output_dict=True)
report_df = pd.DataFrame(report).transpose()

# Round values and print neatly
report_df = report_df.round(3)
display(report_df)
```



	precision	recall	f1-score	support
glioma	0.686	0.473	0.560	300.000
meningioma	0.498	0.497	0.498	306.000
notumor	0.788	0.798	0.793	405.000
pituitary	0.645	0.837	0.729	300.000
accuracy	0.662	0.662	0.662	0.662
macro avg	0.654	0.651	0.645	1311.000
weighted avg	0.664	0.662	0.656	1311.000

```
y_true_binarized = label_binarize(y_true_class,
classes=np.arange(n_classes))
```

```
# Compute macro average metrics
```

```
macro_precision = precision_score(y_true_class, y_pred_class,
average='macro')
```

```
macro_recall = recall_score(y_true_class, y_pred_class,
average='macro')
```

```
macro_f1 = f1_score(y_true_class, y_pred_class, average='macro')
```

```
macro_auc = roc_auc_score(y_true_binarized, y_pred, average='macro',
multi_class='ovr')
```

```
# Append MobileNetV2 results
results_df.loc[len(results_df)] = [
    "resnet101",
    test_acc,
    test_loss,
    round(macro_precision, 4),
    round(macro_recall, 4),
    round(macro_f1, 4),
    round(macro_auc, 4)
]
```

```
# Display the updated results
display(results_df)
```

	Model	Test Accuracy	Test Loss	Precision	Recall
F1-Score \					
0	VGG16	0.791762	0.622208	0.7854	0.7815
0.7778					
1	ResNet50	0.649123	0.968826	0.6300	0.6364
0.6290					
2	DenseNet121	0.865751	0.351833	0.8606	0.8585
0.8554					
3	EfficientNetB0	0.308924	1.378495	0.0772	0.2500
0.1180					
4	MobileNetV2	0.861937	0.362561	0.8592	0.8540
0.8522					
5	InceptionV3	0.881007	0.306638	0.8752	0.8743
0.8736					
6	Xception	0.883295	0.310204	0.8772	0.8778
0.8759					
7	NASANetMobile	0.834477	0.425686	0.8230	0.8251
0.8186					
8	InceptionResNetV2	0.868040	0.369368	0.8635	0.8605
0.8600					
9	InceptionResNetV2	0.662090	0.917726	0.6544	0.6511
0.6447					
10	resnet101	0.662090	0.917726	0.6544	0.6511
0.6447					

	AUC
0	0.9282
1	0.8358
2	0.9781
3	0.6661
4	0.9755
5	0.9800
6	0.9803
7	0.9664
8	0.9715

```
9    0.8541
10   0.8541
```

```
# Create DataFrame
```

```
results_df = pd.DataFrame({
    "Model": [
        "VGG16", "ResNet50", "DenseNet121", "EfficientNetB0",
        "MobileNetV2",
        "InceptionV3", "Xception", "NASNetMobile",
        "InceptionResNetV2", "ResNet101"
    ],
    "Test Accuracy": [0.791762, 0.649123, 0.865751, 0.308924,
0.861937,
                        0.881007, 0.883295, 0.834477, 0.868040,
0.662090],
    "F1-Score": [0.7778, 0.6290, 0.8554, 0.1180, 0.8522,
0.8736, 0.8759, 0.8186, 0.8600, 0.6447],
    "AUC": [0.9282, 0.8358, 0.9781, 0.6661, 0.9755,
0.9800, 0.9803, 0.9664, 0.9715, 0.8541]
})
```

```
# Plotting
```

```
fig, axes = plt.subplots(3, 1, figsize=(14, 12), sharex=True)
```

```
metrics = ["Test Accuracy", "F1-Score", "AUC"]
colors = ["cornflowerblue", "mediumseagreen", "indianred"]
```

```
for i, metric in enumerate(metrics):
    axes[i].bar(results_df["Model"], results_df[metric],
color=colors[i])
    axes[i].set_title(metric, fontsize=14, weight='bold')
    axes[i].set_ylabel("Score", fontsize=12)
    axes[i].set_ylim(0, 1.05)
    axes[i].grid(True, axis='y', linestyle='--', linewidth=0.6)
```

```
plt.xticks(rotation=45, ha='right')
plt.suptitle("CNN Model Performance on Brain Tumor Classification",
fontsize=16, weight='bold')
plt.tight_layout(rect=[0, 0, 1, 0.97])
```

```
# Save the plot
```

```
plt.savefig("cnn_model_comparison.png", dpi=300)
plt.show()
```


CNN Model Performance on Brain Tumor Classification

